Measuring Sustainable Accessibility at the Neighbourhood scale:

A GIS-based multimodal index approach to measure sustainable access to daily opportunities

Master Thesis
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Amsterdam || 26 June 2026
Msc. Metropolitan Analysis Design & Engineering

Abstract

Providing access to daily opportunities (i.e. work, school, employment locations, supermarkets, or parks) through sustainable transport modes is an increasingly prominent goal in transport planning. To support this goal, transport planners require accurate accessibility measures to improve how easily people can reach daily opportunities using only sustainable transport modes in their neighbourhood. However, existing accessibility measures are often either too complex for practical application or too simplified to capture sustainable accessibility to daily opportunities. This thesis addresses this gap by proposing and testing a transparent and decomposable GIS-based method to measure sustainable accessibility at the neighbourhood level in the Netherlands.

Sustainable accessibility is defined in this research as the extent to which people can reach daily opportunities within acceptable travel times by walking, cycling, and multimodal combinations of these active modes with public transport. To operationalize this concept, a Sustainable Accessibility Index (SAI) is developed. This composite index combines the number of reachable opportunities, the reachability of those opportunities within acceptable travel times, and the sufficiency of the reachable opportunities for daily life. The index is calculated from multiple origin points within each neighbourhood and aggregated into a neighbourhood-level sustainable accessibility (SA) score. This makes it possible to calculate an overall neighbourhood SA score, while also being able to decompose this score into underlying layers. These specific layer SA scores could help show whether low accessibility in the neighbourhood is related to longer travel times, too few reachable opportunities, weak access to specific opportunities, or spatial inequalities within a neighbourhood.

The method is applied to nine neighbourhoods in the Netherlands across different degrees of urbanisation. The results show that sustainable accessibility is generally higher in more urbanised neighbourhoods, but accessibility differences are not only explained by urbanisation. Local opportunity distribution, transport network systems, mode-specific travel times, and origin locations within neighbourhoods also influence the SA scores. Additionally, the results show that the multimodal transport combinations of walking and cycling combined with public transport tend to extend access beyond the immediate neighbourhood. However, in highly- and moderate urbanised neighbourhoods, cycling can outperform the multimodal combination. This suggests that public transport adds the most value in neighbourhoods where walking or cycling alone does not provide sufficient access to daily opportunities. Furthermore, the decomposed results show that neighbourhood averages can hide substantial differences in sustainable accessibility within a neighbourhood. Lastly, the sensitivity analysis indicates that the main spatial and modal patterns are generally stable but exact SA scores and interpretations depend on methodological choices such as travel-time thresholds, the way opportunities are counted, and aggregation rules of the SAI.

The main value of the SAI is that it makes sustainable accessibility visible, measurable, and explainable. The resulting index could help transport planners identify what kind of accessibility problem exists and select targeted intervention to improve everyday sustainable accessibility.

Keywords: *Sustainable accessibility, Multimodal accessibility, Composite accessibility index, GIS-based modelling*

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1. INTRODUCTION

1.1 Background

Accessibility

The transport planning paradigm is increasingly more concerned with enabling people to reach the places and activities required for everyday life rather than with simply the movement of people and goods from A to B. Trips to work, education, shops, services, and social activities are therefore better understood not simply as mobility needs, but as access needs. In this sense, mobility is a means to an end, whereas accessibility concerns the ability to reach desired destinations and participate in daily life (Litman, 2025; Miller, 2018; Preston & Rajé, 2007).

The most used definition of accessibility in transport planning literature is the definition by Geurs and van Wee (2004, p. 128): "the extent to which land use and transport systems enable individuals or goods to reach destinations or activities by means of a (combination of) transport mode(s)". This definition highlights the interaction between two core components: the land-use system and the transport system. The land-use system shows the spatial distribution of activities, such as jobs, services, and amenities, are located. The transport system determines how easily these opportunities can be reached (Straatemeier & Bertolini, 2020).

Accessibility is an important goal to strive for in governance. High levels of accessibility can support economic productivity and competitiveness by enabling people to participate in the labour market and allowing businesses to reach workers, consumers, and suppliers (International Transport Forum, 2017). Conversely, low accessibility can contribute to social exclusion when people are unable to reach employment, education, services, or other essential activities (Lucas, 2012; Preston & Rajé, 2007). Accessibility is therefore relevant not only to transport efficiency, but also to broader economic and social outcomes in society.

Sustainability

In recent decades, accessibility has increasingly been discussed in relation to environmental sustainability. Cities face the dual challenge of accommodating population growth while reducing emissions and other environmental impacts, without compromising people's ability to reach daily activities (Lättman et al., 2018; UN-Habitat, 2009). Transport is central to this challenge, as private car use contributes significantly to CO_2 emissions and other environmental pressures (European Environment Agency., 2022; May et al., 2003). As a result, accessibility cannot be considered separately from the environmental impacts of the transport modes used to achieve it.

European urban policies and Dutch mobility strategies have increasingly responded to this challenge by promoting sustainable mobility (European Commission, 2021; Nieuwenhuijsen & Khreis, 2016). Sustainable mobility generally refers to transport systems and modes that reduce the environmental impacts of travel while maintaining access to activities and opportunities (Fatorachian & Kazemi, 2025; Zhou et al., 2023). Walking, cycling, and public transport are commonly identified as central sustainable transport modes as they produce lower emissions than private car use, particularly when assessed per passenger kilometre (Etukudoh et al., 2024; Limburg & Nijënstein, 2021; Mouratidis et al., 2023). Shared mobility (sharing of vehicles

such as GreenWheels and OV bikes) and micromobility (lightweight vehicles going under 25 km/h such as (e-)bikes and steps) may also contribute to sustainable transport systems, but they are not consistently represented in the main travel datasets used in Dutch research and policy (CBS, 2024; Hamersma & Roeleven, 2024).

Another strategy in sustainable mobility planning is the development of multimodal transport systems. Multimodality refers to the integration of multiple transport modes within a single trip, such as cycling to a railway station or walking to a bus stop (Fu et al., 2024; Litman, 2024; Nes, 2002). Such combinations can substantially extend the range of destinations that people can reach without relying on a private car.

Sustainable Accessibility

The relationship between accessibility and sustainability has become increasingly important in European urban planning. Many cities have introduced car-restrictive policies, including low-traffic zones, pedestrianised areas, and car-free or car-light neighbourhoods, to reduce emissions and improve urban liveability (Hrelja & Rye, 2023; Mouratidis, 2025; Nieuwenhuijsen & Khreis, 2016; Selzer, 2021). In the Dutch Randstad region, cities such as Amsterdam, Rotterdam, Utrecht, and The Hague have incorporated sustainability and accessibility goals into mobility visions and redevelopment plans. These policies aim to enable residents to participate in daily life using sustainable transport modes rather than relying primarily on private cars (Gemeente Amsterdam, 2025; Gemeente Den Haag, 2021; Gemeente Rotterdam, 2020; Provincie Utrecht, 2024).

At the national level, the Dutch Mobility Vision also emphasises “future-proof accessibility”. This refers to a transport system that provides access to essential services in a sustainable way, thereby contributing to both environmental and social goals (Ministerie van Infrastructuur en Waterstaat, 2023). Sustainable accessibility has therefore become an important concept in transport planning as it connects access to opportunities with the environmental performance of the transport system (Bertolini et al., 2005; Cheng et al., 2007; Gil Solá et al., 2018; Le Clercq & Bertolini, 2003).

1.2 Problem Statement and Research Gap

Policy ambitions to create car-free, car-light, or sustainable urban environments do not automatically ensure that residents can live well without a private car. Previous research by Snellen and Hilbers (2005) shows that some developments presented as sustainable or car-free still experience high levels of car use when local amenities are limited, public transport quality is insufficient, or cycling networks are incomplete. In such cases, residents may continue to rely on private cars to compensate for gaps in the sustainable transport network, particularly when certain activities cannot be reached conveniently by walking, cycling, or public transport (Borgers et al., 2008; Scheurer, 2001; Selzer & Lanzendorf, 2022).

This issue is particularly relevant at the neighbourhood level, where many sustainable urban development strategies are designed for and implemented (Khatibi et al., 2025). Neighbourhoods are the scale at which residents experience the practical relationship between land use, transport infrastructure, and access to daily needs. A neighbourhood may be designed with sustainability goals in mind, but if essential services or sustainable transport connections are insufficient, residents may still depend on private cars for everyday activities.

A related problem is the lack of coordination between land-use planning and transport planning in urban areas (Bertolini et al., 2005; Hull, 2008; Silva et al., 2017; van Geet et al., 2019). This can create a mismatch between urban design ambitions and the actual accessibility provided by the transport system. Therefore, reducing car use requires more than limiting car access or parking supply. New and existing neighbourhoods must also provide sufficient access to essential services through sustainable transport modes.

Over the years, accessibility has been widely researched and a plethora of accessibility measures have been proposed (Boisjoly & El-Geneidy, 2017; Curl et al., 2011; Geurs & van Wee, 2004, 2023; Handy & Niemeier, 1997; Hansen, 1959). However, many theoretically advanced approaches, such as utility-based and space-time measures, are complex and data intensive (Handy & Clifton, 2001; Mazzulla & Pirrone, 2024; Van Wee, 2021). This limits their usability for planning practice, where accessibility metrics need to be understandable, reproducible, and applicable with available data. As a result, accessibility measurement often involves a trade-off between theoretical richness and practical usability (Bertolini et al., 2005; Straatemeier, 2008).

Despite growing agreement on the importance of sustainable accessibility, defining and measuring it remains challenging. Three main challenges can be identified. First, the concept of sustainable accessibility is ambiguous and context dependent, which means that it must be defined clearly for the specific purpose of a research (Litman, 2025; Miller, 2018; Wu & Levinson, 2020). Second, many accessibility measures focus on single modes, such as comparing cycling accessibility with car accessibility, rather than assessing multimodal combinations of sustainable transport modes (Siddiq & D. Taylor, 2021; Van Wee, 2021). Third, scholars have called for accessibility metrics that are transparent, reproducible, and interpretable for planning practice (Bertolini et al., 2005; Handy & Clifton, 2001).

Therefore, this research addresses a methodological research gap which states that there is a need for a simple, transparent, and reproducible measure of sustainable accessibility that accounts for multimodal sustainable transport at the neighbourhood level. Such a measure can help planners compare neighbourhoods and assess whether residents can reach daily opportunities without relying on a private car.

1.3 Research Aim and Questions

The aim of this research is to develop and test a methodology that measures sustainable accessibility at the neighbourhood level in a transparent and reproducible way. Moreover, a metric that specifically measures how well residents can access daily opportunities using only sustainable transport modes is developed. This metric is then used to compare sustainable accessibility across different neighbourhood contexts in the Netherlands.

The main research question is:
How can sustainable accessibility to daily opportunities be measured at the neighbourhood level using multimodal sustainable transport?

To answer this question, four sub-questions are formulated.

SQ1: How can sustainable accessibility be operationalised into a composite index that remains transparent and interpretable for planning practice?

In the first sub-question the Sustainable Accessibility Index (SAI) is developed. The emphasis is on creating a method that is technically grounded while remaining clear and usable for practitioners. This research question will be answered in Chapter 2.

SQ2: What does the Sustainable Accessibility Index reveal about accessibility levels in neighbourhoods with different degrees of urbanisation?

In the second sub-question the SAI is applied to case study areas. The purpose is to compare accessibility patterns across different spatial contexts and assess how well each neighbourhood supports access to daily opportunities through sustainable transport modes. This question will be investigated in Chapter 4 and further interpreted in Chapter 5.

SQ3: How sensitive is the proposed Sustainable Accessibility Index to key methodological assumptions, such as travel time limits, component calculation changes, and aggregation choices?

In this sub-question the robustness of the SAI is assessed by testing how results change under different methodological choices. This helps assess whether the conclusions remain stable when key assumptions are adjusted. This research question will be explored in Chapter 4 and 5.

SQ4: What do the results and sensitivity analyses imply for the interpretation and use of the Sustainable Accessibility Index in neighbourhood-level planning practice?

Finally, in the last sub-question how the SAI can be used by planners and policymakers is discussed. For example, how to diagnose accessibility gaps, compare neighbourhoods, and the cautions for the interpretation of the SAI are discussed. Chapter 5 and 6 will answer the last research question.

1.4 Research Outline

The remainder of this report is structured as follows. In Chapter 2, the theoretical foundation of the research is established. Key concepts and components that inform the development of the index are defined. Then, the research design and methodology are presented in Chapter 3. Here, the development of the Sustainable Accessibility Index, including the selected indicators, data sources, case study selection, calculation procedure, and analytical methods are explained. In Chapter 4, the empirical results of the index application are presented. In Chapter 5, the interpretation of the results and their implications for neighbourhood-level sustainable accessibility planning are discussed. Subsequently, the findings in relation to existing accessibility literature and the methodological limitations of the research are discussed in Chapter 6. Additionally, the recommendations for planning practice are discussed in Chapter 6. Finally, in Chapter 7 the conclusion is presented.

2. THEORETHICAL FRAMEWORK

In this chapter, the theoretical framework to understand sustainable accessibility and important facets to create a composite index are introduced. First, in Section 2.1 sustainable accessibility is defined and positioned within the broader accessibility literature. Then, accessibility measurement approaches are reviewed and the criteria used to select an appropriate measure for this research are established in Section 2.2. Finally, the core components of accessibility are discussed and synthesized into the conceptual framework used for the sustainable accessibility index in Section 2.3.

2.1 Sustainable Accessibility: Conceptual Definition

As discussed in Chapter 1, accessibility is understood as the extent to which land-use and transport systems enable people to reach activities, destinations, or opportunities. Accessibility is not produced by transport infrastructure alone or by the spatial distribution of opportunities alone, but by the interaction between where activities are located and how they can be reached.

This relational interpretation distinguishes accessibility from mobility. Mobility relates to movement, whereas accessibility concerns what can be reached through movement (Litman, 2025). A transport system could allow for high levels of travel without necessarily improving access to everyday opportunities. Conversely, accessibility could improve not only through faster movement, but also through land-use distribution that locate essential activities closer to residential areas. This distinction is central to accessibility-based planning because it shifts the focus from movement itself to the practical ability to participate in social, economic, and everyday activities (Handy & Niemeier, 1997; Miller, 2020).

A sustainability perspective adds an additional dimension. High accessibility based primarily on private car use conflicts with objectives related to emissions reduction, air quality, noise, land consumption, public space, and social inclusion. Banister's sustainable mobility paradigm argues that transport planning should move beyond conventional assumptions of traffic growth and travel-time minimisation, and should instead support more sustainable patterns of movement through integrated land-use planning, shorter distances, active travel, and public transport (Banister, 2008). Similarly, Bertolini et al. (2005) argue that sustainable accessibility offers a framework for integrating land use and transport planning since it evaluates what can be reached, and by which means, rather than treating traffic performance as an end in itself.

Building on this literature, sustainable accessibility in this research is defined as:

> **The ease with which individuals can reach essential opportunities for daily life within acceptable travel times using only sustainable transport modes.**

There are two main elements in this definition that require further clarification, namely ease and acceptable travel times. Ease, refers to the degree of difficulty involved in reaching essential opportunities. It is a concept that can be assessed through factors such as travel time, distance and available transport modes and ultimately means the effort that is required to reach destinations. In this research, ease is mainly interpreted through the travel time required to reach essential opportunities by sustainable transport modes.

Acceptable travel times are used to distinguish destinations that are technically reachable with meaningfully accessible destinations. Long travel times could make a destination unrealistic to use. Therefore, acceptable travel times refer to travel time thresholds within which opportunities can reasonably be considered accessible for daily life. The travel time thresholds used for this research are based on data gathered from the Dutch Institute for Transport Policy Analysis and the Statistics Netherlands, and will be elaborated on in the next chapter.

The chosen definition for sustainable accessibility contains four conceptual choices. First, the focus is on residents, because the research evaluates the accessibility provided by neighbourhood environments. Second, the focus is on essential daily opportunities, because the analysis concerns everyday access rather than solely one or all possible destinations. Third, the relevant transport modes are walking, cycling, the multimodal combinations of active modes with public transport, because this research is concerned with sustainable transport access compared to the less sustainable private car mode. Fourth, the definition includes behaviourally plausible travel times, because destinations that are technically reachable but require excessive travel times do not contribute to everyday accessibility.

This definition also establishes the neighbourhood-level perspective adopted in this research. Neighbourhoods are not assumed to contain all relevant opportunities within their administrative boundaries. Rather, they are treated as residential origin points from which access to daily opportunities can be assessed. This allows the Sustainable Accessibility Index (SAI) to evaluate how far different neighbourhood environments support sustainable access.

2.2 Measuring Accessibility: Approaches and Selection Criteria

The way accessibility is measured, determines how accessibility will be perceived, assessed and interpreted. Since accessibility can be measured in a variety of methods, the chosen measure should match the objective of the study. In this research, the selected method should be able to measure sustainable accessibility that allows for comparison between different types of neighbourhoods.

In the following section, the main accessibility measurement approaches are reviewed. Then, the selection criteria used to evaluate the approaches are defined. Finally, the choice of measurement is justified.

2.2.1 Main Accessibility Measurement Approaches

Accessibility measures are often categorised as location-based or person-based measures (see reviews on accessibility measures from these authors: Dianin et al., (2022); Geurs & van Wee, (2023); Miller, (2020); Vale et al., (2016); Wu & Levinson, (2020). Table 1 summarises the main accessibility measurement approaches and describes their relevance for this research based on the reviews of accessibility measures as described above.

Table 1: Accessibility measurement approaches and relevance for this research.

Measurement family	Measurement approach	Main idea	Strengths	Limitations	Relevance for this research
Infrastructure-based		Assess transport system performance, such as network speed, congestion, service frequency, or infrastructure supply.	Relatively easy to calculate and useful for evaluating transport network performance.	Do not directly measure access to opportunities because they often omit the land-use component.	Not suitable as the main approach because this research evaluates access to daily opportunities, not transport performance alone.
Location based	Travel-cost or contour measures	Assess whether a destination or area can be reached within a specified time, distance, or cost threshold.	Simple, intuitive, and easy to communicate; useful for service standards and policy targets.	Sensitive to threshold choice; treat destinations just inside and outside the threshold differently.	Relevant because the research uses behaviourally plausible travel-time limits, but threshold sensitivity must be tested.
	Cumulative opportunity measures	Count the number of opportunities reachable within a specified time or distance threshold.	Easy to calculate and interpret; directly link transport and land-use systems; widely used in accessibility planning.	Assume all opportunities within the threshold are equally accessible; may ignore opportunity quality, competition, and gradual distance decay.	Forms the main basis of the SAI because it supports transparent neighbourhood comparison.
	Gravity or potential measures	Weight opportunities by travel impedance, so closer destinations contribute more than distant destinations.	Avoid hard binary cut-offs and represent distance decay more gradually.	Require choosing and calibrating an impedance function; results may be less intuitive for non-specialist users.	Partly relevant. Time-based weighting can introduce a degree of distance decay while preserving interpretability.

Person-based	space-time measures	Represent accessibility through individual activity schedules, time budgets, and feasible activity spaces.	Strong representation of temporal and individual constraints; useful for detailed equity and behavioural analysis.	Data intensive and difficult to translate into a simple neighbourhood-level index.	Not used as the main approach. Temporal constraints are simplified through travel-time thresholds and weighting.
Utility-based		Estimate accessibility as the utility or welfare value of available destination and mode choices, often using discrete choice models.	Strong behavioural and economic foundation; useful for welfare appraisal.	Require detailed behavioural data and model estimation; results can be difficult to communicate.	Not suitable for the main index because the aim is a transparent, reproducible, planning-oriented measure.
Composite indices		Combine several indicators or dimensions into one index or structured dashboard.	Suitable for multidimensional concepts; can combine modes, opportunities, and temporal assumptions.	Risk opaque weighting, double counting, and subjective aggregation if not clearly justified.	Suitable for the SAI if the index remains theoretically grounded, decomposable, and sensitivity-tested.

Table 1 highlights the central methodological trade-off. More complex measures may better capture behaviour, welfare, or individual constraints, but they often require detailed data and may be difficult to communicate. Simpler measures are more transparent and easier to operationalise, but they simplify behavioural, temporal, and individual dimensions.

The proposed SAI focuses on a location-based composite approach as a deliberate compromise between these demands. It is location-based because it evaluates access from residential origins to spatially distributed opportunities. It is composite since sustainable accessibility cannot be adequately represented by a single mode, destination category, or travel-time assumption using only one indicator.

2.2.2 Selection Criteria for this Research

Four criteria guide the selection of the SAI measurement approach.

First, the measure must have a clear theoretical basis. It should represent the interaction between land use and transport, and it should be sensitive to variation in opportunity distribution, travel time, and temporal constraints. This criterion follows from Geurs and van Wee's (2004) argument that accessibility measures should capture the core components of accessibility rather than transport performance alone.

Second, the measure must be operationally feasible. This research compares neighbourhood-level accessibility across several sustainable modes and opportunity categories. A method requiring extensive individual behavioural data, stated preferences, or complete activity schedules would therefore be inappropriate for the purpose of producing a reproducible planning-oriented measure (Handy & Niemeier, 1997; Miller, 2020).

Third, the measure must be interpretable and communicable. Accessibility indicators are more likely to influence planning when their meaning can be understood by researchers, planners, and policymakers. This is especially important as accessibility measures often face implementation barriers when they are perceived as too complex, insufficiently transparent, or poorly connected to planning decisions (Boisjoly & El-Geneidy, 2017; Silva et al., 2017).

Fourth, the measure must support transparent composite-index construction. Since the SAI combines multiple dimensions, it must follow the logic of composite indicator design: a clear theoretical framework, justified indicator selection, explicit weighting and aggregation choices, and sensitivity analysis. The OECD handbook emphasises that composite indicators are only analytically useful when methodological choices are transparent and tested for robustness (Nardo et al., 2005). Saisana et al. (2005) similarly argue that uncertainty and sensitivity analysis are necessary for assessing the reliability of composite indicators.

In summary, the literature indicates that accessibility measures for this research should meet two main requirements. First, an accessibility measure should be transparent and results should be easy to understand and communicable to users. Second, the selection of indicators, methodological choices such as normalization, weighting and sensitivity tests should be transparently discussed.

2.2.3 Justification of the Selected Approach

The SAI is designed as a transparent, location-based composite index. This approach is not intended to provide a complete representation of all dimensions of accessibility. Rather, it is selected because it offers an appropriate balance between conceptual validity, operational feasibility, and planning interpretability.

A location-based measure is appropriate as the research evaluates the accessibility provided by neighbourhood environments from residential origins. A composite structure is appropriate since sustainable accessibility depends on multiple dimensions, including transport mode, opportunity type, travel-time acceptability, and opportunity sufficiency.

2.3 Conceptual Framework Sustainable Accessibility

Accessibility is multidimensional. Geurs and van Wee (2004) distinguish four interrelated components: the land-use component, the transport component, the temporal component, and the individual component. These components provide the conceptual basis for the SAI as Geurs and van Wee clarify in their 2004 study what a comprehensive accessibility measure would ideally capture and where simplifications are required when applying accessibility measurement.

2.3.1 Components of Sustainable Accessibility

Land-Use Component

The land-use component refers to the quantity, quality, and spatial distribution of opportunities. These opportunities may include jobs, shops, schools, healthcare facilities, green spaces, and other destinations that support everyday life. Accessibility tends to increase when relevant opportunities are located closer to residential origins or when the spatial organisation of land uses allows them to be reached efficiently.

The land-use component also concerns the distribution of demand, which is not modelled in this research due to the complexity of including distribution of demand. Access to a school, healthcare facility, or public transport service may depend not only on spatial proximity, but also on capacity, availability, and competition among users. For instance, a school can spatially be accessible, but if the school has no available places, then the access is limited. Similarly, a general practitioner facility can be nearby, but the practical accessibility depends on the number of doctors, appointment availability, waiting times, or whether they accept new patients.

Transport component

The transport component refers to the effort required to travel between origins and destinations. This effort may include travel time, monetary cost, reliability, comfort, physical effort, transfer burden, and perceived safety.

Temporal Component

The temporal component concerns the timing of both travel and activities. Accessibility depends not only on the existence of destinations and routes, but also on whether destinations can be reached within realistic time budgets. A service may be spatially close but practically inaccessible if travel times are excessive or public transport frequencies are limited.

Individual Component

The individual component refers to differences in people's needs, abilities, resources, and preferences. These differences shape which opportunities are relevant, which modes are available, and what travel times are acceptable. Cycling accessibility, for example, depends not only on the cycling network but also on whether a person is physically able to cycle, has access to a bicycle, and feels safe using the infrastructure. Public transport accessibility may likewise depend on affordability, service knowledge, physical accessibility, and transfer complexity.

2.3.2 Scope of the Accessibility Concept in This Research

The four components above define the conceptual scope of the SAI. The index focuses on physical access to daily opportunities through sustainable transport modes. It does not directly measure perceived accessibility, affordability, digital access, safety perceptions, service quality, realised travel behaviour, or accessibility for specific social groups.

This scope is a deliberate methodological choice. Recent accessibility research increasingly emphasises equity, subjective experience, and social differentiation (Allen & Farber, 2020; Jamei et al., 2022; Pot et al., 2025). These dimensions are important, but incorporating them would require a broader research design than feasible within this research time period. The SAI

should therefore be interpreted as a measure of potential sustainable accessibility: it estimates what the spatial and transport environment makes reachable under defined assumptions.

2.3.3 Conceptual Framework for the Sustainable Accessibility Index

For this research, the following conceptual framework is created (Figure 1). It translates the theoretical discussion and four components of accessibility framework from Geurs and van Wee (2004) into a conceptual framework for sustainable accessibility. The conceptual framework is visualized in Figure 1. Next, the operationalisation of the components are discussed and summarized in Table 2.

The framework is based on the following proposition: a neighbourhood has high sustainable accessibility when residents can reach a sufficient range of essential daily opportunities by sustainable transport modes within realistic travel times.

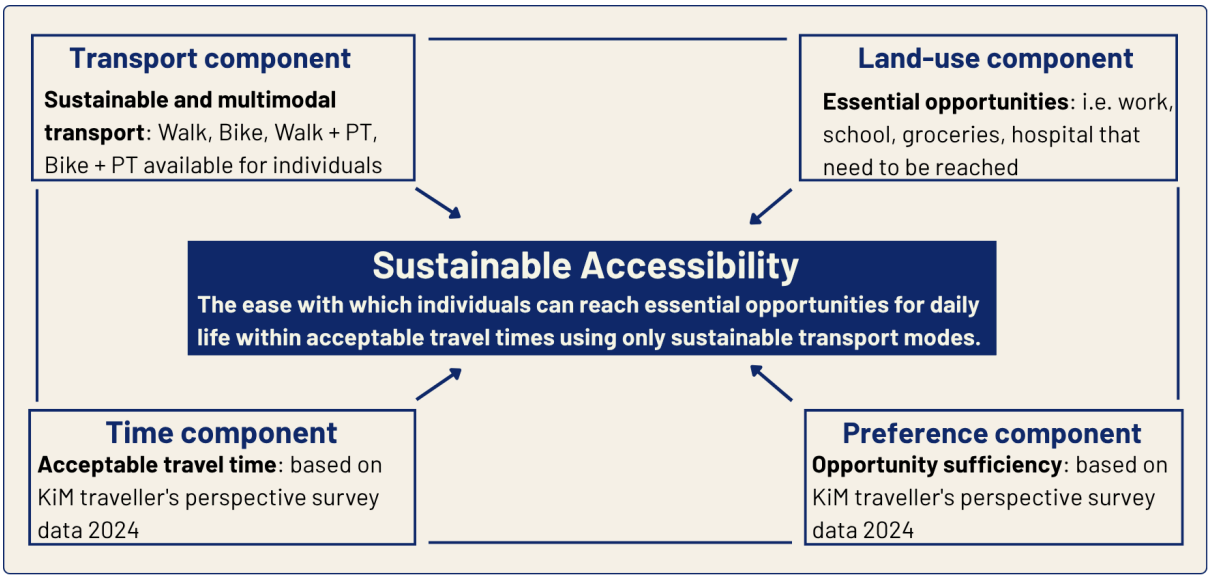


Figure 1: Conceptual framework of sustainable accessibility.

Land-Use Component

The land-use component is addressed in the SAI by focusing on different opportunities types that can be important in daily life. The SAI measures potential accessibility rather than realised service use. It estimates whether opportunities are spatially reachable by sustainable modes, but it does not determine whether these opportunities have sufficient capacity for all potential users nor models the demand side of the land-use system since modelling these dimensions requires detailed data on supply, demand, and service capacity.

Transport Component

In this research, the transport component is represented through travel time by sustainable transport modes. This focus follows directly from the sustainability orientation of the research. The relevant transport system is not the car network, but the network formed by walking, cycling, public transport, and multimodal chains that combine active travel with public transport. This distinction is analytically important since car-based accessibility and sustainable-mode accessibility may produce very different interpretations of the same

neighbourhood. A neighbourhood may be highly accessible by car while offering substantially weaker access for residents travelling by walking, cycling, or public transport.

Temporal Component

This research incorporates the temporal component through travel-time thresholds and time-based weighting.

Travel time thresholds refer to the set maximum travel time for each mode and help in deciding whether an opportunity counts as accessible. For example, a destination can be counted as accessible if it can be reached within a selected threshold which can be 5, 10, 30 or 60 minutes depending on the type of opportunity and transport mode.

Time-based weighting is a method where opportunities that are closer can weigh more strongly to the accessibility score than opportunities that are reached later. This reflects the principle that opportunities are not equally useful regardless of travel time. A daily destination reachable within a short trip contributes more meaningfully to everyday accessibility than one reachable only through a long journey.

At the same time, the use of thresholds introduces a methodological choice since these cumulative accessibility results may be sensitive to the selected cut-off (Tomasiello et al., 2023). For example, if a supermarket is reachable within 14 minutes and the travel time threshold is 15 minutes, then the supermarket counts as accessible. However, a similar supermarket reachable within 16 minutes would not count as accessible even though the differences in travel time is small. For this reason, the robustness of travel-time assumptions is examined through sensitivity analysis in the empirical part of the research.

Preference Component

Fully incorporating individual heterogeneity would require detailed behavioural, demographic, and perceptual data. This research therefore adopts a baseline-user assumption rather than including an individual component as described in Geurs and van Wee (2004) proposition. The SAI estimates the accessibility offered by the built environment and transport system to a standard resident who is able to walk, cycle, and use public transport. This improves comparability across neighbourhoods but limits the ability of the index to represent specific groups, such as children, older adults, disabled residents, low-income households, or people without access to a bicycle. These limitations are addressed in the discussion chapter.

The individual component is taken into account in a more general sense in which general time acceptability and quantity preference function are included in the SAI development.

Table 2: Translation of accessibility theory into the Sustainable Accessibility Index.

Theoretical component	Meaning in accessibility theory	Translation in the SAI	Main simplification
Land-use component	Spatial distribution, quantity, and quality of opportunities.	Essential daily opportunity categories.	Opportunity quality, capacity, and competition are not fully modelled.

Transport component	Travel effort between origins and destinations.	Travel time by walking, cycling, public transport, and multimodal chains.	Cost, comfort, safety, reliability, and perceived burden are not fully included.
Temporal component	Time budgets, opening hours, and timing of travel.	Behaviourally plausible travel-time thresholds and time-based weighting.	Full activity schedules and opening hours are not modelled in detail.
Individual component	Differences in needs, abilities, resources, and preferences.	Baseline user: abled person and access to the following modes walk, cycle, and public transport.	Group-specific accessibility is not directly modelled. General time acceptability and quantity preference function are included based on KiM/CBS data.

3. METHODOLOGY

In this chapter, sustainable accessibility as defined in the conceptual framework is translated into a measurable approach. First, the overall research design is presented in Section 3.1. Then, in Section 3.2 and 3.3 the data sources used and the case study selection process are discussed. Next, in Section 3.4 the step-by-step calculation procedure is explained. Finally, the sensitivity and validation strategies adopted in this research are presented in Section 3.5.

3.1 Research Design and Methodological Workflow

The methodology follows established guidelines for composite-index construction. The OECD handbook on composite indicators recommends that index development should begin with a clear theoretical framework, followed by indicator selection, data preparation, normalisation, weighting, aggregation, and sensitivity analysis (Nardo et al., 2005). Salzman (2003) similarly emphasises that composite indicators require transparent methodological choices because index outcomes can be affected by indicator selection, scaling, weighting, and aggregation. See Figure 2 for the phases and steps of the Sustainable Accessibility Index (SAI) index calculation.

Following this logic, the SAI is constructed in five steps (see Table 3).

Table 3: Steps to construct the Sustainable Accessibility Index.

Step	Purpose	Output
1. Conceptual definition	Define sustainable accessibility and its components.	Theoretical basis from Chapter 2
2. Indicator selection	Select origins, destinations, modes, and behavioural parameters.	Input indicators for the SAI
3. Data preparation	Clean, classify, and spatially align datasets.	Prepared origin, destination, and network data
4. Index construction	Calculate temporal acceptability, opportunity sufficiency, and aggregate scores.	SA scores by origin, mode, and neighbourhood
5. Sensitivity analysis	Test the effect of key methodological assumptions.	Robustness assessment

The methodological workflow (Figure 2) shows the exact steps taken to calculate the SA scores at the neighbourhood level (step 4 from Table 3). The grey boxes represent the input data, whereas the dark blue boxes show the steps taken in each phase of the SAI index calculation. The light blue boxes show the index construction from origin specific Sustainable Accessibility (SA) scores to the neighbourhood-level SA scores. Additionally, the green boxes represent which type of sensitive analyses is performed and the yellow boxes illustrate which visualizations are made from each step of the index construction.

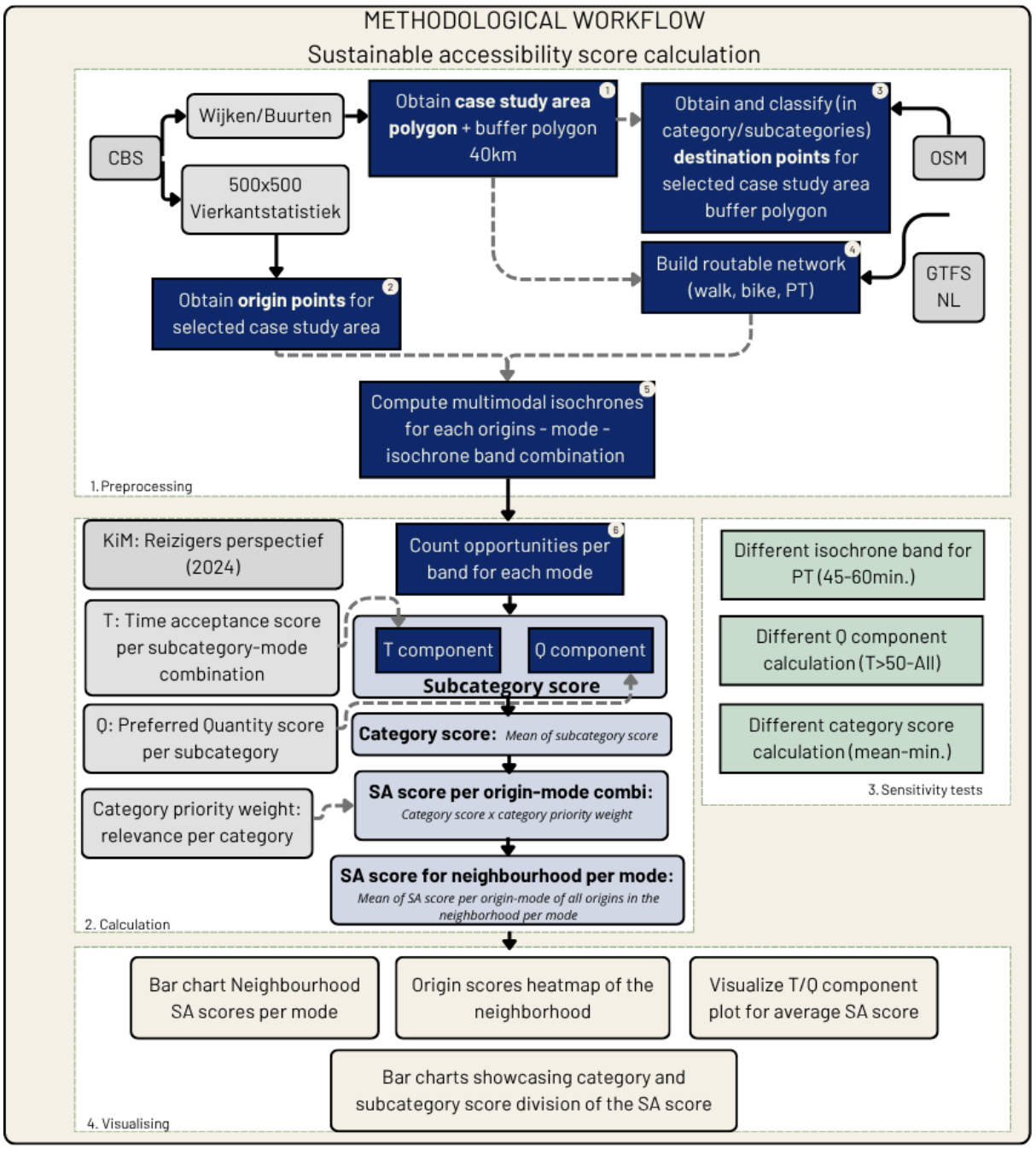


Figure 2: SAI index calculation workflow.

The study uses a comparative case study design. The same SAI framework is applied to neighbourhoods with different levels of urbanisation. This allows the methodology to be evaluated across contrasting spatial and transport contexts and enables comparison of sustainable accessibility patterns between case study areas.

3.2 Case Study Selection

Several case selection strategies were possible, namely a full national analysis or a different analysis scale such as street-level. However, these alternatives would have required substantially more routing computation and were not feasible within the scope of this research. Therefore, three cross-case studies with a total of nine neighbourhoods in the Netherlands were selected to test whether the SAI can be applied and interpreted across contrasting spatial and transport system contexts. The cases were selected from three degrees of urbanisation: highly urbanised, moderately urbanised, and non-urbanised/rural neighbourhoods (CBS, n.d.-b).

The neighbourhood size was also considered during the case selection. Very large neighbourhoods with more than 20 origin points were avoided to ensure enough computational power was available. Table 4 presents the nine selected case study neighbourhoods, including their corresponding municipality and urbanisation levels. Figure 3 provides a spatial overview of these case study areas.

For the remainder of this report, the selected neighbourhoods are referred to by their respective municipality names, since these are more concise and generally more recognisable than the neighbourhood names.

Table 4: Case studies details.

Urbanisation degree and colour	Selected Municipalities	Selected Neighbourhoods	Purpose in the research design
1: Highly urbanised	Amsterdam, Den Haag, Tilburg	Omval/Overamstel, Bezuidenhout, Bouwmeester	Represents dense urban settings with relatively high opportunity density and stronger sustainable transport provision.
3: Moderately urbanised	Zutphen, Dronten, Smallingerland	Noordveen, Dronten Midden, De Drait	Represents medium-density towns with mixed accessibility conditions.
5: Low-urbanised or rural	Heusden, Hulst, Schagen	Heesbeen, Kuitaart, Tuitjenhorn	Represents lower-density settings with more dispersed opportunities and potentially weaker public transport provision.

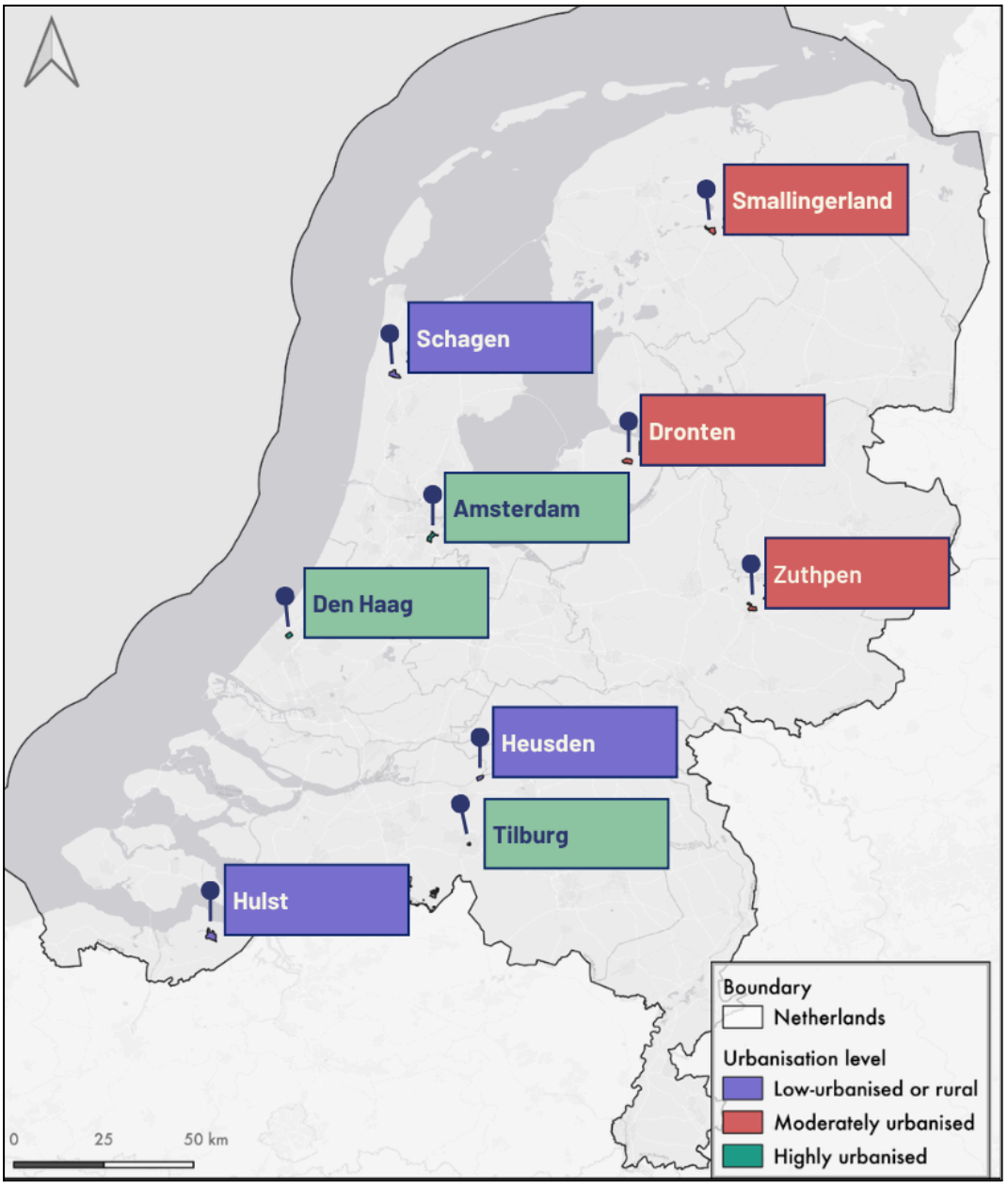


Figure 3: Case study areas overview map.

The cases were selected purposely to examine whether the SAI can be applied across contrasting spatial contexts. The comparison is not intended to provide a statistically representative sample of all Dutch neighbourhoods. Instead, the aim is to test and demonstrate the index across different combinations of density, land-use patterns, and sustainable transport provision.

3.3 Data Sources and Preprocessing

The SAI calculation requires four types of data: study area boundaries, origin points, destination opportunities, and transport network data (see Table 5 for a summary of the input data). In addition, behavioural parameters are used to convert travel times and opportunity counts into accessibility scores.

Table 5: Overview of data inputs.

Data input	Source	Role in the SAI
Neighbourhood boundaries	CBS Wijken en Buurten 2025 - Downloaded on 22 March 2026 from (CBS, n.d.) .	Defines the case study areas.
Residential origins	CBS 500 m × 500 m population grid 2024 - Downloaded on 22 March 2026 from (CBS, n.d.-a).	Defines origin points for accessibility calculation.
Destination opportunities	OpenStreetMap - Downloaded on 22 March 2026 from (Geofabrik, n.d.).	Defines essential daily opportunities.
Walking and cycling network	OpenStreetMap - Downloaded on 22 March 2026 from (Geofabrik, n.d.).	Enables active-mode routing.
Public transport network	General Transit Feed Specification (GTFS) feeds - Downloaded on 22 March 2026 from (OVapi, 2026).	Enables public transport and multimodal routing.
Behavioural parameters	KiM traveller perspective survey - (Hamersma & Roeleven, 2024).	Provides time-acceptability, opportunity-sufficiency, and category-relevance parameters.
Routing engine	R5 via r5py - (Fink et al., 2022).	Computes travel times between origins and destinations.

3.3.1 Neighbourhood Boundaries

The boundaries of the selected neighbourhoods were obtained from the CBS Wijken en Buurten dataset (CBS, 2025). These boundaries define the spatial extent of each case study area.

Additionally, 40km buffer zones around the neighbourhoods were extracted as well to use as spatial extent for transport network and destination data to include opportunities outside the neighbourhood boundary that may still be reachable and relevant to residents. 40km is chosen as the maximum buffer distance since it is slightly above the average trip distance of 35km, thus ensuring that the average trip distance can be met in the calculation (CBS, 2024; Faber & Huang, 2025).

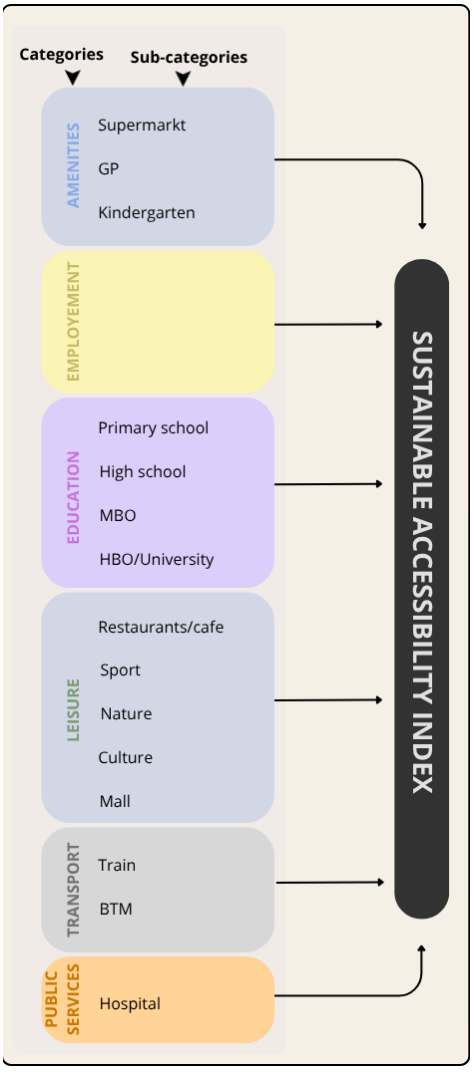
Alternative buffer distances of 35, 45, 50km based on the maximum speed and average travel time of the train, which is the mode with the furthest reach to consider, were tested but showed no significantly different result.

3.3.2 Origin Points

To calculate representative origin points, the statistics Netherlands (CBS) 500m x 500m grid data was selected for its uniform size of the cells (CBS, n.d.-a). The centroids of the grid cells were calculated in python and chosen to represent a local residential origin from which sustainable accessibility is assessed. Using grid-cell centroids provides a balance between spatial detail and computational feasibility. In contrast, the 100m x 100m grid data was not used due to the fine spatial contrast requiring substantially more computational power.

3.3.3 Destination Opportunities

OpenStreetMap (OSM) is a widely used open-source geospatial database that is regularly updated and easily accessible to anyone thus making it a practical and transparent choice for this research to construct the destination dataset. The relevant destination categories to this research aim were carefully selected based on specific OSM tags.



A local OSM.pbf file for the Netherlands was used for the extraction of OSM destinations, obtained from Geofabrik and processed using the Osmium python package. The national OSM file was clipped to the case study buffer area polygon. This approach was chosen over repeated online queries because it provides a reproducible workflow, avoids dependency on live API availability, and allows all relevant OSM object types to be processed consistently and run once for any case study area.

A key methodological choice concerned which OSM geometry types to include in the tag filtering. The first option was to use only OSM point features, whilst this would provide clean point locations, it would omit many destinations that are mapped as polygons such as school grounds, hospitals, parks, and sport centres. The second option was to include all geometry options (points, polygons, lines). This option would maximize completeness, however, lines often represent infrastructure rather than clearly identifiable destinations which risks introducing more noise in the data and additional unnecessary computing time. The third option, which was chosen in this research, was to include OSM points and polygons. With this choice, point features capture explicitly mapped points of interests (POIs) while the polygon features capture the larger destinations that are often mapped as areas.

Figure 4: Destination opportunity categories.

The polygonal features were converted into representative points to make them compatible with the destination-based accessibility analysis. After the conversion, the polygon-derived points were deduplicated based on the mapped OSM points. The mapped points were prioritised as those usually represent the POI location more directly, thus the polygon-derived points were removed when they were located within 20 metres of a mapped point with a comparable tag and or name structure. This threshold and comparison of similar tag or name structure was chosen as the assumption is that this combination is likely to identify duplicates caused by the same destination being mapped as a point and as an area.

Then, the extracted POIs were assigned into the main categories and subcategories (see Figure 4) for the full categorization of destination opportunities. The categories reflect essential daily opportunities identified in the theoretical framework and linked to common trip purposes from the Dutch National Travel Survey (ODiN) in 2024 (CBS, 2024). To filter the desired essential daily opportunities, the following OSM tags for each category were used based on the open-source OSM wiki guide website:

- **Amenities** include everyday facilities that support general needs. In this research, this category includes supermarkets, kindergartens, and general practitioners. Supermarkets were identified through shop tags such as shop=supermarket, shop=convenience, and shop=grocery. Kindergartens were identified through amenity=kindergarten. General practitioners were identified using amenity=doctors and healthcare=doctor.
- **Education** includes formal educational institutions. Universities were classified as HBO/University, whilst colleges were classified using the closest available OSM proxy for MBO-level education. Primary schools and high schools were classified using isced:level were possible. ISCED levels 0-1 were interpreted as primary schools, levels 2-3 as high schools, level 4 as MBO, and level 5-8 as higher education. When ISCED information was missing, additional tags such as school, education, operator: type, and name were used to identify terms such as mbo, hbo, university manually.
- **Leisure** includes destinations associated with recreation, social activities, culture, sport and nature. Restaurants and cafes were classified as restaurant-related leisure destinations using the amenity=restaurant and amenity=cafe tags. Cultural destinations were identified using the tags amenity=theatre, amenity=cinema, tourism=museum, and tourism=gallery. Nature-related destinations were identified through leisure=park, and leisure=nature_reserve tags. Sports destinations included leisure=sports_centre, fitness_centre, sports_hall, stadium, and swimming_pool tags. Shopping malls were identified using the tag shop=mall.
- **Public services** only included hospitals. Police and fire stations, and municipality buildings were not taken into account due to absence in ODiN CBS data and time constraints, but would be a good addition to this category. Hospitals were identified using amenity-hospital and healthcare=hospital.
- **Transport stops** include public transport access points divided into train, and bus/tram/metro subcategories. Train stations were identified using railway tags railway=station, and railway-halt, unless additional tags indicated that the stop was part of the metro or tram system. But, tram, and metro stops were grouped into the BTM category, and identified using the tags highway=bus_stop, amenity=bus_station, railway=tram_stop, railway=subway_entrance, and station=subway.

- **Employment** destinations represent locations that serve as proxies for concentrations of jobs or work-related activity in a minimalistic way. This category does not contain information on exact employment counts, rather it identifies land uses and establishments where employment is likely to occur. This also means that digital employment and working from home is not taken into account as there are not yet tags in OSM that represent this type of work. The employment destinations were selected using office=*, craft=*, and industrial=*. The office=* key captures office-based activities such as companies, administrative services, and institutional workspaces. The craft=* key captures smaller-scale production and services destinations, such as workshops and specialised trades. The industrial=* key represents industrial production-related activity. These tags were grouped into an all-employment subcategory because the ODiN and KiM data does not allow for any further refinement of employment in those subcategories.

3.3.4 Transport Network and Travel-Time Calculation

In this section, the transport network and travel-time calculation choices that were made in the preprocessing phase are explained. Several methodological choices were made in this process, such as which transport modes to include, which routing engine to apply, which speeds and distances per mode to use.

Travel times were calculated for four transport modes that can be considered more sustainable than private cars as a transport mode (see Table 6).

Table 6: Transport modes included in the analysis.

Mode notation	Mode description
W	Walking
B	Cycling
W + PT	Walking combined with public transport
B + PT	Cycling combined with public transport

The multimodal transport network was constructed using OpenStreetMap data for walking and cycling routes and General Transit Feed Specification (GTFS) public transport timetable data from OVapiNL (OVapi, 2026). Travel times were calculated using the routing engine of R5 through the Python package r5py (Fink et al., 2022). This routing engine includes pedestrian paths, cycle routes, roads available for cycling, public transport services, and access or egress legs for multimodal trips. Other popular routing engines (i.e. TravelTime API and Valhalla) were considered, but lacked the option to calculate multimodal trips distances.

In the routing engine the following parameters were set to calculate sustainable accessibility:

I. The walking speed was set at 4 km/h based on recent research (Geržinič et al., 2026) and cycling speed at 19 km/h based on Nationwide Data Collection (NDC) Nederland research on cycling speed in the Netherlands (NDC, 2025).

Fixed walking and cycling speeds were used to ensure comparability across the case study areas. This choice simplifies the individual component as the user gets standardized.

II. For public transport routing, departure time was standardised to a weekday morning peak period. The selected departure time was set to Monday 23 March 2026 between 08:30–09:30h.

    Public transport accessibility varies by time of the day since service frequencies and transfers change across the timetable. A weekday morning peak period was selected as it could represent a regular travel period for commuting, education, and many daily activities. This choice provides a consistent comparison across case study areas. The implication is that the results represent only the specific Monday morning accessibility and not the accessibility during the evening, weekend or off peak periods.

III. The isochrone bands per mode:
    - Walk [5, 15, 30]min
    - Bike [5, 15, 30]min
    - Walk + PT [15, 30, 45]min
    - Bike + PT [15, 30, 45]min

    The isochrone bands were defined by estimating the maximum travel time for each mode based on ODiN data by converting those trip distances into minutes using the mode specific average speeds. This way of estimating the isochrone bands ensures that the routing limit is informed by observed travel distance patterns from ODiN travel survey data by the Netherlands Statistics rather than selected arbitrarily (CBS, 2024).

For each origin, mode, and destination subcategory, the routing procedure produced either a travel time in minutes or an unreachable value if no route was found within the maximum routing limit per mode.

3.3.5 Behavioural Parameters

To measure reachable destinations within acceptable travel times and quantities, behavioural parameters were required. Several alternatives were available. For instance, a fixed travel-time threshold for each mode based on accessibility literature was considered as this would be simple and transparent. However, that method could not account for differences between activity types. Another option that was considered, was to assign equal weights to all destination categories and opportunity counts as that would maximize simplicity but this would ignore the differences between how relevant different types of opportunities are for everyday life.

In this index, three simplified behavioural parameters were used, namely acceptable travel-time parameters for the temporal acceptability component, opportunity sufficiency parameters for the quantity component, and category relevance weights for the origin-level SA score.

The Netherlands Institute for Transport Policy Analysis (KiM) traveller perspective research data of 2024 was selected since this research provides Dutch context specified data that could be linked to travel-time acceptability, opportunity sufficiency and category relevance. With this

data as parameters, this research is able to incorporate behavioural assumptions in the SAI whilst remaining transparent. The implication is that the index reflects general traveller perspective values rather than the preference of specific individual or social group meaning that differences in age, income, ability, household type, or subjective perception is not directly taken into account.

3.4 Construction of the Sustainable Accessibility Index

In this section, the construction of the SAI is explained. The SAI is calculated through a hierarchical procedure. The index begins at the most detailed level of origin–mode–subcategory-level and is then aggregated to category-, mode-origin-, and neighbourhood levels scores.

The index has two core components: i. Temporal acceptability (T): which shows whether the nearest relevant destination can be reached within a travel time that people are likely to consider acceptable. ii. Opportunity sufficiency (Q): which shows whether the number of reachable opportunities is sufficient to provide meaningful choice.

The final SA score ranges from 0 to 1, where 0 indicates no sustainable accessibility and 1 indicates high sustainable accessibility under the assumptions of the index.

3.4.1 Notation

Table 7 shows the notation of core elements that are used in the SAI equations. All calculations are made for a given origin o and a transport mode , unless stated otherwise. To keep the equations readable, the subscripts o and m are omitted in the step by step formulas. Additionally, the category index c is omitted in some equations where the subcategory s already belongs to a specific category.

Table 7: Core notation used in the SAI equations.

Symbol	Meaning	Unit or range
o	Origin point	Grid-cell centroid
m	Transport mode	W, B, W + PT, B + PT
c	Main opportunity category	See Figure 4
s	Destination subcategory within category c	See Figure 4
d	Individual destination	Destination point
b	Travel-time band	Minutes
$t_{o,m,d}$	Travel time from origin o to destination d by mode m	Minutes
D_s	Set of destinations belonging to subcategory s	Destination set

$T_{o,m,s}$	Temporal acceptability score	0–1
$Q_{o,m,s}$	Opportunity sufficiency score	0–1
$A_{o,m,s}$	Subcategory accessibility score	0–1
$A_{o,m,c}$	Category accessibility score	0–1
W_c	Weight assigned to category c	0–1
$SAI_{o,m}$	Origin-level SAI for origin o and mode m	0–1
$SAI_{N,m}$	Neighbourhood-level SAI for neighbourhood N and mode m	0–1

3.4.2 Step 1: Minimum Travel Time to Each Subcategory

The first step determines what the minimum travel time is for each origin point to each destination subcategory by each transport mode. This step uses the travel-time outputs from the preprocessed data and parameters described in Section 3.3.4.

Equation 1: Minimum Travel Time

$$t_s = \min \{t_d \mid d \in D_s\}$$

If no destination in subcategory s is reachable, t_s is noted as unreachable, and score 0 is assigned.

Symbol	Explanation
t_s	Minimum travel time from origin o to subcategory s by mode m
t_d	Travel time from origin o to destination d by mode m
D_s	Set of all destinations belonging to subcategory s

This step is needed because the temporal cost of reaching a type of opportunity is usually determined by the closest relevant option. For example, if the nearest supermarket is 8 minutes away by cycling and the second-nearest supermarket is 12 minutes away, the travel-time acceptability of grocery access is based on the 8-minute option. The nearest destination is therefore used to represent the minimum effort required to access that subcategory.

The assumption behind this step is that, for temporal access, the closest relevant opportunity is the most behaviourally relevant. This does not mean that additional destinations are ignored. Additional opportunities are included later in the Q-component, which measures whether there are enough reachable options. Thus, Step 1 isolates the time costs of reaching at least one relevant destination, while later steps assess the sufficiency of choice.

The output of this step is one minimum travel time for each origin–mode–subcategory combination. This value is used in Step 2 to calculate the temporal acceptability score.

3.4.3 Step 2: Temporal Acceptability

The second step converts the minimum travel time into a temporal acceptability score. This score indicates whether the travel time from an origin to the nearest relevant destination is likely to be considered acceptable for a specific activity subcategory and transport mode.

The temporal acceptability score is calculated using behavioural acceptability functions derived from KiM traveller-perspective data (Hamersma & Roeleven, 2024). For each subcategory s and mode m, the KiM data provide a mean acceptable travel time and a standard deviation see Table 4.19 in KiM data. These two parameters are used to describe the variation in acceptable travel-time thresholds among travellers using a cumulative density function. For example, suppose that the closest supermarket for a given origin is 8 minutes away on foot and the KiM data indicate that about 80% of the travellers consider a supermarket trip acceptable when it can be reached within 10 minutes walking. Then, the temporal acceptability score for this origin-mode-subcategory combination is approximately 0.8. This means that the destination is not just reachable, but reachable within a time range that most people (80%) would find acceptable. Conversely, if the nearest supermarket was 18 minutes away on foot, the expected acceptability would be substantially lower.

Equation 2: Temporal Acceptability

$$T_s = f_{m,s}\left(t_s\right)$$

Where the acceptability function is:

$$f_{m,s}(t) = 1 - \Phi\left(\frac{t - \mu_{m,s}}{\sigma_{m,s}}\right)$$

Therefore, the temporal acceptability score is calculated as:

$$T_s = 1 - \Phi\left(\frac{t_s - \mu_{m,s}}{\sigma_{m,s}}\right)$$

$$T_s = 0 \quad \text{if no destination is reachable}$$

Symbol	Explanation
T_s	Temporal acceptability score for origin o, mode m, and subcategory s
$f_{m,s}$	Acceptability function for subcategory s and mode m
t_s	Minimum travel time from origin o to the nearest destination in subcategory s by mode m
$\mu_{m,s}$	Mean acceptable travel time for subcategory s and mode m, derived from KiM data
$\sigma_{m,s}$	Standard deviation of acceptable travel time for subcategory s and mode m, derived from KiM data
$\Phi(\cdot)$	Cumulative density function of the standard normal distribution

The function $f_{m,s}(t)$ returns a value between 0 and 1. It can be interpreted as the estimated share of travellers who would consider a travel time of t acceptable for a given subcategory and mode. A low travel time results in a high temporal acceptability score, because most travellers are expected to accept that travel time. As travel time increases, the score decreases, because fewer travellers are expected to consider longer travel times acceptable.

This formulation uses the complement of the cumulative normal distribution. The model does not calculate the share of travellers whose acceptable travel time is below the observed travel time. Instead, it calculates the share of travellers whose acceptable travel-time threshold is above the observed travel time. This corresponds to the proportion of travellers for whom the trip remains acceptable.

The use of subcategory- and mode-specific acceptability functions is preferable to applying one fixed travel-time threshold to all activities. Acceptable travel time differs by both destination type and transport mode. For example, travellers may accept longer travel times for employment or higher education than for grocery shopping, childcare, or basic healthcare. Similarly, acceptable travel times may differ between walking, cycling, or public transport.

This step ensures that the accessibility index does not treat all reachable destinations as equally meaningful. A destination that is technically reachable but requires a long travel time contributes less to sustainable accessibility than a destination that can be reached within a travel time considered acceptable by most travellers.

The output of this step is a temporal acceptability score between 0 and 1 for each origin–mode–subcategory combination. Higher values indicate that the nearest relevant destination is reachable within a travel time acceptable to a larger share of travellers. Lower values indicate that the travel time is acceptable to only a smaller share of travellers.

3.4.4 Step 3: Reachable Opportunities Counts

In the third step, the quantity dimension of the SAI is introduced. Step 3 counts how many destinations of each subcategory are reachable from each origin by each mode. This forms the basis of the opportunity sufficiency component (Q).

A methodological choice to make in this step is how to count the reachable opportunities. A simple approach is to count all destinations within a fixed travel-time threshold. This would be easy to interpret, but it does ignore differences in acceptable travel time between trip purposes and transport modes. A second approach is to count all the opportunities within the maximum travel time set as parameters for each mode. This captures all opportunities provided by the transport network, but it might overestimate accessibility as this option includes destinations that might require travel times that many people would find unacceptable. A third option would be to apply a continuous distance- or time-decay weighting function to all reachable opportunities. This option allows for differentiation between trip purposes, travel times and modes, however it would also partly duplicate the temporal acceptability component that is already included in the SAI with the T-component. Finally, a general opportunity count, counting destinations reachable in the selected travel times, could be adjusted for service availability by considering available places in schools, or accepting new patients for General Practitioners. However, this requires consistent data on availability and capacity across all opportunity categories, which is not available within the scope of this research.

The options above show that counting opportunities involve a trade-off between behavioural accuracy, transparency, and data availability. Based on these considerations, this research uses a threshold-based alternative that includes a temporal acceptability inclusion rule rather than a more complicated continuous time-decay weighting function (Q-threshold) and the option of counting all reachable opportunities within the maximum routing limit (Q-separate). The Q-threshold variant allows for destinations being differentiated based on travel-travel time acceptability, whereas the Q-separate considers all reachable destinations within the transport network. The Q-threshold variant is the baseline calculation, while Q-separate is used in the sensitivity analysis. See Table 8 for a summary of the differences in variants.

Equation 3: Q-Threshold Opportunity Count

$$n_s^{thr} = \sum_b R_{o,m,s,b} \cdot I\left(Ts, m, b \geq 0.5\right)$$

$$I(\cdot) = 1 \text{ if the condition is true, and } 0 \text{ otherwise}$$

Equation 4: Q-Separate Opportunity Count

$$n_s^{sep} = \sum_{b \in B_s^{reach}} R_{o,m,s,b}$$

Symbol	Explanation
n_s^{thr}	Number of reachable opportunities under the Q-threshold variant
n_s^{sep}	Number of reachable opportunities under the Q-separate variant
$R_{o,m,s,b}$	Number of opportunities in band b
$T_{s,m,b}$	Temporal acceptability weight for subcategory s, mode m, and band b

Table 8: Opportunity-count variants.

Variant	Definition	Interpretation	Role
Q-threshold	Counts opportunities only in bands where temporal acceptability is at least 0.5	Measures quantity within behaviourally acceptable travel times	Main specification
Q-separate	Counts all reachable opportunities within the routing limit	Measures the wider reachable opportunity field	Sensitivity analysis

The Q-threshold variant counts only opportunities located in travel-time bands where temporal acceptability is at least 0.5. This means only opportunities where at least half of travellers are expected to consider acceptable are included in the opportunity count. This variant is used as the main specification because the SAI aims to measure practically acceptable daily access, rather than technical reachability alone.

The Q-separate variant counts all reachable opportunities within the routing limit, regardless of whether their travel-time band reaches the 0.5 acceptability threshold. This variant is included in the sensitivity analysis because it separates opportunity quantity from temporal acceptability. It answers a different question: how many opportunities are reachable through the transport system, even if some require longer travel times than most travellers would consider acceptable.

For example, suppose there are 2 supermarkets accessible by foot within 0-10 minutes and the temporal accessibility is 0.8, 5 supermarkets within 10-20 minutes with a temporal accessibility of 0.55, and 8 supermarkets within 20-30 minutes with a temporal accessibility of 0.35. Under the Q-threshold specification, only the supermarkets that have a temporal acceptability of at least 0.5 are counted, thus in this example the reachable opportunity count is 7 supermarkets. The Q-separate specification counts all reachable supermarkets regardless of the temporal acceptability level, therefore the reachable opportunity count would be 15 supermarkets.

This step is necessary because accessibility is not only about reaching the nearest destination. Having multiple reachable options may improve practical accessibility by increasing choice. For example, access to one high school is better than none, but access to several high schools may provide more meaningful choices. However, the benefit of additional opportunities is not assumed to increase linearly and equally for each subcategory. This is addressed in step 4.

The output of this step is a reachable opportunity count for each origin–mode–subcategory combination.

3.4.5 Step 4: Opportunity Sufficiency

The fourth step converts the number of reachable opportunities into an opportunity sufficiency score. This score ranges from 0 to 1 and represents whether the number of reachable destinations is sufficient for the relevant activity type.

The Q-score reflects diminishing marginal returns. The first reachable opportunity is usually important as it means that a basic need can be met. Additional opportunities may improve choice, but each additional option contributes less than the previous one. For example, if the first reachable supermarket contributes 0.6 to the sufficiency score, the second 0.2, the third 0.1, and the fourth 0.06, then one reachable supermarket results in a Q-score of 0.6, two supermarkets in 0.8, three in 0.9. In this way, this method focuses on whether residents have a sufficient level of access to everyday activities and not every additional destination is rewarded equally.

For all subcategories except employment, *Q* is calculated using marginal weights from a lookup table based on KiM data: Table 4.20 on page 94 (Hamersma & Roeleven, 2024).

Equation 5: Opportunity Sufficiency

$$Q_s = \min\left(1, \sum_{i=1}^{n_s} q_{s,i}\right)$$

$$Q_s = 0 \quad \text{if no opportunity is reachable}$$

Symbol	Explanation
Q_s	Opportunity sufficiency score
n	Number of reachable opportunities from Step 3
$q_{s,i}$	Marginal contribution of the i-th opportunity in subcategory s
$min(1, ...)$	Caps the score at 1

The marginal weights represent how much each additional opportunity contributes to perceived sufficiency. For example, the first reachable supermarket may contribute strongly to the Q-score, while the second and third may add smaller increments. Once the score reaches 1, additional opportunities do not increase the score further, because the subcategory is treated as sufficient.

This step is included to avoid the limitations of simple cumulative opportunity counts. A purely linear count would imply that every additional destination contributes equally to accessibility. The sufficiency approach instead assumes that people need enough options, not an unlimited number of options. This makes the index more suitable for everyday accessibility, where the practical question is whether residents have sufficient access to essential opportunities.

Employment-Specific Sufficiency Function

Employment-related destinations (examples used in this thesis for these types of destinations are described in Chapter 3.3.3) require a separate treatment since employment opportunity counts can be much larger than counts for other daily services. Additionally, the employment category was not considered in the KiM data. A neighbourhood may have hundreds or thousands of reachable employment-related destinations, while categories such as hospitals, schools, or supermarkets have much smaller counts. Therefore, using the same lookup-table approach for employment could fail to capture meaningful variation.

For this reason, employment opportunity sufficiency is calculated using a logarithmic function. This functional form reflects diminishing marginal returns where the first reachable employment opportunities contribute strongly to the sufficiency score, while additional opportunities add progressively less once a large employment field is already accessible.

Equation 6: Employment Sufficiency

$$Q_{o,m,emp} = \min\left(1, \frac{\ln\left(1 + \frac{n_{o,m,emp}}{\alpha}\right)}{\ln\left(1 + \frac{N_{sat}}{\alpha}\right)}\right)$$

Symbol	Explanation
$Q_{o,m,emp}$	Employment opportunity sufficiency score
$n_{o,m,emp}$	Number of reachable employment opportunities
N_{sat}	Saturation point at which employment access is treated as sufficient

α	Shape parameter controlling the steepness of diminishing returns
ln	Natural logarithm

The α was set to = 0.01 and the N_{sat} was set to 500. Thus the formula becomes:

$$Q_{o,m,emp} = \min\left(1, \frac{\ln\left(1 + \frac{n_{o,m,emp}}{0.01}\right)}{\ln\left(1 + \frac{500}{0.01}\right)}\right)$$

The saturation parameter N_{sat} was set to 500. This value was selected as an empirically informed benchmark based on exploratory inspection of employment-related destination counts in several large urban neighbourhoods. In dense urban areas, including high-access neighbourhoods in Amsterdam, reachable office-related destination counts were generally observed to fall within approximately 350 to 450 destinations. The saturation value was therefore set slightly above this observed upper range. This ensures that the function does not classify typical high-access urban neighbourhoods as fully saturated too early, while still defining a realistic national benchmark at which employment access can be considered quantitatively sufficient.

The shape parameter α was set to 0.01. This low value was chosen to preserve a strong diminishing-return structure in the employment sufficiency function. The function assumes that the difference between having no reachable employment opportunities and having some reachable employment opportunities is highly important. However, once a neighbourhood already has access to a large number of employment destinations, each additional destination contributes only a small amount to the sufficiency score. This reflects the idea that employment access is not purely linear. For example, increasing accessible opportunities from 0 to 50 is more meaningful than increasing them from 400 to 450.

Together, N_{sat} = 500 and α = 0.01 define a function that is sensitive to low and moderate levels of employment access, while preventing very large employment counts from dominating the accessibility index. The function reaches its maximum value of 1 once the number of reachable employment-related opportunities is equal to or greater than 500. The logarithmic form preserves the diminishing-return logic where early increases in reachable employment opportunities raise the sufficiency score substantially, while additional opportunities contribute progressively less once the accessible employment field becomes large.

The output of this step is an employment opportunity sufficiency score between 0 and 1 for each origin–mode combination.

3.4.6 Step 5: Subcategory Accessibility Score

The fifth step combines temporal acceptability and opportunity sufficiency into one subcategory accessibility score. This score represents the accessibility of a specific type of destination, such as supermarkets, general practitioners practices, primary schools, or musea, from a given origin by a given mode.

Equation 7: Subcategory Score

$$A_s = \frac{T_s + Q_s}{2}$$

Symbol	Explanation
A_s	Accessibility score for subcategory s
T_s	Temporal acceptability score
Q_s	Opportunity sufficiency score

The mean is used because T and Q represent complementary dimensions of accessibility. A high subcategory score requires both an acceptable travel time and a sufficient number of reachable opportunities. If a destination is close but there are very few options, the Q-score reduces the subcategory score. If many opportunities exist but they require long travel times, the T-score reduces the subcategory score.

Equal weighting of the two components is used for transparency and because there is no strong empirical basis for assigning greater importance to temporal acceptability or opportunity sufficiency. This choice also keeps the index easy to interpret. The subcategory score is the average of time-based accessibility and quantity-based sufficiency.

The assumption behind this step is that temporal access and opportunity choice are equally important at the subcategory level. This is a simplifying assumption. Alternative specifications, such as unequal weighting or a minimum function, can be tested in sensitivity analysis if there are theoretical or policy reasons to prioritise one component.

The output of this step is a subcategory accessibility score between 0 and 1.

3.4.7 Step 6: Category Scores

The sixth step determines how the subcategory scores can be aggregated into broader opportunity category scores. This step is required as every category consists of different subcategories. For example, the Education category includes primary schools, high schools, MBO, and HBO/University, while the Leisure category includes restaurants, cultural destinations, sport facilities, nature areas, and shopping malls. Thus, this step presents the methodological choice of how to combine the different subcategories into one category-level score that can represent each category fairly.

Several aggregations options were considered. First, a mean of the subcategory-score could be taken from all the subcategories within that category. This method is transparent and easy to interpret, however if outliers occur in the data, the resulting category score can be unclear to interpret since the detail of which specific subcategory influenced the averaged score is lost in the calculation. Second, a weighted mean where some subcategories contribute more strongly than others, could be an option. Nevertheless, this option would require a clear empirical or theoretical basis for assigning the subcategory-level weights. Finally, a minimum function could be used, where the weakest subcategory determines the category score. This is useful to identify whether a category contains a weak subcategory, but this might be too strict for a baseline specification as poorly accessible subcategories can strongly influence the entire category score resulting in an unrealistic low sustainable accessibility score.

Based on these considerations, two aggregation methods were chosen: a mean method and a minimum method.

Equation 8: Category Score Mean

$$A_{o,m,c}^{mean} = \frac{1}{|S_c|} \sum_{s \in S_c} A_s$$

Equation 9: Category Score Minimum

$$A_{o,m,c}^{min} = \min_{s \in S_c} A_s$$

Symbol	Explanation
$A_{o,m,c}^{mean}$	Category score using mean aggregation
$A_{o,m,c}^{min}$	Category score using minimum aggregation
S_c	Set of subcategories belonging to category *c*
A_s	Subcategory accessibility score
$s \in S_c$	Aggregation is performed over all subcategories contained in category *c*

Table 9: Category aggregation methods.

Method	Formula	Interpretation	Role
Mean	$A_{o,m,c}^{mean} = \frac{1}{\|S_c\|} \sum_{s \in S_c} A_s$	Average accessibility across subcategories.	Main specification
Minimum	$A_{o,m,c}^{min} = \min_{s \in S_c} A_s$	Weakest subcategory determines the category score.	Sensitivity analysis

The mean method is used in the main specification because it provides a clear average score for each category. It allows stronger subcategory performance to partly compensate for weaker subcategory performance. For example, if access to primary schools is strong but access to HBO/University is weaker, the Education category score reflects the average condition across the included Education subcategories.

The minimum method is used in sensitivity analysis since it reflects a stricter sufficiency principle. Under this method, a category performs well only if all included subcategories are reasonably accessible. This is useful for examining whether high average scores hide weak links in essential services.

This step is added because the index must remain interpretable. Reporting every subcategory separately would provide detail but reduce clarity. Aggregating subcategories into main categories creates a structure that is easier to communicate while still preserving the possibility of disaggregation.

The output of this step is a category-level score for each origin, mode, and category.

3.4.8 Step 7: Origin-Level SAI

The seventh step combines category scores into an origin-level SA score for each transport mode. This score represents the sustainable accessibility from a specific residential origin point by a specific mode.

Equation 10: Origin-Level Score

$$SAI_{o,m} = \sum_{c} W_c \cdot A_{o,m,c}$$

Symbol	Explanation
$SAI_{o,m}$	SA score for origin o and mode m
W_c	Weight assigned to category c
$A_{o,m,c}$	Category accessibility score

Category weights reflect the relative relevance of different opportunity categories. For example, categories that are relevant to most residents or are essential for daily life may receive higher weights than categories that are less frequent or more group-specific. In the main specification, these weights are based on KiM category-relevance data: Table 6.1 on page 145 (Hamersma & Roeleven, 2024).

Table 10: Category weights.

Category	Weight
Amenities	0.20
Employment	0.22
Education	0.05
Leisure	0.14
Transport stops	0.15
Public services	0.25

This step is necessary since sustainable accessibility is multidimensional. A resident's ability to live without a private car depends not only on access to one destination type, but on access to a range of daily opportunities. The origin-level SAI summarises this multidimensional accessibility into one score while retaining the possibility of examining the underlying category scores. Additionally, the category priority weights can be shifted depending on the context and research needs for users using this SA index calculation. However, this is not tested as it falls outside of the current research scope.

The SAI is calculated separately for each mode (W, B, W + PT, and B + PT). Modes are not combined into a single best-mode score because one aim of the research is to compare how different sustainable modes shape accessibility. Keeping the modes separate allows the analysis to show, for example, whether cycling alone performs better than the multimodal combination of cycling and public transport, or whether public transport substantially improves accessibility in lower-density areas.

The output of this step is an origin-level SA score for each mode.

3.4.9 Step 8: Neighbourhood-Level SA Scores

The final step aggregates origin-level SA scores to the neighbourhood level. This produces the main neighbourhood-level SA score used for the cross-case comparisons in Chapter 4.

Equation 11: Neighbourhood-Level Score

$$SAI_{N,m} = \frac{1}{n_o} \sum_{o \in N} SAI_{o,m}$$

Symbol	Explanation
$SAI_{N,m}$	Neighbourhood-level SA for neighbourhood N and mode m
n_o	Number of origin points in neighbourhood N
$SAI_{o,m}$	Origin-level SA score

The mean treats each sampled origin equally. This is useful in this research where the aim is to compare the accessibility of spatial locations within a neighbourhood.

The output of this step is a neighbourhood-level SA score for each mode.

3.4.10 Summary of the SAI Calculation Procedure

The main elements of the equations are summarized in Table 11. Table 12 shows a summary of the step by step calculation procedure.

Table 11: Overview of variable reference table for SAI calculation.

Symbol	Description	Range / unit	Source or calculation
o	Origin point	Grid centroid	CBS population grid
N	Neighbourhood	Spatial polygon	CBS Wijken en Buurten
m	Transport mode	W, B, W + PT, B + PT	Research design

c	Opportunity category	Food, healthcare, education, etc.	Category classification
s	Opportunity subcategory	Supermarket, GP, park, etc.	Category classification
d	Individual destination	Point location	OSM
b	Travel-time band	Minutes	Routing output / lookup table
t_d	Travel time from origin to destination	Minutes	R5 / r5py
t_s	Minimum travel time to subcategory	Minutes	$\min\{t_d \mid d \in D_s\}$
T_s	Temporal acceptability	0–1	KiM-derived time function
n_s	Reachable opportunity count	Integer ≥ 0	Count of reachable destinations
Q_s	Opportunity sufficiency	0–1	KiM-derived quantity function
A_s	Subcategory accessibility score	0–1	$(T + Q) / 2$
$A_{o,m,c}$	Category accessibility score	0–1	Mean or minimum of subcategories
W_c	Category weight	0–1	KiM-derived category priority weights
$SAI_{o,m}$	Origin-level SAI	0–1	Weighted category average
$SAI_{N,m}$	Neighbourhood-level SAI	0–1	Mean or population-weighted mean across origins

Table 12: Summary of SAI calculation steps.

Step	Calculation	Purpose	Output
1. Minimum travel time	$t_s = \min\{t_d \mid d \in D_s\}$	Identifies the nearest reachable opportunity in each subcategory.	Minimum travel time
2. Temporal acceptability	$T_s = f_{m,s}(t_s)$	Converts travel time into behavioural acceptability.	T score
3. Opportunity count	n_s^{thr} or n_s^{sep}	Counts reachable opportunities.	Opportunity count

4. Opportunity sufficiency	$Q_s = \min\left(1, \sum_{i=1}^{n_s} q_{s,i}\right)$	Converts opportunity count into sufficiency score.	Q score
5. Subcategory score	$A_s = \frac{T_s + Q_s}{2}$	Combines time and quantity dimensions.	Subcategory score
6. Category score	$A_{o,m,c}^{mean}$ or $A_{o,m,c}^{min}$	Aggregates subcategories into main categories.	Category score
7. Origin-level score	$SAI_{o,m} = \sum_c W_c \cdot A_{o,m,c}$	Aggregates category scores at each origin.	Origin-level SAI
8. Neighbourhood-level score	$SAI_{N,m} = \frac{1}{n_o} \sum_{o \in N} SAI_{o,m}$	Aggregates origins within each neighbourhood.	Neighbourhood-level SAI

3.5 Sensitivity Analysis

Sensitivity analysis is included in this research to assess how robust the Sustainable Accessibility Index (SAI) is to key methodological assumptions. This is important because accessibility measures, especially cumulative and composite indicators, can be influenced by choices about travel-time thresholds, opportunity definitions, weighting, and aggregation. As discussed in Chapter 2, the OECD handbook on composite indicators emphasises that sensitivity analysis is necessary to evaluate whether index results are stable under alternative plausible assumptions (Nardo et al., 2005).

The sensitivity analysis directly addresses SQ4, which asks how sensitive the proposed SAI is to key methodological assumptions. In this study, the sensitivity analysis is used as a way to critically evaluate whether the proposed methodology behaves in line with the theoretical assumptions on which the SAI is based. For this reason, a set of analytical hypotheses is formulated that provide the basis for the sensitivity analysis.

The hypotheses serve two purposes. First, these hypotheses should help clarify what patterns the SAI should be able to reveal to capture sustainable accessibility in a conceptually meaningful way. Second, the hypotheses provide a basis for selecting and interpreting the sensitivity tests. The sensitivity tests examine whether these expected patterns remain stable when key methodological choices are changed, such as the maximum travel-time threshold for public transport, the way opportunities are counted, and the aggregation method used to go from subcategories to categories SA scores.

3.5.1 Literature-Based Analytical Hypotheses

This section formulates analytical hypotheses that are derived from the theoretical framework and relevant literature and against which the empirical results can be interpreted.

H1: More Urbanised Neighbourhoods are Expected to Have Higher Sustainable Accessibility

More urbanised neighbourhoods are expected to have higher sustainable accessibility because they generally combine denser opportunity distributions, shorter travel distances, finer-grained street networks, and stronger public transport provision (International Transport Forum, 2019; Pot et al., 2023). This hypothesis follows from the land-use and transport components of accessibility: access improves when more opportunities are located within shorter travel times and when transport networks connect origins and destinations more effectively (Geurs & van Wee, 2004; Litman, 2025). However, this hypothesis should not be interpreted deterministically. Urban density alone does not guarantee sustainable accessibility. Accessibility depends on the interaction between land-use patterns, transport networks, mode availability, and temporal constraints. Vale and Lopes research (2023), for example, shows that urban access differs across European cities, indicating that accessibility is also shaped by the variety of urban form rather than by metropolitan size or opportunity density alone.

Therefore, more urbanised neighbourhoods are expected to show higher sustainable accessibility than less urbanised neighbourhoods, but the strength of this pattern is expected to vary by transport mode and opportunity category.

H2: Multimodal Trip Chains are Expected to Increase Sustainable Accessibility

Multimodal chains that combine walking or cycling with public transport are expected to increase sustainable accessibility relative to walking or cycling alone. Walking and cycling provide local access, but their range is limited by travel time and physical effort. Public transport can extend the range of reachable destinations, but its usefulness depends partly on access, starting leg of the journey, and egress, final leg of the journey from destination transit stop to the final destination (Krygsman et al., 2004). In the context of sustainable accessibility, multimodal chains are especially relevant because they allow active travel and public transport to function as an integrated alternative to car-based travel.

Therefore, multimodal combinations of walking or cycling with public transport are expected to increase the number and diversity of reachable opportunities, particularly in areas where essential opportunities are not located within short walking or cycling distances.

H3: Neighbourhood Averages are Expected to Conceal Internal Spatial Variation

Neighbourhood-level averages are expected to conceal variation between origins within the same neighbourhood. Accessibility is spatially specific since it is calculated from particular origins to spatially distributed opportunities through specific transport networks. Residents in different parts of the same neighbourhood may have different levels of potential access, even when they are represented by the same neighbourhood-level average. For example, a physical barrier such as a river without a bridge or train railway could cross through neighbourhoods causing different accessibility levels depending from where accessibility is measured. This way of thinking has made some scholars call for using more spatially disaggregated outputs in accessibility measures as it can help identify local accessibility gaps (Geurs & van Wee, 2004; Papa et al., 2015).

To conclude, neighbourhood-level averages are expected to hide internal spatial variation. Origin-level analysis is therefore expected to reveal differences in sustainable accessibility within neighbourhoods.

H4: Results are Expected to be Sensitive to Methodological Assumptions

SAI results are expected to be sensitive to methodological assumptions, particularly travel-time thresholds, weighting choices, opportunity definitions, and aggregation rules. This hypothesis is well established in accessibility measurement literature. Cumulative opportunity measures are transparent and easy to communicate, but they require the selection of travel-time thresholds. Tomasiello et al. (2023) note that the choice of a single travel-time threshold can influence the resulting scores.

Performing sensitivity tests are also expected because the SAI is a composite index. The OECD handbook emphasises that composite indicators depend on methodological choices concerning indicator selection, normalisation, weighting, and aggregation (Nardo et al., 2005). Saisana et al. (2005) argue that uncertainty and sensitivity analysis are necessary for assessing the reliability and quality of composite indicators. Thus, including a sensitivity analysis is necessary to assess whether the main empirical patterns are robust.

To conclude, SA scores are expected to change under alternative travel-time thresholds, opportunity definitions, and aggregation methods.

3.5.2 Sensitivity Tests

Based on the four hypotheses, three sensitivity tests are applied. See Table 13 for an overview of the sensitivity test applied.

Table 13: Overview sensitivity tests applied in this research.

Test	Main parameter	Sensitivity variant	Purpose
Travel-time threshold	Base travel-time thresholds derived from ODiN/KiM travel-time data.	Extended threshold for multimodal modes of W + PT and B + PT.	Test whether accessibility patterns change when longer travel times are allowed.
Opportunity count	Q-threshold: counts opportunities only in acceptable travel-time bands.	Q-seperate: counts all reachable opportunities within the routing limit.	Tests whether separating quantity from time acceptability changes sustainable accessibility scores.
Category aggregation	Mean of subcategory scores.	Minimum Q component subcategory score.	Tests whether results change when poor access to one subcategory cannot be compensated by stronger access to others.

Travel-Time Threshold Sensitivity

The first test examines the effect of changing the maximum travel-time threshold. The main specification uses the base thresholds defined in Section 3.4. Alternative scenarios extend the

travel-time limits, particularly for public-transport-based modes, to test whether allowing longer trips changes the relative accessibility of the case study neighbourhoods.

This test is important because a longer threshold may increase the number of reachable opportunities, especially in less urbanised areas where destinations are more dispersed. However, longer trips may be less meaningful for everyday accessibility if they exceed behaviourally plausible travel times. Comparing base and extended thresholds therefore helps evaluate whether results are driven by the selected time limits.

Opportunity-Count Specification

The second test compares two ways of counting reachable opportunities for the opportunity sufficiency component Q. The main specification uses the Q-threshold variant, which counts only opportunities located within travel-time bands where acceptability is at least 0.5. The sensitivity variant uses the Q-separate approach, which counts all reachable opportunities within the routing limit, regardless of their time-acceptability score.

This distinction tests whether the index should count only opportunities that are reasonably acceptable in travel-time terms, or whether opportunity quantity should be treated independently from temporal acceptability. The comparison is especially relevant for neighbourhoods where many opportunities are technically reachable but require relatively long travel times.

Aggregation Method Sensitivity

The third test examines how the index changes when a different aggregation rule is used at the category level. The main specification uses the category mean, where subcategory scores are averaged within each main category. The sensitivity variant uses the category minimum, where the lowest subcategory score determines the category score.

The mean method reflects average performance across related subcategories. The minimum method applies a stricter sufficiency principle, where a category is considered weak if any essential subcategory is poorly accessible. This test is relevant because composite indicators can hide poor performance in one dimension when other dimensions score highly. Testing the minimum method therefore helps examine whether high SA scores depend on compensation between subcategories.

3.5.3 Role of Sensitivity Analysis in This Research

The sensitivity tests are intended to show how the index behaves under alternative plausible assumptions. Therefore, the main results in Chapter 4 are presented alongside sensitivity results where relevant. Whether the main empirical patterns remain stable across specifications is discussed in Chapter 5.

The sensitivity analysis helps answer whether the SAI is robust enough for planning interpretation. If the broad patterns remain consistent across variants, the index can be interpreted with greater confidence, whereas if the results change substantially, this indicates that the index is sensitive to specific methodological assumptions and should be interpreted more cautiously.

3.6 Validation and Interpretation Strategy

The SAI is a proposed constructed index, so its results need to be assessed for credibility and practical meaning. Full validation against observed travel behaviour or resident satisfaction is beyond the scope of this research, because the research does not include individual-level travel survey data for the selected neighbourhoods. Instead, the validation strategy focuses on three forms of plausibility assessment: theoretical validation, comparative validation, and interpretability validation.

The validation strategy evaluates whether the SAI behaves in line with the theoretical framework and the proposed hypotheses and whether the results can be interpreted in a useful way for neighbourhood-level planning.

3.6.1 Theoretical Validation

The theoretical validation assesses whether the SAI behaves in line with the conceptual framework developed in Chapter 2. The outcome of the index should respond logically to the components of sustainable accessibility. For example, neighbourhoods with more nearby essential opportunities, shorter sustainable-mode travel times, and stronger public transport connections should generally receive higher SA scores than neighbourhoods with fewer opportunities and longer travel times.

This validation step is linked to hypothesis 1, which states that more urbanised neighbourhoods are expected to show higher sustainable accessibility because they often combine denser opportunity distributions with stronger sustainable transport provision. However, this hypothesis is not treated as a fixed rule. If a less urbanised neighbourhood performs well for certain modes or categories, this may indicate favourable local land-use patterns or strong transport connections. Thus, the SAI is evaluated on whether it reveals meaningful exceptions and not solely on whether it can confirm expected patterns.

3.6.2 Comparative Validation Across Modes and Urbanisation Levels

The comparative validation examines whether the SAI produces plausible differences across case study contexts and transport modes. This includes comparing results between highly urbanised, moderately urbanised, and low-urbanised or rural neighbourhoods. It also includes comparing walking, cycling, walking + PT, and cycling + PT scores.

This step is linked to hypothesis 2, which states that multimodal chains are expected to increase sustainable accessibility compared with walking or cycling alone. If the index shows higher SA scores for multimodal combinations, especially in areas where local opportunities are dispersed, this supports the plausibility of the measure. Whereas, if the multimodal effect is weak or absent, the results may indicate poor public transport connectivity, weak first- or last-mile access, or limited reachable opportunities within acceptable travel times.

Thus, the comparative validation examines whether the index explains why certain neighbourhoods perform better or worse by decomposing results by mode and opportunity category.

3.6.3 Internal Spatial Validation

The internal spatial validation examines whether neighbourhood-level averages conceal variation between origin points within the same neighbourhood. To assess this, the results are reported not only as neighbourhood averages but also, through origin-level distributions, ranges, standard deviations, or maps. This allows the analysis to identify whether a neighbourhood's average score reflects broadly shared accessibility or whether high accessibility is concentrated in only part of the area.

This is linked to hypothesis 3, which states that average neighbourhood scores are expected to mask internal spatial variation. This step is important for planning interpretation. A neighbourhood may have a moderate average SA score while still containing pockets of poor accessibility. Conversely, a low average score may conceal specific areas with relatively good access due to proximity to a station, local centre, or cluster of services.

3.6.4 Robustness Validation

The robustness validation uses the sensitivity tests to assess whether the main findings remain stable when calculating the SAI with alternative methodological specifications. The three sensitivity tests evaluate the main SAI result patterns on extending travel-time thresholds, different opportunity counting specifications and aggregation rule differences. Stable findings across these variants would suggest that the SAI results are relatively robust whereas substantial changes would suggest that specific findings should be interpreted with caution and that the relevant methodological choice has a strong influence on the SAI outcome.

This validation step is linked to hypothesis 4, which states that the SAI results are expected to be sensitive to methodological assumptions. The robustness validation makes the interpretation more transparent by showing which conclusions are consistent across the specifications and which are more methodological assumption dependent.

3.6.5 Limitations of the Validation Strategy

The validation strategy has several limitations. First, the SAI measures potential accessibility rather than realised travel behaviour. Therefore, it cannot confirm whether residents actually use the reachable opportunities identified by the index. Second, the index does not directly measure subjective perceptions, safety, affordability, comfort, or personal constraints. Third, without resident survey data, the analysis cannot validate the index against experienced accessibility. Fourthly, the results are not discussed with planning or mobility professionals to provide face validation and gain insights in the interpretability and accuracy of the SAI index scores due to time constraints within this research.

These limitations mean that the SAI should be interpreted as a structured planning indicator rather than as a complete measure of lived accessibility. The value of the SAI lies in making sustainable accessibility measurable, comparable, and decomposable at the neighbourhood level.

4. RESULTS

This chapter presents the empirical results of the Sustainable Accessibility Index (SAI) by cross comparing the results from three case study sets. The results are shown in five sections. First, the overall neighbourhood-level scores by transport mode are reported in Section 4.1. Next, in Section 4.2 the origin-level variation within neighbourhoods are presented. Then, the category and subcategory patterns are stated in Section 4.3. Furthermore, in Section 4.4 the temporal acceptability component T from the opportunity sufficiency component Q are separated. Subsequently, in Section 4.5 the sensitivity analysis results are reported. Finally, in Section 4.6 the main findings are summarized.

4.1 Neighbourhood-Level SA Scores by Mode

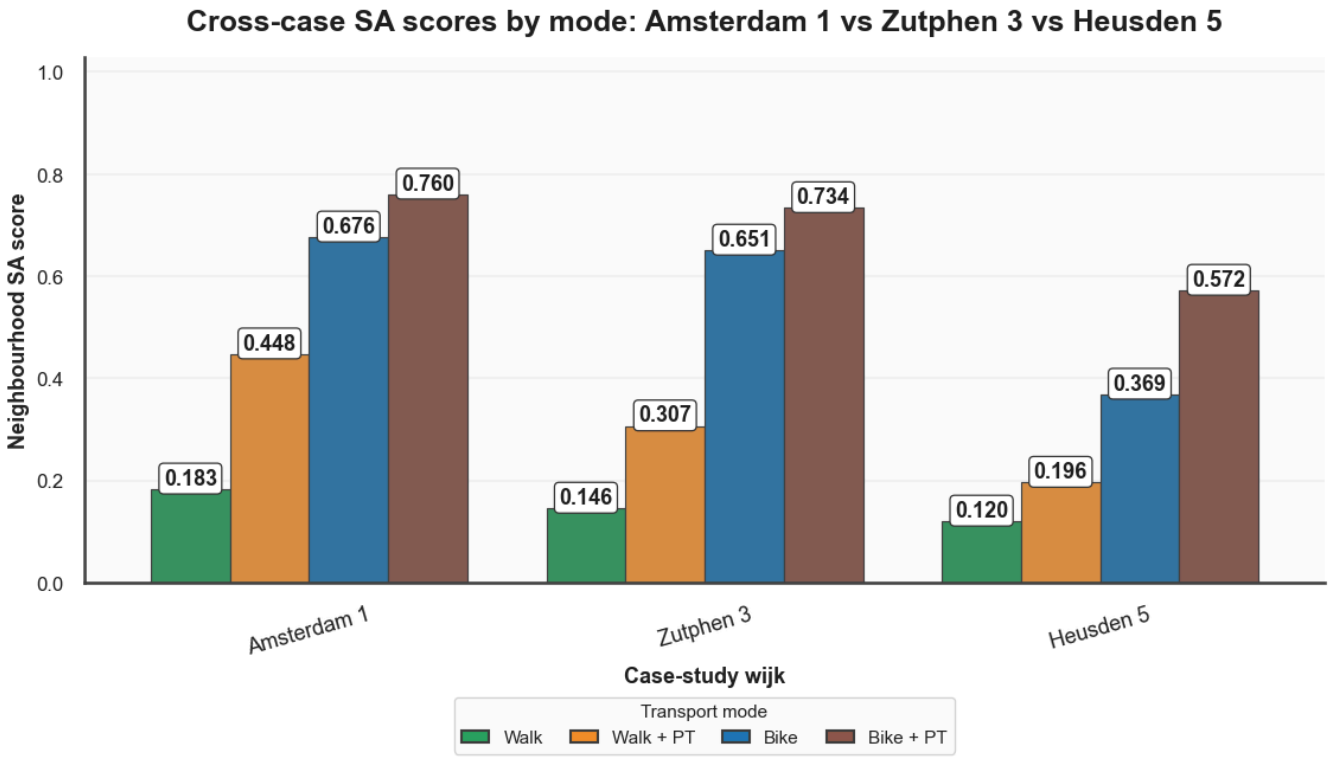
This section presents the neighbourhood-level SA scores by transport mode. The comparison is organised into three cross-case study sets, each containing one case study of each urbanisation level. This allows the results to be compared across urbanisation levels and between transport modes. Figure 5 shows the cross-case study set consisting of Amsterdam, Zupthen, and Heusden. The second cross-case study set with Den Haag, Smallingerland, and Hulst is shown in Figure 6. Figure 7 shows the last cross-case study set of Tilburg, Dronten, and Schagen.

Across all three cross-case study sets, two general patterns are visible. First, the results show an urbanisation gradient where highly urbanised neighbourhoods generally score higher SA scores than moderately urbanised neighbourhoods and the rural cases. Second, the results show a clear modal hierarchy where walking produces the lowest SA scores in all cases, while cycling and cycling combined with public transport produce the highest scores.

In the Amsterdam–Zutphen–Heusden comparison, Amsterdam records the highest score across all four modes. Zutphen follows respectively. Heusden records the lowest scores. The difference between Amsterdam and Zutphen is relatively small for cycling and cycling combined with public transport (PT). The largest difference in this cross case study set is between the two more urbanised cases and Heusden, particularly for cycling.

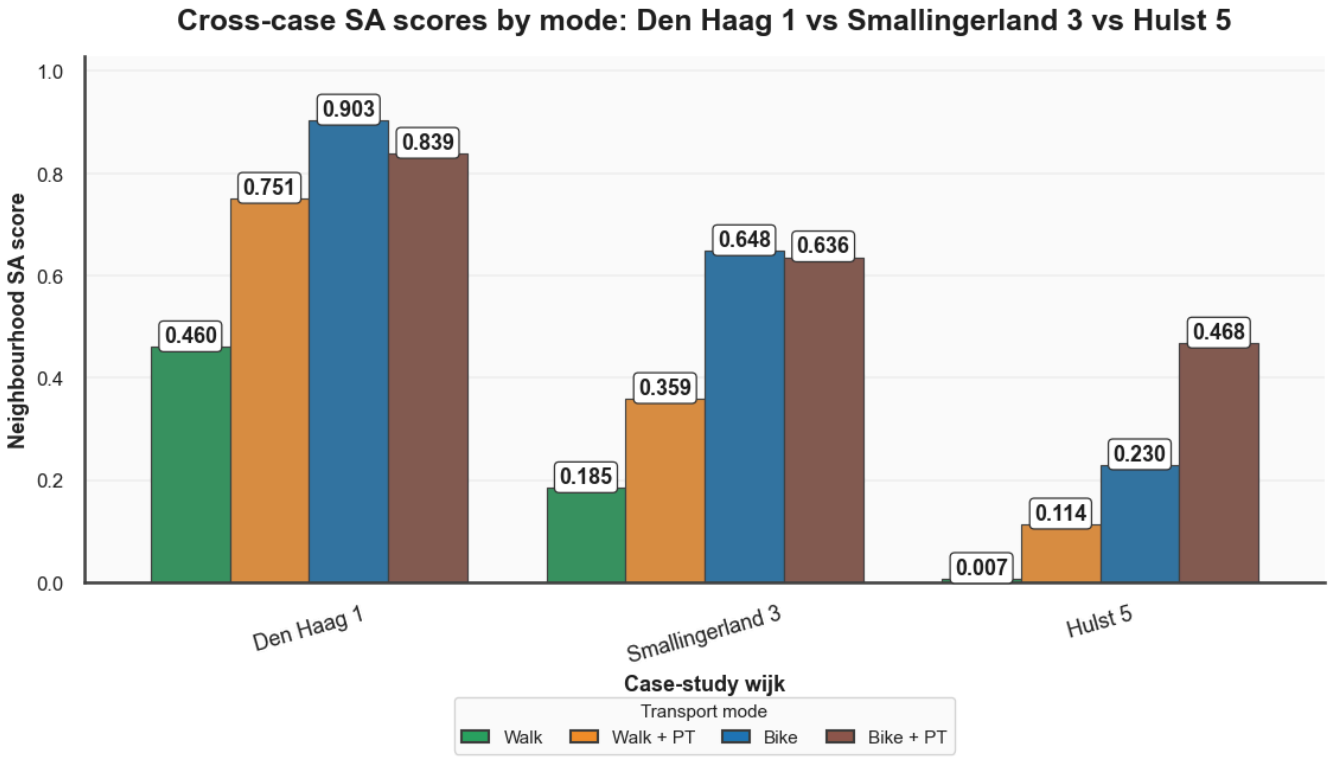
In the Den Haag-Smallingerland-Hulst comparison, Den Haag stands out with high SA scores across all modes. The walking score of Den Haag is noticeably high compared to the other highly urbanised cases. Additionally, the cycling mode outperforms the cycling + PT mode in both Den Haag and Smallingerland. Furthermore, Hulst shows a near zero SA score for walking.

The same urbanisation gradient is visible in the Tilburg-Dronten-Schagen comparison. Again, in this cross-case study set the cycling mode outperforms the cycling + PT mode for the two urbanised case study areas.



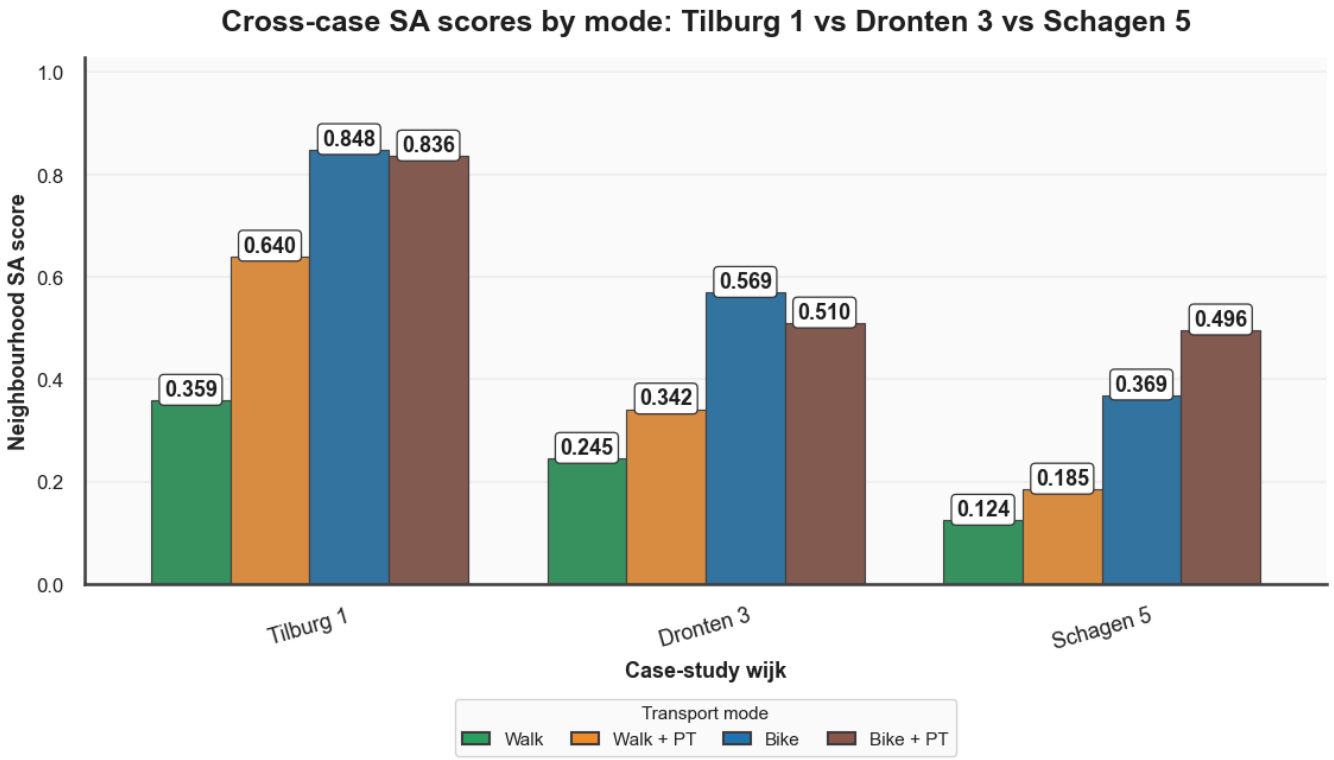
Note: within each case-study wijk, the four bars compare the four transport modes. The bar height is the neighbourhood SA score on a 0-1 scale, so higher bars mean better sustainable accessibility. White labels show the exact score, and the colors always map to the same mode across all figures.

Figure 5: Baseline SA scores for cross case study set 1: Amsterdam-Zuthpen-Heusden.



Note: within each case-study wijk, the four bars compare the four transport modes. The bar height is the neighbourhood SA score on a 0-1 scale, so higher bars mean better sustainable accessibility. White labels show the exact score, and the colors always map to the same mode across all figures.

Figure 6: Baseline SA scores for cross case study set 2: Den Haag-Smallingerland-Hulst.



Note: within each case-study wijk, the four bars compare the four transport modes. The bar height is the neighbourhood SA score on a 0-1 scale, so higher bars mean better sustainable accessibility. White labels show the exact score, and the colors always map to the same mode across all figures.

Figure 7: Baseline SA scores for cross case study set 3: Tilburg-Dronten-Schagen.

4.2 Origin-Level Variation Within Neighbourhoods

The origin-level boxplots and maps show how neighbourhood-level averages are distributed across origins. This is important because neighbourhood means can conceal internal variation. To improve the readability of the remaining results chapter, only cross-case study 1 (Amsterdam - Zutphen - Heusden) is presented (see Figure 8). The case specific visualizations of the other case studies can be found in Appendix B.

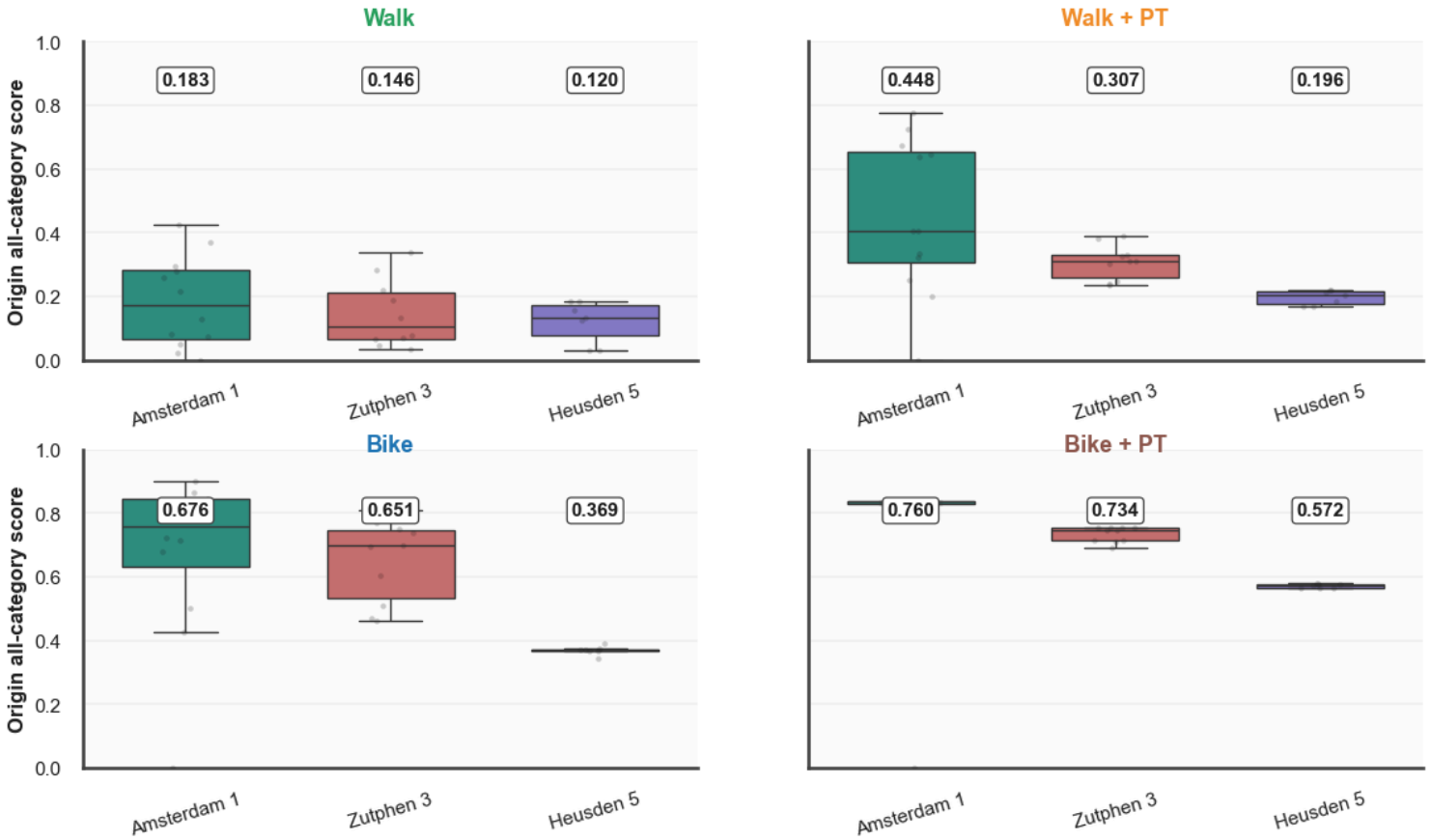
Out of the main cross-case comparison, Amsterdam shows the largest internal range. The spatial maps show clear differences between lower- and higher-scoring origins, particularly for walking and walking + PT (see Figure 9). Zutphen has lower walking scores and more compact multimodal scores. This indicates that the cycling + PT score is both high and relatively consistent across sampled origins. Heusden has low walking scores, and a narrow cycling + PT range. The rural case therefore has lower overall accessibility, but its cycling + PT scores are relatively uniform across the sampled origins.

The other case study areas confirm that origin-level variation differs substantially between cases. Den Haag and Tilburg have high and compact cycling distributions. Hulst shows the strongest walking limitation. Dronten has a wider walking range, but compact cycling and cycling + PT distributions.

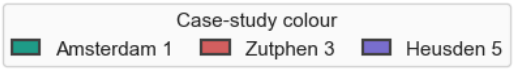
Overall, the origin-level results show that neighbourhood averages do not fully describe accessibility conditions within neighbourhoods. In some cases, high neighbourhood scores are

broadly distributed across origins. In others, averages conceal substantial differences between origins.

Cross-case origin score comparison by mode: Amsterdam 1 vs Zutphen 3 vs Heusden 5



Note: Each panel shows one transport mode. The box shows the spread of origin all-category scores for the three case-study wijken, the dots are individual origins, and the white labels are the mean origin scores for each case. Case colors stay consistent across the figure, and higher values mean stronger accessibility outcomes on the 0-1 scale.



Origin-level sustainable accessibility by mode | Amsterdam_1

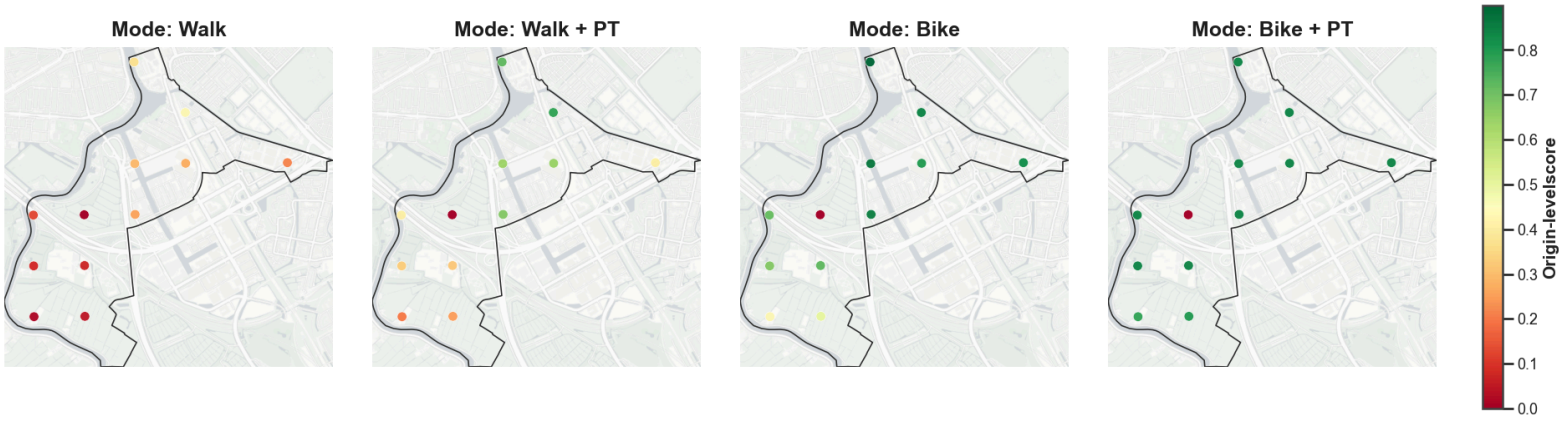


Figure 8: Origin-level SAI distributions by case and mode.

Figure 9: Spatial distribution of origin-level SA scores for Amsterdam.

4.3 Category and Subcategory Results

The category and subcategory results show how the opportunity domains contribute to the overall SA scores. In general, two patterns are visible across the nine cases. First, Employment and Leisure are often highly accessible. Second, Education and Public services are often the least accessible destinations.

For the selected Amsterdam neighbourhood, walking accessibility is the highest for Employment and Leisure destinations, whereas Public services is the least accessible destination (see Figure 10 and 11). At subcategory level, Sport, Nature, and Restaurants are the most accessible destinations by foot. The weakest destinations by foot are Primary school, Hospital, and HBO/University. The low score of 0 for primary school is counterintuitive but logical as there are no primary schools within this proximity in Omval/Overamstel neighbourhood which can be explained by the fact that this neighbourhood is still in development to a residential area. For the Amsterdam case study, the results also show that mode extension substantially changes the category profile. Walking + PT improves access to several destinations, such as Employment, Leisure, Education, and Public services. The accessibility by bike shows even higher accessibility scores for the destinations relating to Employment, Leisure, Transport stops, and Education. The multimodal combination of cycling + PT records the highest accessibility scores over the categories.

Zutphen shows similar high category scores overall but lower walking accessibility scores. Cycling + PT produces a stronger overall accessibility score. This suggests that multimodal travel improves the sustainable accessibility in Zutphen.

Heusden shows the clearest accessibility gaps. Walking accessibility is limited to Employment and Transport stops destinations, while Education and Public services both score 0. The multimodal combination of cycling + PT does raise the accessibility scores of Employment and Leisure. However, access to Transport stops remain very low, and Education accessibility score remains lower than in the urban and medium-urban cases.

Across the other six cases, Employment and Leisure are generally among the strongest categories, especially under cycling and cycling + PT. Den Haag and Tilburg show the most balanced category profiles. For cycling accessibility, Den Haag and Tilburg score around or above 0.8 in all categories. By contrast, Hulst, Schagen, and Heusden show repeated zero or near-zero scores for Education, Public services, Hospital, Train, HBO/University, and Kindergarten destinations under walking and walking + PT modes.

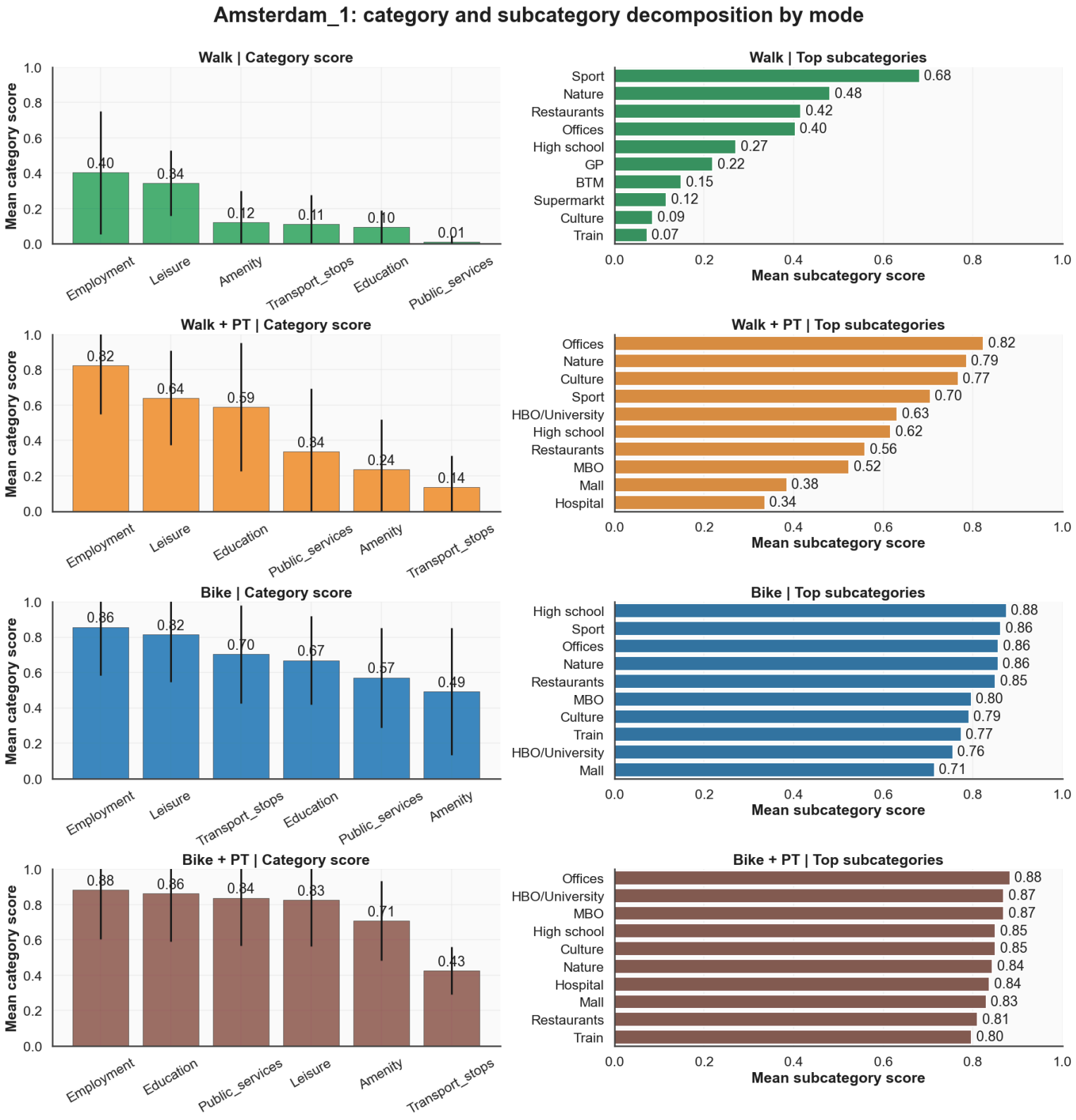


Figure 10: Best category and subcategory-level SA scores by mode for Amsterdam.

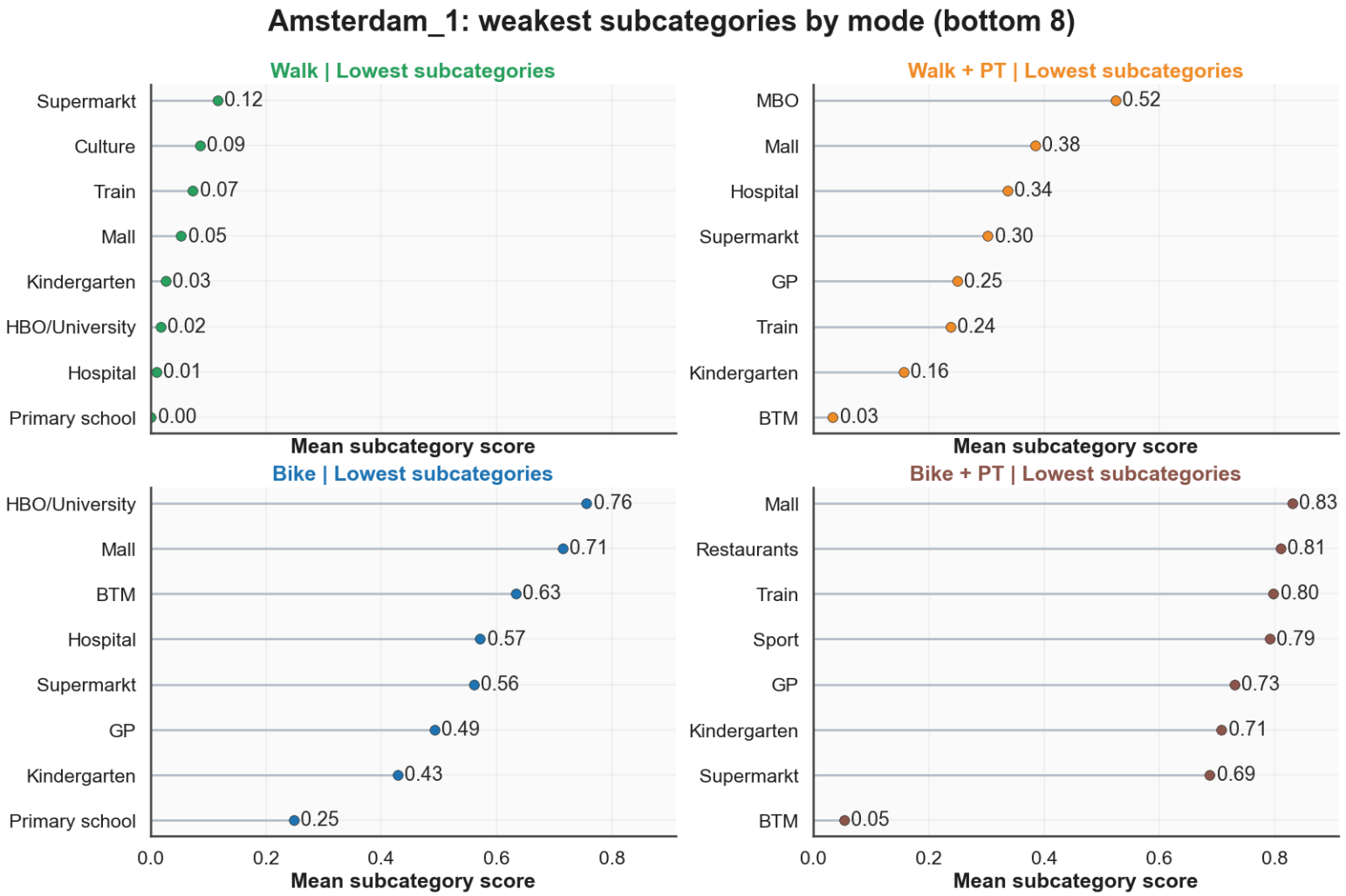


Figure 11: Weakest subcategory-level SA scores by mode for Amsterdam.

4.4 Temporal Acceptability and Opportunity Sufficiency Components

The T/Q component results separate the role of temporal acceptability T from opportunity sufficiency Q. This separation shows whether low sustainable accessibility scores are associated mainly with long travel times, insufficient opportunity quantity, or both. Figure 12 can be interpreted as follows: the upper left quadrant (purple) shows categories where T is low but the Q is high meaning that the travel time to that category is relatively long but the quantity reachable is high. The upper right quadrant (green) suggests that both the travel time and the quantity of that category is high indicating a highly sustainable accessible category. The lower left quadrant (orange) indicates that the category is not quick to access and not many alternatives available. Finally, the lower right quadrant (blue) implies that the travel time to the category is relatively quick and acceptable but there are not many alternatives of the category reachable.

In general, low scores are not associated with a single mechanism. In some categories, low scores are primarily linked to low temporal acceptability. In other cases, they are linked to

insufficient opportunity quantity. Rural and medium-urban cases more often show zero or near-zero Q scores, while urban cases more often show high T and Q values once cycling or public transport is included. The T/Q plots therefore show that similar overall SA scores can result from different underlying component patterns.

In Amsterdam (see Figure 12), low walking scores in Public services and Education are associated with both low temporal acceptability and low opportunity sufficiency. Employment and Leisure have higher T and Q values under the mode walking, and most categories move into stronger T/Q positions under the modes of cycling and cycling + PT. Zutphen shows a similar but lower overall pattern, with Public services remaining weak under walking and walking + PT. Cycling improves T values for several categories, but Education and Amenity remain relatively low for the Q component. The multimodal combination of cycling + PT produces high T/Q scores for Employment and Leisure, while Transport stops and Education remain the weakest components. For Heusden, Education and Public services have the weakest T/Q position under walking and walking + PT modes. In the cycling mode, Education remains weak, but Amenity improves slightly. The multimodal combination of cycling + PT improves Employment and Leisure T/Q values substantially, but Transport stops remain quantity-constrained. Indicating that there are no sufficient amount of transport stops within acceptable travel times.

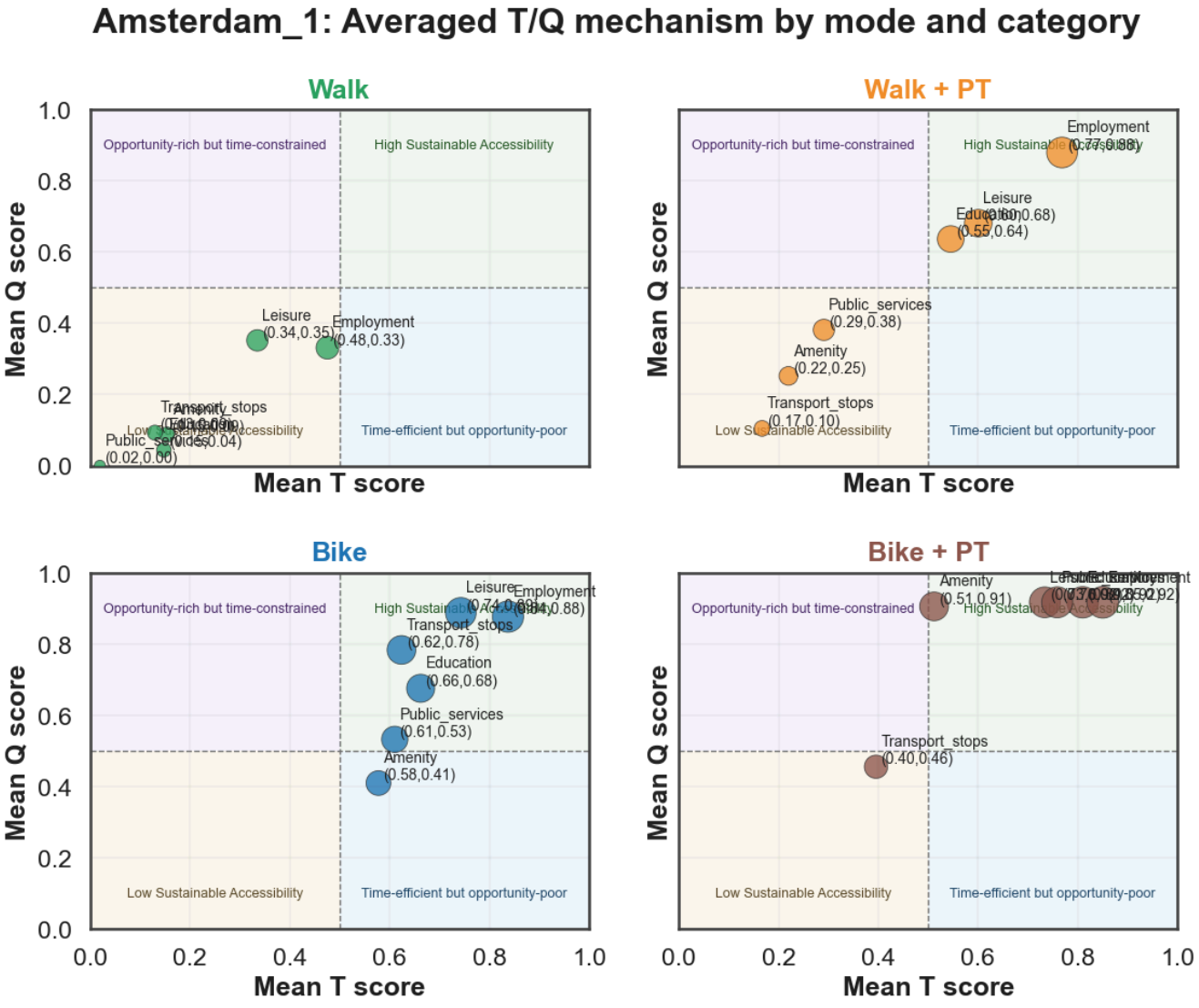


Figure 12: T/Q component plot by mode, and category for Amsterdam.

4.5 Sensitivity Analysis Results

This section examines how sensitive the Sustainable Accessibility Index (SAI) is to alternative methodological assumptions. The purpose of the sensitivity analysis is twofold. First, it assesses the extent to which absolute SA scores depend on selected modelling choices. Second, whether the main empirical patterns remain robust when these assumptions are changed is evaluated.

In this section, sensitivity refers to the magnitude of changes in SA scores under alternative specifications. Robustness refers to the extent to which substantive patterns, such as origin-level rankings and cross-case ordering, remain stable despite these changes.

The baseline specification is: **Maximum travel time of 45 minutes for public transport – Threshold-based opportunity counting – Mean based aggregation across subcategories**. This base specification uses a 45-minute public transport (PT-) profile, counts opportunity quantities only within acceptable travel-time bands, and applies mean aggregation from subcategories to categories score.

In the sensitivity tests, three alternative calculation methods are compared against the same baseline. The first sensitivity tests replace the mean aggregation with a minimum aggregation, thus giving more weight to the weakest-performing subcategory. The second test counts all the reachable opportunities instead of only those within acceptable travel times. The last one combines both changes by counting all reachable opportunities and applying the minimum aggregation. Table 14 summarises these specifications and provides the technical labels used in the sensitivity analysis.

Table 14: Sensitivity tests methods and description.

In-text reference	Abbreviation	Description
Threshold-counted opportunities with minimum aggregation.	Qthr_catmin	Test whether the SA scores change when the weakest subcategory has more influence in aggregation.
All opportunities with mean aggregation.	Qsep_catmean	Test whether the SA scores change when all reachable opportunities are counted.
All opportunities with minimum aggregation.	Qsep_catmin	Test the combined effect of broader opportunity counting and stricter aggregation.

The analysis is structured in four steps. First, the effect of extending the PT-profile from 45 to 60 minutes is tested, whilst keeping the model base specification unchanged. Second, the effects of the opportunity counting and aggregation assumptions are assessed at neighbourhood level. Third, the origin-level robustness is examined by comparing each alternative specification with the baseline in terms of SA score changes, rank-order stability, and the share of origins with substantial deviations in SA scores. Fourth, the cross-case ranking

stability per mode is assessed. Together, these steps allow for the distinguishingment between changes in absolute scores and changes in modal and spatial ranking of the results.

4.5.2 Sensitivity to Public Transport Travel Time Profile

The first sensitivity test isolates the effect of extending the PT-profile from maximum travel time of 45 to 60 minutes. This is assessed by comparing the baseline specification of the maximum travel time of 45 minutes PT-profile with the equivalent 60 minutes PT-profile baseline specification. This comparison keeps the opportunity-counting method and category aggregation method the same. Therefore, any score difference can be attributed specifically to the change in the PT-profile.

The pattern of changes being positive or zero are consistent across the case study areas (see Table 15). This indicates that the effect of extending the PT-profile on the neighbourhood-level Sustainable Accessibility (SA) score is small and the opportunities contributing to the baseline PT scores are already captured within the 45-minute PT-profile.

Walking and cycling scores do not change, as expected, because these modes are not affected by the PT-profile. Additionally, it was expected that extending the PT-profile would maintain or increase the SA scores given that longer travel time to opportunities logically could expand the set of reachable opportunities. For the PT-based modes, the changes are limited since Walking + PT increases by an average of 0.004 across the nine cases, while cycling + PT increases by an average of 0.002.

The largest walking + PT increase occurs in Zutphen. This is followed by Heusden, Hulst, Dronten, and Schagen. Amsterdam increases only slightly, while Den Haag shows no change. For cycling + PT, the largest increases are found in Smallingerland, and Schagen, followed by Heusden and Dronten.

Table 15: Sensitivity of SA scores to extending the public transport time profile from 45 to 60 minutes per case study.

Case study area	**Δ Walk + PT**	**Δ Delta Bike + PT**
Amsterdam	+0.002	0
Zutphen	+0.011	+0.002
Heusden	+0.008	+0.003
Den Haag	0	0
Smallingerland	+0.001	+0.004
Hulst	+0.006	0
Tilburg	0	0
Dronten	+0.005	+0.003
Schagen	+0.004	+0.004
Mean	**+0.004**	**+0.002**

Note: The values represent PT60_Qthr_catmean - PT45_Qthr_catmean SA scores. The deltas are calculated from the underlying values and rounded to three decimals.

4.5.3 Sensitivity to Opportunity-Counting and Aggregation Assumptions

The second sensitivity test examines the effect of changing the opportunity-counting and category aggregation assumptions. This section assesses how the SA scores change when the internal construction of the SAI is modified.

The three alternative methods are compared against the same maximum travel time PT-profile baseline specification (threshold-based opportunity counting - mean-based aggregation across subcategories). This means that the maximum 45-minute PT-profile with the alternative specifications is compared with the same maximum 45-minute PT-profile baseline specification. Similarly, the maximum 60-minute PT-profile with the alternative specifications is compared with the related maximum 60-minute PT-profile baseline specification. For a reminder of the specifications that are tested, see Table 14.

At neighbourhood level, the results show three patterns. First, changing from the mean category aggregation to the minimum category aggregation tends to lower the SA scores. Second, changing from threshold-counted opportunities to all reachable opportunities tends to raise the SA scores. Third, the combined method, which counts all reachable opportunities but applies minimum aggregation, produces mixed results.

For the maximum 45-minute PT-profile cases (see Table 16), the *threshold-based opportunity counting - minimum based aggregation* specification lowers scores in every case. The largest mean reductions in SA scores occur in Dronten, Smallingerland, Tilburg, and Schagen. The smallest reduction occurs in Hulst. This indicates that minimum aggregation is a stricter aggregation rule than mean aggregation. Under minimum aggregation, the weakest subcategory within a category has a stronger influence on the category score. As a result, cases with uneven access across subcategories are penalised more strongly.

Changing from *threshold-based opportunity counting* to *all opportunity counting* while retaining mean aggregation raises scores in every case. For the maximum 45-minute PT-profile cases, the *all opportunities counting - mean aggregation* specification produces on average an increase in SA scores. The largest increases occur in Schagen, Hulst, Heusden, Zutphen, and Amsterdam. Den Haag has the smallest increase. This shows that counting all reachable opportunities produces higher SA scores than counting only opportunities within acceptable travel-time bands. The effect is particularly visible in cases where additional opportunities are technically reachable but fall outside stricter acceptability thresholds.

For the specification of *all opportunities counting - minimum aggregation* mixed results are found. The average changes range from -0.080 in Smallingerland and -0.064 in Dronten to +0.063 in Amsterdam and +0.056 in Hulst. This method therefore does not uniformly increase or decrease scores. Instead, it combines the upward effect of counting all reachable opportunities with the downward effect of applying minimum aggregation.

Table 16: Neighbourhood-level sensitivity under maximum of 45-minute public transport threshold.

Case	Threshold-based opportunity counting – minimum based aggregation reflection weakest subcategory performance	All opportunities – mean based aggregation across subcategories	All opportunities – minimum based aggregation reflecting weakest subcategory performance
Amsterdam	-0.074	+0.122	+0.063
Zutphen	-0.086	+0.129	+0.041
Heusden	-0.090	+0.131	-0.004
Den Haag	-0.094	+0.064	+0.016
Smallingerland	-0.147	+0.101	-0.080
Hulst	-0.050	+0.139	+0.056
Tilburg	-0.119	+0.106	+0.034
Dronten	-0.156	+0.106	-0.064
Schagen	-0.111	+0.143	+0.006

Note: Values indicate the average neighbourhood-level SA score change relatively to the 45-minute PT baseline specification. Positive values indicate higher scores than the baseline, whereas negative values indicate lower scores.

For the extended PT-profile of maximum 60-minutes (see Table 17), the same general pattern remains visible. The *threshold-based opportunity counting - minimum based aggregation* specification again lowers scores in all cases. The *all opportunities counting - mean aggregation* specification again raises scores in all cases. The *all opportunities counting - minimum aggregation* specification produces on average an increase in SA scores, but the direction of change remains mixed across cases. Smallingerland still decreased, while the other cases increased.

Table 17: Neighbourhood-level sensitivity under maximum of 60-minute public transport threshold.

Case	Threshold-based opportunity counting – minimum based aggregation reflection weakest subcategory performance	All opportunities – mean based aggregation across subcategories	All opportunities – minimum based aggregation reflecting weakest subcategory performance
Amsterdam	-0.074	+0.132	+0.079
Zutphen	-0.085	+0.160	+0.080
Heusden	-0.089	+0.175	+0.057
Den Haag	-0.094	+0.064	+0.016
Smallingerland	-0.147	+0.138	-0.014

Hulst	-0.050	+0.154	+0.073
Tilburg	-0.119	+0.108	+0.037
Dronten	-0.153	+0.153	+0.017
Schagen	-0.110	+0.148	+0.013

Note: Values indicate the average neighbourhood-level SA score change relatively to the 60-minute PT baseline specification. Positive values indicate higher scores than the baseline, whereas negative values indicate lower scores.

4.5.4 Origin-Level Robustness and Spearman Correlation

The previous subsections examined sensitivity at neighbourhood level. However, neighbourhood averages conceals variation between individual origins as shown in Chapter 4.2. Therefore, origin-level robustness was assessed using three indicators: i. Mean delta, ii. Spearman rank correlation, and iii. The share of origins with an absolute score change greater than 0.02. The delta (Δ) measures the average size of the score change. This means that higher values indicate greater average variation from the baseline, while lower values indicate more stable SA scores. The spearman correlation measures whether the rank ordering of origins remains stable. In this case, higher values indicate more stable rankings across origins, while lower values indicate that the origins SA scores change more strongly relative to the baseline. The share of origins with $|\Delta|>0.02$ indicates how widespread the changes are across the given case study origins set. This means that higher values indicate that changes are more widespread across origins, while lower values indicate that most origins remain close to the baseline SA score. Together, these indicators capture both the magnitude and the spatial extent of sensitivity to alternative specifications. A low mean delta, high spearman correlation, and a low share of origins with $|\Delta|> 0.02$ indicates strong origin-level robustness relative to the baseline specification.

The origin-level robustness analysis shows that the alternative specifications affect a large share of origins, but the magnitude change of the SA score is minimal (maximum of 1 percent point) and the ranking is generally preserved (see Table 18). Overall, the results show that the magnitudes of the SA scores are most affected using the *all opportunities counting - mean aggregation* specification specification and the rank stability is the weakest using the *all opportunities counting - minimum aggregation* specification.

The *threshold-based opportunity counting - minimum based aggregation* specification (Qthr-catmin) produces a mean absolute delta of 0.103 under PT45 and 0.102 under PT60. The mean Spearman correlation with the baseline is 0.936 under PT45 and 0.938 under PT60. This indicates that minimum aggregation generally lowers scores, but largely preserves the relative ordering of origins.

The *all opportunities counting - mean aggregation* specification (Qsep-catmean) produces the largest positive score changes, especially under PT60. The mean Spearman correlation remains high under PT45, but decreases under PT60. This suggests that counting all reachable opportunities generally raises scores, although only with 1 percent point, while preserving much of the origin-level order.

The *all opportunities counting - minimum aggregation* specification (Qsep-catmin) produces the smallest average change compared to the other specification. Additionally, this specification shows the lowest rank stability as well.

Table 18: Origin-level robustness summary.

PT45 profile - Specification	Mean |Δ|	Mean Spearman ρ	Mean share |Δ| > 0.02
Qthr_catmin	-0.103	0.936	0.903
Qsep_catmean	+0.116	0.964	0.918
Qsep_catmin	+0.008	0.931	0.622

PT60 profile - Specification	Mean |Δ|	Mean Spearman ρ	Mean share |Δ| > 0.02
Qthr_catmin	-0.102	0.938	0.907
Qsep_catmean	+0.137	0.909	0.922
Qsep_catmin	+0.081	0.875	0.690

Note: Mean |Δ| indicates the average change in origin-level scores compared with the baseline specification. Positive values indicate higher scores than the baseline, while negative values indicate lower scores. Spearman ρ indicates the stability of the origin ranking relative to the baseline. The share of |Δ| > 0.02 indicates the proportion of origins where the absolute score change exceeds 0.02. Close to null |Δ| values, higher spearman ρ values, and lower shares |Δ| indicate stronger origin-level robustness.

4.5.5 Cross-Case Modality Ranking Stability

The final robustness check examines whether the cross-case ranking of the nine cases remains stable under alternative methodological specifications. This is assessed separately for each PT profile and mode. For each mode, the nine cases are ranked according to their SA scores under the baseline specification and their alternative specification. The similarity between the two rankings is measured using a Spearman's rank correlation coefficient. Additionally, the maximum absolute rank change indicates the largest number of ranking positions of an individual case that shifts relative to the baseline rank.

The cross-case modality ranking results show that the relative ordering of the nine cases is stable across the specifications given that the spearman coefficient is in almost all cases above 0.9 and the maximum rank changes is maximum 3 out of the 9 cases (see Table 19). This indicated that for most specification-mode combinations, the cases remain stable relative to the baseline ranking.

The main deviations are found in the cycling related modes and in the specification that combines all opportunity counting with the minimum aggregation. In contrast, modes related to walking produce generally a more stable ranking. This indicates that the relative ordering is less affected for walking-related modes than for cycling-related modes.

Table 19: spearman rank stability and maximum rank change across case study areas.

PT45 profile - Specification	Walking	Walking + PT	Bike	Bike + PT
Qthr_catmin	0.950 (2)	0.983 (1)	0.967 (1)	0.929 (2)
Qsep_catmean	0.950 (2)	0.983 (1)	0.983 (1)	0.862 (3)
Qsep_catmin	0.900 (2)	0.950 (2)	0.883 (3)	**0.845 (3)**

PT60 profile - Specification	Walking	Walking + PT	Bike	Bike + PT
Qthr_catmin	0.950 (2)	0.983 (1)	0.967 (1)	0.946 (2)
Qsep_catmean	0.950 (2)	1.000 (0)	0.983 (1)	0.946 (2)
Qsep_catmin	0.900 (2)	0.917 (2)	**0.883 (3)**	0.962 (1)

Note: The values before the parentheses are spearman rank correlation coefficients (ρ) between the baseline specification and each alternative specification, calculated across the nine case study areas. Values in parentheses indicate the maximum absolute rank change. For example, (3) means that at least one case study area moves three rank positions compared with the baseline ranking. Higher ρ values and lower values in parentheses indicate greater rank stability. The bolded values are the least unstable specifications with the most rank changes across the case study areas.

4.6 Summary of the Results

General Results

The results show four main empirical patterns across the nine neighbourhoods with different degrees of urbanisation.

First, the overall SA scores broadly follow the urbanisation gradient. Urban cases generally score higher than medium-urban cases, and medium-urban cases generally score higher than rural cases. This pattern is visible across all four modes when the nine cases are averaged by urbanisation degree.

Second, cycling and cycling + PT produce substantially higher scores than walking and walking + PT in most cases. The largest modal gains are found in lower-scoring cases, where walking accessibility is limited but cycling or cycling + PT reaches more opportunities.

Third, neighbourhood averages conceal intra-neighbourhood variation. For example, Amsterdam has a relatively high cycling + PT score but also includes very low scoring origins as can be seen in the boxplot variation. In contrast, Zutphen and Heusden have more compact cycling + PT distributions meaning that the most origin specific SA scores are close to the average neighbourhood SA score.

Fourth, accessibility is uneven across opportunity categories. Employment and Leisure are often the strongest categories, while Education, Public services, Transport stops, and specific subcategories such as Hospital, Train, HBO/University, and Kindergarten are frequently weak in medium-urban and rural cases.

Sensitivity Analysis

The results show three main empirical patterns.

The sensitivity analysis produces three main findings. i. Extending the public transport profile from 45 to 60 minutes has only a limited effect on SA scores. The average increases for PT-based modes are small, and walking and cycling scores remain unchanged. This indicates that the baseline results are not strongly dependent on the selected PT profile duration. ii. The results are more sensitive to the treatment of opportunity quantities and category aggregation. Counting all reachable opportunities generally increases scores, while applying minimum category aggregation generally decreases scores. The combined method produces mixed results because it combines these two opposing effects. iii. Robustness differs between absolute scores and rankings. At origin level, most alternative specifications preserve much of the baseline rank ordering, but some cases show stronger sensitivity, particularly Dronten and Smallingerland under PT60. At cross-case level, the broad ordering of urban, medium-urban, and rural cases is mostly stable, although some mode-specific rankings change under the combined Qsep_catmin specification.

Overall, the sensitivity analysis supports the robustness of the main empirical patterns, especially the order of the urbanisation gradient and the relative advantage of cycling and cycling + PT. However, it also shows that the magnitude of SA scores is method-dependent, particularly with respect to opportunity-counting assumptions and category aggregation. Therefore, these findings should be considered when interpreting the results and when comparing individual neighbourhoods or origins.

5. INTERPRETATION OF RESULTS

In Chapter 4 the empirical results of the Sustainable Accessibility Index (SAI) were presented. In this chapter, those results are interpreted in relation to the theoretical framework and hypotheses.

The interpretation is structured around the four hypotheses introduced in Chapter 3. In Section 5.1 the urbanisation gradient hypothesis is discussed. Next, the role of multimodal accessibility in providing better sustainable accessibility is interpreted in Section 5.2. Afterwards, in Section 5.3 the neighbourhood internal spatial variation hypothesis is discussed. Then, the category and component variation are discussed in the same section. Next, in Section 5.4 the sensitivity analysis is interpreted. Finally, all the interpretations are summarized in Section 5.5.

5.1 Interpreting the Urbanisation Gradient

The results broadly support hypothesis 1, which states that more urbanised neighbourhoods are expected to have a higher sustainable accessibility score. Across the nine cases, the mean scores follow a consistent degree 1 > degree 3 > degree 5 ordering for all modes. More urbanised areas are generally more likely to combine higher opportunity density with shorter travel distances and stronger sustainable transport networks.

However, the case-level results also show that the urbanisation degrees do not fully explain the SAI outcomes. For example, Zutphen, a moderate urban neighbourhood, performs close to the Amsterdam neighbourhood for cycling and cycling combined with public transport (PT). These deviations are important as it shows that the SAI is not simply measuring urbanisation density. Instead, it captures the combined effect of local opportunity distribution, network reach, mode-specific travel times, and the position of origins within each neighbourhood.

Nevertheless, critical assessment is needed to ensure that the chosen neighbourhoods are fairly comparable. Choosing neighbourhoods with the same urbanisation degree can still yield very different results as neighbourhoods characteristics can still differ. For example, the neighbourhood chosen for Amsterdam with the same urbanisation level as Den Haag and Tilburg neighbourhoods had a lower SA score which could have been due to the fact that the chosen neighbourhood is still developing.

To conclude, the expected gradient is visible at the aggregate level, but local variations reveal differences that would be hidden by urbanisation classification alone. This supports the value of the SAI as a diagnostic measure. The SAI identifies not only whether more urbanised areas generally perform better, but also where specific neighbourhoods perform above or below what their urbanisation degree might suggest.

5.2 Interpreting Multimodal Accessibility

The results also partly support hypothesis 2, which states that multimodal chains are expected to increase sustainable accessibility compared with single active modes. Walking + PT scores are higher than walking scores in every case. This indicates that public transport extends

accessibility beyond the immediate pedestrian environment, particularly where walking access is limited.

The evidence for cycling + PT is more conditional. In the Amsterdam, Zutphen, Heusden, Hulst, and Schagen neighbourhoods, cycling + PT improved the Sustainable Accessibility (SA) score compared to cycling alone. This effect is particularly visible in the rural cases, where cycling + PT substantially raises the SA scores relative to cycling as a mode. In these cases, the combined mode appears to compensate partly for lower local opportunity density by extending the accessible opportunity field.

However, cycling alone performs equally well or better than cycling + PT in the selected neighbourhoods of Den Haag, Tilburg, Smallingerland, and Dronten. This suggests that the marginal benefit of adding public transport is limited where direct cycling already provides access to a sufficient set of destinations. The multimodal hypothesis should therefore be interpreted conditionally. Multimodal integration improves sustainable accessibility where public transport connects origins to otherwise unreachable opportunities, but it does not automatically outperform cycling in dense or compact contexts where many destinations are already reachable by bicycle.

5.3 Interpreting Neighbourhood Internal Spatial Variation

Intra-Neighbourhood Variation

The origin-level maps and distributions support hypothesis 3, which states that neighbourhood averages are expected to conceal internal spatial variation. The Amsterdam case is the clearest example. Although its neighbourhood-level cycling + PT score is relatively high, the origin-level results include very low-scoring origins, and the walking and walking + PT distributions show substantial differences between origins. This means that a single neighbourhood average can overstate accessibility for residents located in less favourable parts of the neighbourhood.

Other cases show different intra-neighbourhood variation. Zutphen and Heusden have relatively compact cycling + PT distributions, meaning that the combined mode produces more even accessibility across the origins. Hulst shows a different pattern. Here, walking accessibility is not only low on average, but low across almost all origins.

This finding is important for planning practice. A neighbourhood-level SA score is useful for comparison, but it should not be used alone. Origin-level maps and distributions could be useful to identify where accessibility deficits are located and whether low accessibility is spatially concentrated or widespread.

Category and Component Variation

The category and subcategory results show that sustainable accessibility is unevenly distributed across opportunity domains. Employment and Leisure are repeatedly among the strongest categories, while Education, Public services, and specific subcategories such as Hospital, Train, and HBO/University are more frequently weak.

The temporal acceptability and opportunity sufficiency components help explain why these differences occur. Low category scores are not always caused by the same mechanism. In some cases, low scores reflect poor temporal acceptability, meaning that destinations exist but require travel times that fall outside the defined acceptability range. In other cases, low scores reflect opportunity insufficiency, meaning that few or no relevant destinations are reachable

within the scoring structure. This distinction is important because the same overall SA score can represent different accessibility problems. A neighbourhood with long travel times to many destinations requires a different planning response from a neighbourhood with nearby destinations but insufficient variety or service coverage. The decomposition into T and Q strengthens the diagnostic value of the SAI as it shows whether low accessibility is mainly a transport problem, a land-use problem, or a combination of both.

The results also show that strong performance in one category can coexist with weak performance in another. For example, a neighbourhood may score well for Employment or Leisure while still performing poorly for Education, Public services, or Transport stops. This supports the argument that sustainable accessibility should not be interpreted only through a single composite score. The category and component results are needed to understand whether accessibility is broadly distributed across daily needs or concentrated in a limited number of opportunity domains.

5.4 Sensitivity, Robustness, and Methodological Interpretation

The sensitivity analysis supports hypothesis 4, which states that the SAI results are expected to be sensitive to methodological assumptions. The results show that the SAI is robust in its broad cross-case pattern, but sensitive in precise score levels and in some origin-level rankings. The urban cases generally remain stronger than the medium-urban and rural cases, and rural cases continue to show the weakest walking accessibility. However, the magnitude of scores and the explanation of why a neighbourhood performs well or poorly depend on the calculation method.

Out of the three sensitivity tests, the public transport threshold extension has a relatively small numerical effect. Extending PT maximum travel-time profile from 45 to 60 minutes produces only minor increases under the threshold-based baseline method. This suggests that, for these cases, the main differences between accessibility to opportunities in neighbourhoods are not driven by an additional 15 minutes public transport option. For this reason, the 45 minute profile is a good baseline choice as the extended travel time does not substantially change the results or improve the diagnostic value of the SAI. Additionally, the 45 minute profile can be considered a more suitable baseline since this methodology focuses on everyday sustainable accessibility and not unnecessarily larger regional travel time assumptions.

The second sensitivity test, the opportunity counting specification, has a stronger effect. When all reachable opportunities are counted through all opportunities counting specification, the SA scores increase in every case. This indicates that many neighbourhoods have additional opportunities that are technically reachable but not within the stricter acceptable travel times. The two methods for opportunity counting answer different questions. *Threshold-based opportunity counting - minimum based aggregation* specification measures opportunities that are reachable within behaviourally acceptable travel times, while the *all opportunities counting - mean based aggregation* specification measures the wider opportunity field reachable with the transport system. The difference between the two can therefore be interpreted as the gap between everyday acceptable access and broader regional access. For this methodology, the opportunity counting with a threshold is the better choice since the aim of this method is to measure what is reasonably accessible in daily life rather than to measure what is technically all accessible.

Additionally, the aggregation rule also has a clear methodological effect. The mean aggregation specification gives a balanced score across subcategories while the minimum aggregations gives more weight to the weakest subcategory and therefore lowers the SA scores when one part of a category destinations performs poorly. This distinction is theoretically important because it changes the meaning of the index as the minimum aggregation transforms the SAI from an average-performance measure into a sufficiency-oriented measure. In this methodology, the minimum aggregation is too strict to pass as the main SAI specification since the meaning changes the index into a weakest-link accessibility measure.

To conclude, the baseline specification is suitable as the main methodological version of the SAI as it aligns the best with the research aim of measuring sustainable accessibility to everyday opportunities within acceptable travel times. The baseline specification uses a maximum of 45 minutes PT-profile, threshold based opportunity counting and the mean aggregation of subcategories. This specification gives the most balanced picture of how well neighbourhoods perform across categories without making the SA score too strict or too broad. Additionally, the alternative specifications are valuable to use as additional tests or to answer specific planning questions where stricter aggregation is needed or specific users are considered that have specific daily opportunities that need to be reached.

5.5 Summary of Interpretation

The findings suggest that the SAI can be used as a diagnostic planning framework for neighbourhood-level sustainable accessibility planning but only when the methodological assumptions are made explicit. The SAI can compare neighbourhoods, but the main planning value lies in explaining why accessibility differs in terms of which modes provide access, which opportunity categories are less well served, where in the neighbourhood spatially access deficits occur, and whether accessibility is related to opportunity sufficiency or acceptable travel times.

Furthermore, the sensitivity analysis identified three parameters (travel-time thresholds, opportunity counting, aggregation logic) that require explicit agreement before the SAI is applied to ensure uniformity and comparability across implementation of this methodology. The baseline specification as advised in this research is most suitable for comparing everyday sustainable accessibility as it counts the opportunities within acceptable travel-time conditions and uses mean aggregation to summarize the category performance.

The alternative specification can be used for different planning goals. The extended public transport maximum travel time of 60 minutes is useful for testing whether longer public transport reach adds meaningful accessibility, although the found isolated effect is small in this study. The all opportunities count and mean aggregation specification is useful to identify broader regional opportunity potential. The specifications using minimum aggregation are more suitable for sufficiency-oriented planning questions as these calculation specifications can expose weak subcategories that mean aggregation can hide.

This flexibility is a key contribution of the method. The SAI can support different planning goals such as average accessibility, behaviourally acceptable access, wider regional opportunity reach, and weakest-link sufficiency. The results show that the broad empirical results are stable enough to be meaningful, while the sensitivity analysis shows that the policy message can change depending on the specification used.

6. DISCUSSION

Chapter 5 interpreted the empirical findings in relation to the hypotheses. This chapter discusses the wider academic and methodological implications of the study. First, the findings are positioned in relation to accessibility theory and sustainable mobility literature in Section 6.1. Then, the methodological contributions of the Sustainable Accessibility Index (SAI) are discussed in Section 6.2, before suggesting practical recommendations in Section 6.3. Finally, in Section 6.4 the limitations of this research and directions for future research are reflected on.

6.1 Positioning the Findings in Relation to Accessibility Theory

The goal of this research was to find a way to measure sustainable accessibility on a neighbourhood level to essential opportunities for different sustainable transport modes. The findings are broadly consistent with established accessibility theory. The results showed that the Sustainable Accessibility (SA) scores follow the general urbanisation gradient, with highly urbanised neighbourhoods achieving higher sustainable accessibility scores than medium-urban and rural neighbourhoods. This pattern reflects the central accessibility principle that access depends on the interaction between land-use patterns and transport systems. Areas with a higher concentration of opportunities and stronger sustainable transport networks are generally more likely to provide access within shorter travel times (Geurs & van Wee, 2004; Hansen, 1959).

At the same time, the results also show that urban density alone does not fully determine sustainable accessibility. Some medium-urban cases performed close to high urban cases for cycling and cycling + PT. This finding supports arguments in sustainable accessibility literature that suggests density and proximity are not sufficient planning goals to improve sustainable accessibility. Instead, it must be measured through the combined effects of opportunity distribution, mode-specific network reach, temporal acceptability, and origin location. For instance, Gil Solá et al. (2018) show in their research that planners regard proximity to everyday activities as central to sustainable accessibility but also that implementation raises questions as to which opportunities matter and whose access is prioritised. Similarly, Vilhelmson & Elldér (2021) suggest that proximity to essential opportunities is strongly shaped by changing the spatial distribution of amenities.

The second finding is that accessibility gaps are often category-specific. Employment and leisure are generally the strongest categories, in particular for cycling and in the multimodal combination with public transport. On the other hand, education, public services, train access, and kindergartens are often among the weakest categories, especially in rural cases and by foot. Weaker cases are not only less accessible overall, but perform in certain categories particularly worse. The implication is that planning for sustainable accessibility requires attention to specific types of opportunities that are missing or poorly connected, rather than only increasing general density with opportunities that are not needed. This aligns with the argument of Elldér et al. (2022) that built environment indicators such as density, diversity and design, could overlook the role that specific neighbourhood amenities have in shaping everyday travel.

The third finding is related to the role of cycling and multimodality. The results confirm that combining active modes with public transport can extend the accessibility. This supports the

theoretical hypothesis that the multimodal combination of using public transport and active modes can increase the range of accessibility (De Stasio et al., 2011). However, the results also indicate that multimodal combinations do not always outperform single modes, especially for several highly urban cases. In those cases cycling alone provides similar or even better accessibility scores than cycling combined with public transport. This finding supports the results found by Knap et al. (2023) which showed that cycling can provide strong access to daily services in the urban area of Utrecht.

The fourth finding is that neighbourhood-level averages conceal the internal spatial variation of SA scores. Origins in the same neighbourhood can still differ in their proximity to services, public transport stops, cycling routes, which are not visible in the average neighbourhood-level scores. This means that neighbourhoods can appear reasonably accessible on average while still containing some origins in the neighbourhood with much weaker sustainable accessibility. Research by Qu et al. (2026) found a similar result where testing public transport access using high-resolution accessibility proxies gave more nuanced accessibility interpretation compared to using zonal accessibility proxies.

6.2 Methodological Contribution

The main methodological contribution of this research is the development of a transparent, multimodal, and decomposable index for assessing sustainable accessibility at the neighbourhood level. The SAI produces not only an overall sustainable accessibility score, but also allows for seeing the composition of the score across transport modes, opportunity categories, origins, and the two components of temporal acceptability and opportunity sufficiency.

The first contribution of this methodology is the ability to disaggregate the neighbourhood-level SA score into temporal acceptability score and opportunity sufficiency score, thereby improving the diagnostic value of the index as it helps identify whether the main problem is related to the transport system, land-use system, or a combination of both.

The second contribution relates to the specific focus on sustainable modes and multimodality. Measuring sustainable accessibility for walking, cycling, and the multimodal combination of those two modes with public transport has not yet been done to this extent and makes this index suitable for multimodal focused sustainable accessibility planning.

Another contribution is that the SAI makes internal variation within neighbourhoods visible. The methodological value lies in combining average scores with the origin-level maps and distribution scores.

From a practical perspective, the SAI should be used as a layered diagnostic framework rather than a single ranking tool. The baseline specification can provide an initial comparison between neighbourhoods but the interpretation requires looking at the underlying mechanisms (such as modes scores, origin-level variation, opportunity categories, time acceptability score, opportunity sufficiency scores) that produced the neighbourhood-level SA score. In this way, the SAI can support neighbourhood-level sustainable accessibility planning by showing what kind of accessibility problem is present.

6.3 Recommendations

The methodological discussion in Section 6.2 shows that the SAI is most useful when applied as a diagnostic framework rather than a single ranking tool. The following recommendations are formed to formulate a practical guidance to use the SAI for neighbourhood-level sustainable accessibility planning.

1. Use the Baseline Specification as the Standard Comparison Score
The SAI should be calculated with the recommended baseline specification of maximum travel time of 45 minutes for public transport, threshold-based opportunity counting, and mean aggregation across subcategories. This specification is most appropriate for this research aim as it focuses on everyday sustainable accessibility within acceptable travel-time conditions whilst still providing a balanced score across opportunity categories. Using one fixed baseline is important because the sensitivity analysis showed that using the alternative specifications changes the interpretation of the SA scores. Without intentionally stating the baseline specification, the comparability of the SA scores between neighbourhoods becomes less reliable.

2. Interpret the Baseline Score Using the Underlying Mechanisms
Planners should examine the underlying mechanisms, such as mode scores, category and subcategory scores, origin-level variation, and the time acceptability score and quantity sufficiency score, when interpreting the neighbourhood-level SA score. This is necessary as the findings in this research showed that similar overall neighbourhood scores can represent different accessibility problems. A low score can result from long travel times, or too limited reachable opportunities, or weak access to specific opportunities or spatially concentrated deficits within the neighbourhood due to natural barriers such as a train railway. Therefore, the practical value of the SAI lies in being able to explain the potential causes for poor sustainable accessibility rather than only identifying that accessibility is low.

3. Use Alternative Specifications to Test for Specific Planning Needs
The sensitivity variants specifications can be used as additional ways to calculate SA scores or with specific planning goals in mind. The all opportunities specification is useful to identify wider regional opportunity and total accessibility potential. The minimum aggregation specification is useful to expose the weak-link subcategories in the neighbourhood.

4. Match Interventions to the Diagnosed Accessibility Problem
Planners should match the intervention based on the type of accessibility problem identified with the SAI. If low accessibility is mainly caused by poor temporal acceptability, then planning interventions could focus on reducing travel times through improving cycling routes, walking connections, public transport services, or multimodal integration. If low accessibility is caused by poor opportunity sufficiency for a certain subcategory, planning interventions could target the spatial distribution of those opportunities rather than general accessibility improvement. This makes the SAI useful to select more precise planning interventions tailored to the context.

Together, these recommendations suggest that the SAI should be used in a layered way. The baseline score provides a common reference point, while the disaggregated layers and sensitivity specification variants can help provide the explanation behind the neighbourhood-level SA score. This allows planners to use the SAI to compare neighbourhoods without reducing sustainable accessibility to a single number that is unclear to interpret.

6.4 Limitations and Future Research Directions

Several limitations should be considered when interpreting the results of this research.

1. Data Quality and Completeness
This research relies on open spatial and transport datasets, including OpenStreetMap and public transport timetable data. These sources make the analysis reproducible, but they may contain omissions or classification discrepancies. For example, some amenities may be missing from OpenStreetMap, incorrectly tagged, or represented differently across areas. This is especially relevant in smaller and rural areas, where a single missing amenity can influence the subcategory score substantially. Future research could improve this by validating OSM destinations against municipal data bases or use more advanced OSM filtering techniques.

Furthermore, public transport data also represent scheduled services rather than real-time performance. The analysis does not account for delays, cancellations, crowding, reliability, or service disruption. As a result, the SAI measures scheduled potential accessibility rather than perceived public transport reliability. This limitation is important because a route that appears accessible in the timetable may be less useful if service quality is poor. Future alterations of the SAI could incorporate reliability indicators, such as service frequency, real-time performance data as this would make the public transport component more sensitive to the quality of the service experienced by users.

2. Opportunity Quality and Capacity
The SAI treats destination points primarily as reachable opportunities. It does not fully account for differences in quality, capacity, affordability, opening hours, or suitability. For example, a primary school and a MBO may both be represented as education opportunities, even though their functions differ. Similarly, a supermarket's presence does not necessarily indicate affordability, product range, or capacity. This simplification is necessary due to time, data, and ability constraints. The SAI measures potential access to opportunities, not whether they fully meet all qualitative requirements. Future research could incorporate destination weights based on service size, capacity, quality, or opening hours.

3. Temporal and Behavioural Complexity is Simplified
The SAI is based on selected routing conditions and behavioural parameters. It does not fully capture variation by time of day, day of week, seasonal conditions, or trip chaining. Accessibility may be lower in evenings, weekends, or periods with reduced public transport service. Similarly, the index does not model complex daily schedules, combined trips, or individual activity patterns.

The behavioural parameters used for travel-time acceptability and opportunity sufficiency improve realism, but they remain approximations. They represent generalised preferences rather than the preferences of all residents in each neighbourhood. Therefore, the SAI should be interpreted as a structured estimate of potential accessibility rather than a direct measure of experienced accessibility. Future research could test multiple time-windows or compare SAI results with observed travel behaviour.

Additional Future Research Directions
Five broader directions for future research follow from these limitations.

First, to consider a more broad definition of sustainability, future research could look into ways to include ecological, economical, and social sustainability perspectives. For example, future research could develop group-specific versions of the index. This could include accessibility profiles for older adults, children, disabled residents, low-income households, or people without access to a bicycle. Such work would strengthen the equity relevance of the SAI and show whether general sustainable accessibility scores hide disadvantages for specific groups.

Second, future studies could incorporate quality, reliability, and capacity indicators. This would make the index more sensitive to differences between services that are currently treated as equivalent. For example, healthcare facilities could be weighted by service type, public transport could be adjusted for reliability, and walking or cycling networks could include safety or comfort indicators.

Third, future research could compare objective SAI scores with perceived accessibility. Perceived accessibility research shows that measured access and experienced access do not always align (Lättman et al., 2018). Surveying residents in the case study areas would help assess whether the SAI reflects how people experience access in daily life.

Fourth, the SAI could be used for scenario analysis. Future work could estimate how accessibility changes after a new public transport service, cycling connection, housing development, school, supermarket, or healthcare facility is introduced. This would make the index useful not only for diagnosis, but also for project appraisal and monitoring.

Finally, there is potential to translate the SAI into a practical planning tool. A GIS or Python-based tool could allow municipalities to calculate SA scores, map accessibility gaps, and compare scenarios. However, such a tool should preserve the transparency of the method and allow users to inspect the assumptions behind the final score.

7. CONCLUSION

In this research, it is examined how sustainable accessibility to daily opportunities can be measured at the neighbourhood level. This research is a response to a practical and methodological need for a transparent sustainable accessibility measure that is theoretically sound yet easy to interpret.

To address this need, a Sustainable Accessibility Index (SAI) was developed and tested. This index combines travel-time acceptability, opportunity sufficiency, sustainable transport modes, and multiple categories of daily opportunities into a structured and decomposable measure. The central research question in this research is:

How can sustainable accessibility to daily opportunities be measured at the neighbourhood level using multimodal sustainable transport?

This research shows that sustainable accessibility can be measured through a transparent, and decomposable composite index that combines sustainable transport modes, essential opportunities, travel-time acceptability, and opportunity sufficiency. This way, the SAI measures whether essential opportunities are reachable within acceptable travel times rather than all technical reachable opportunities.

The main contribution of the SAI lies in its capacity to make sustainable accessibility differences explainable. The index produces an overall sustainable accessibility score for a neighbourhood, but the main value is that those scores can be decomposed into more detailed explanations. This makes it possible to identify whether the sustainable accessibility differences are related to a specific mode reach, local opportunity provision, or internal spatial variation. Additionally, the sensitivity analysis shows that the alternative specifications can be used to answer different planning questions.

The main practical recommendation is that the SAI is most valuable as a layered diagnostic planning framework, rather than a singular ranking tool, as different specifications reveal distinct dimensions of sustainable accessibility. This methodological flexibility could enable planners to move beyond the neighbourhood-level comparisons by identifying whether interventions should target local service provision, active mobility infrastructure, public transport integration, or specific weak opportunity categories.

To conclude, this research demonstrates that sustainable accessibility can be made measurable without reducing the measure into a single non-interpretable score. Sustainable accessibility requires more than improving the transport system or increasing land-use density. It requires an understanding in what places are reachable and under which conditions. The SAI contributes to this goal by making sustainable accessibility visible, measurable, and open to critical interpretation by offering multiple calculation methods that can be tailored to specific planning goals.

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APPENDIX

A. Intermediate outputs from SAI calculation procedure

This appendix presents the intermediate outputs generated during the SAI calculation procedure for the Amsterdam case study using python scripts and input data as described in the main report. The materials follow the methodological workflow. The aim of this appendix is to provide transparency on how the SAI is operationalised and illustrate the transformation from the raw data to the aggregated sustainable accessibility score. The python scripts are available upon request.

I. Spatial input

The first step in the preprocessing phase of the SAI calculation is to isolate the neighbourhood boundary with a 40km buffer zone. The input data for this step is the neighbourhood boundary which can be extracted from CBS Wijken en Buurten dataset (CBS, 2025) using Python code. Then, the origin points within the selected neighbourhood are extracted from CBS CBS 500 m × 500 m population grid (CBS, n.d.-a) by calculating the centroids of the population grid that falls within the neighbourhood boundary.

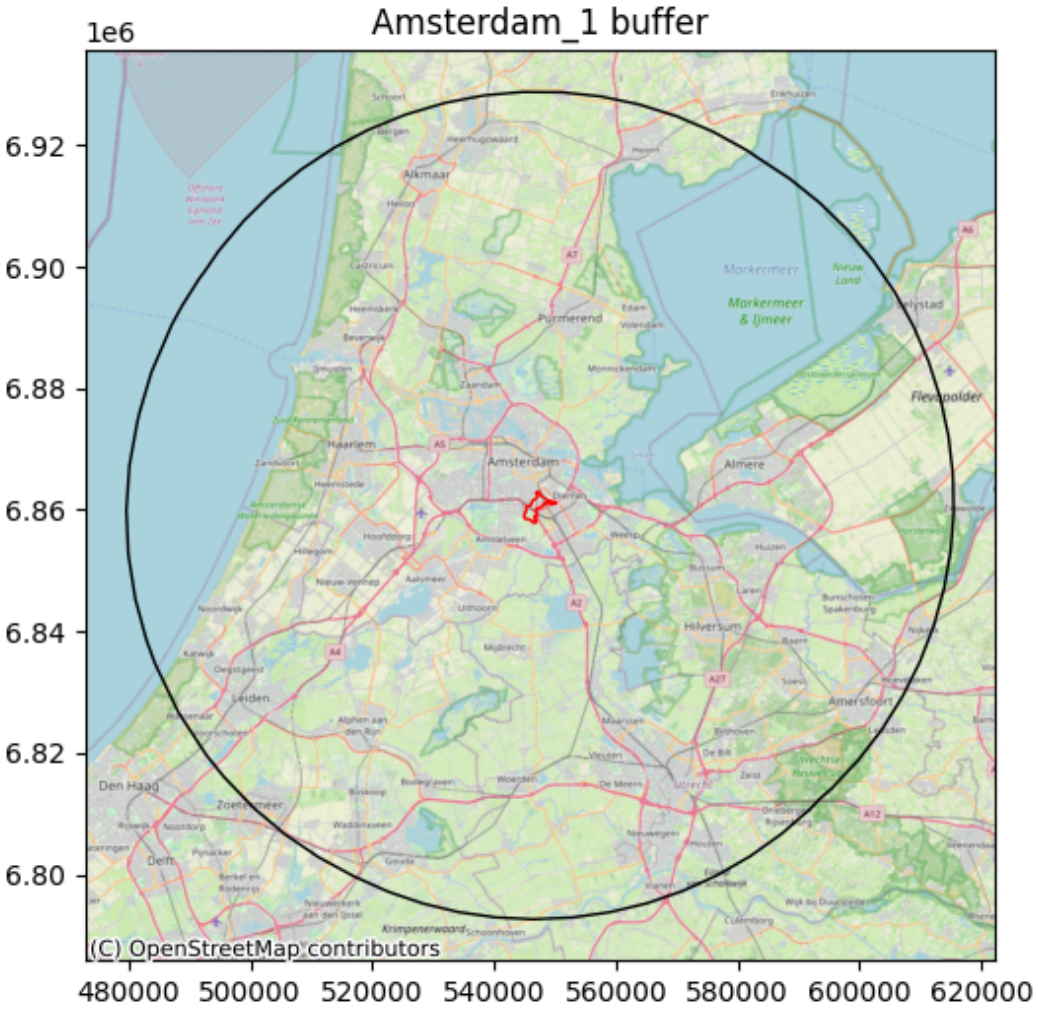


Figure A1: Buffer-zone map Amsterdam.

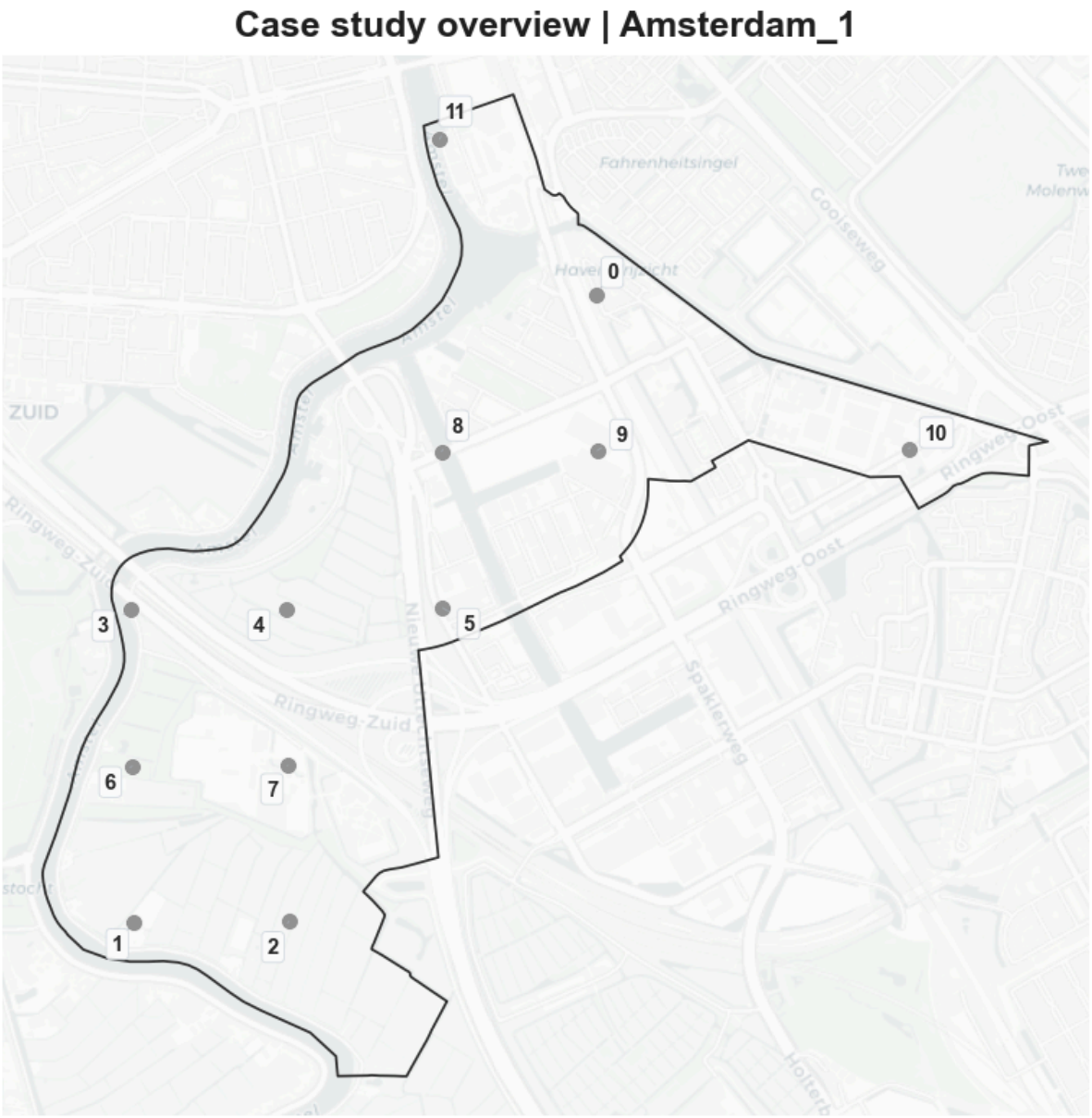


Figure A2: Neighbourhood map of Amsterdam case study area with preprocessed origin points.

II. Opportunity map

The second step in the preprocessing phase of the SAI calculation can be visualised in an opportunity map where all POI's within the neighbourhood with the 40km boundary are categorised and spatially visualized. For this step, POI's should be extracted from OSM, then categorized into the desired subcategories/categories, and finally visualized in an opportunity map.

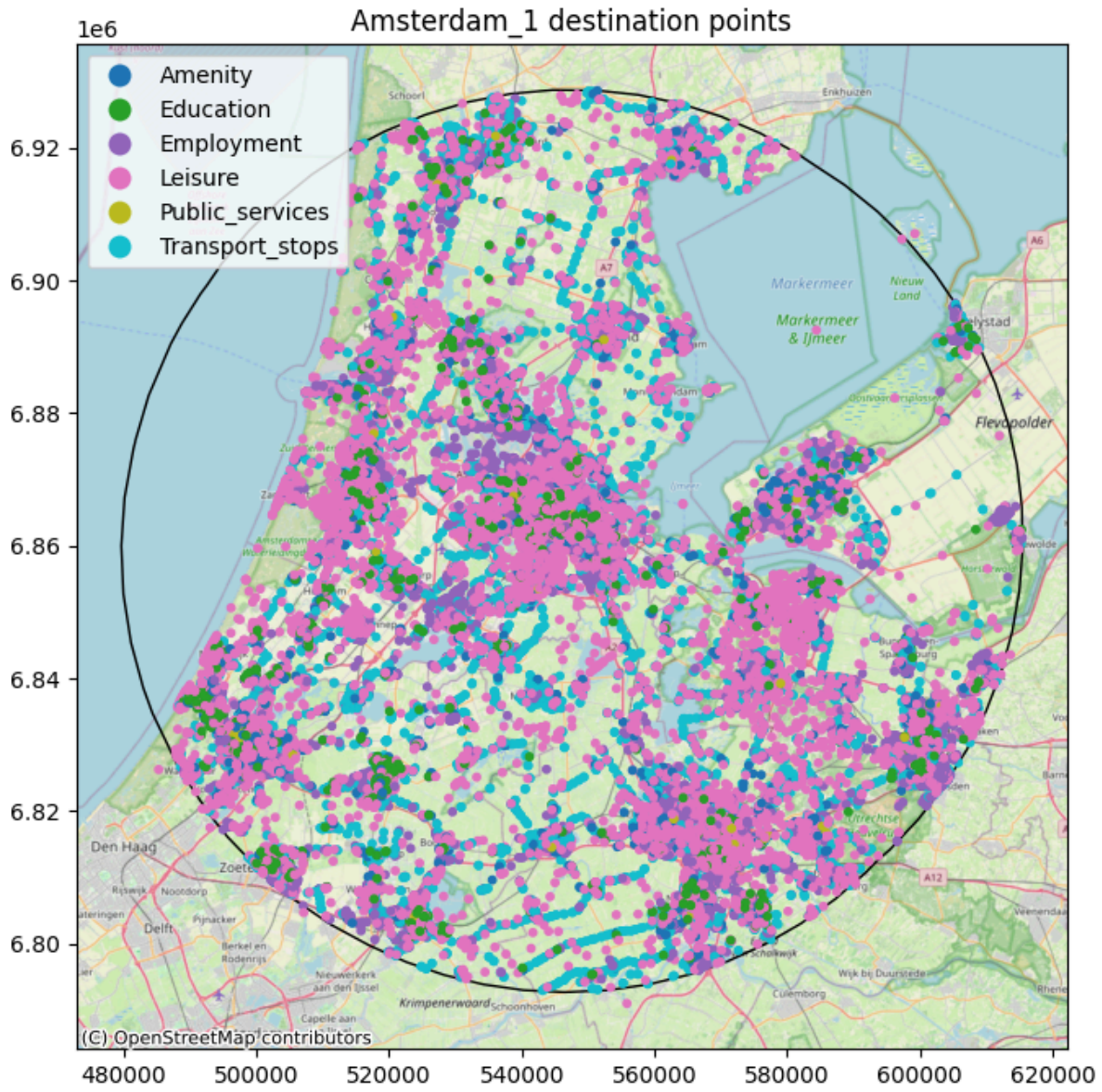
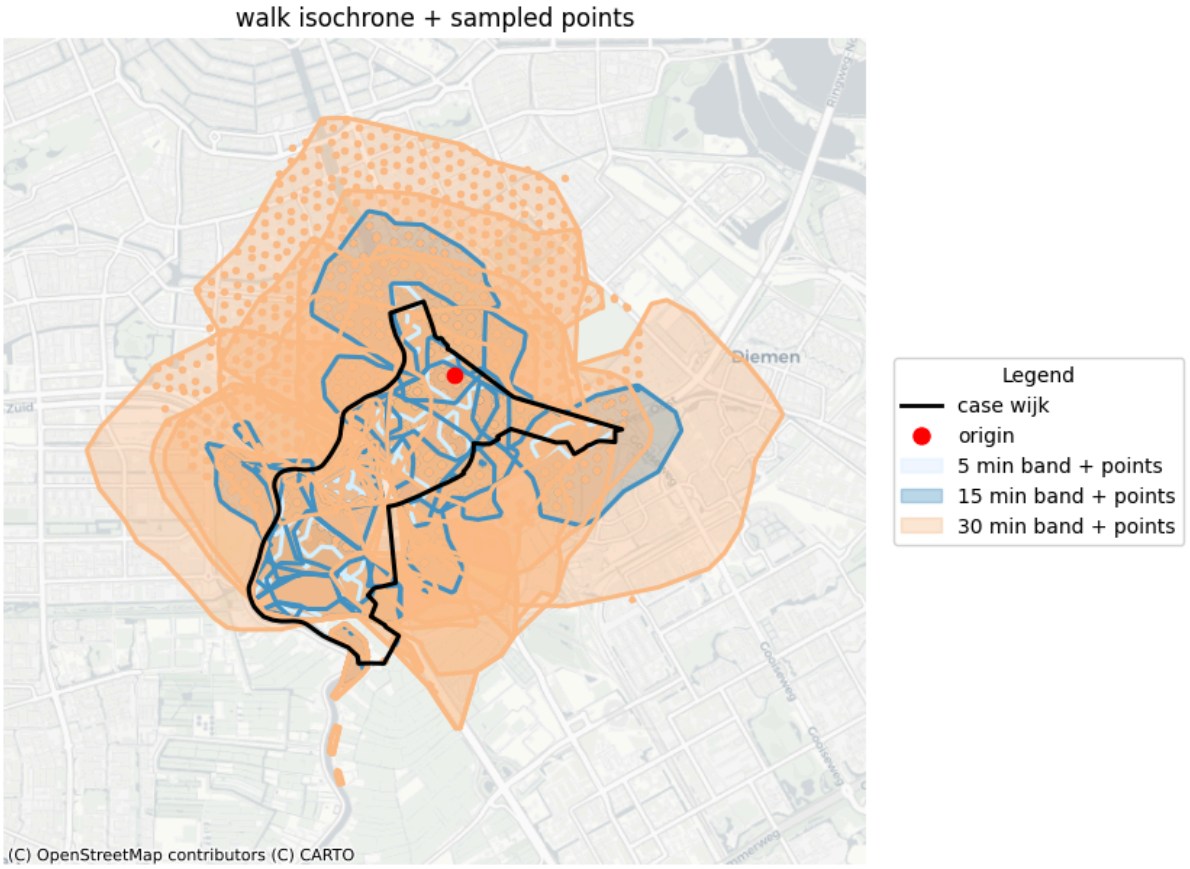


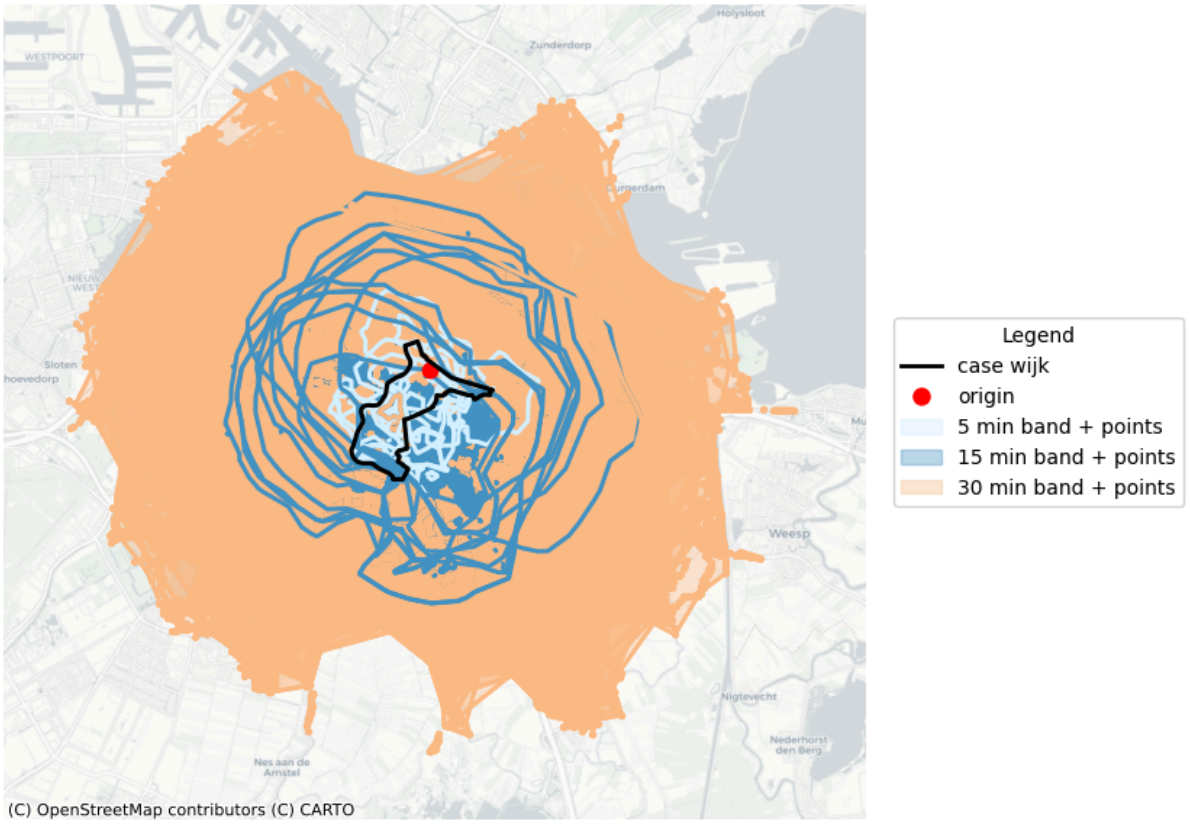
Figure A3: Opportunities map Amsterdam.

III. Accessibility modelling

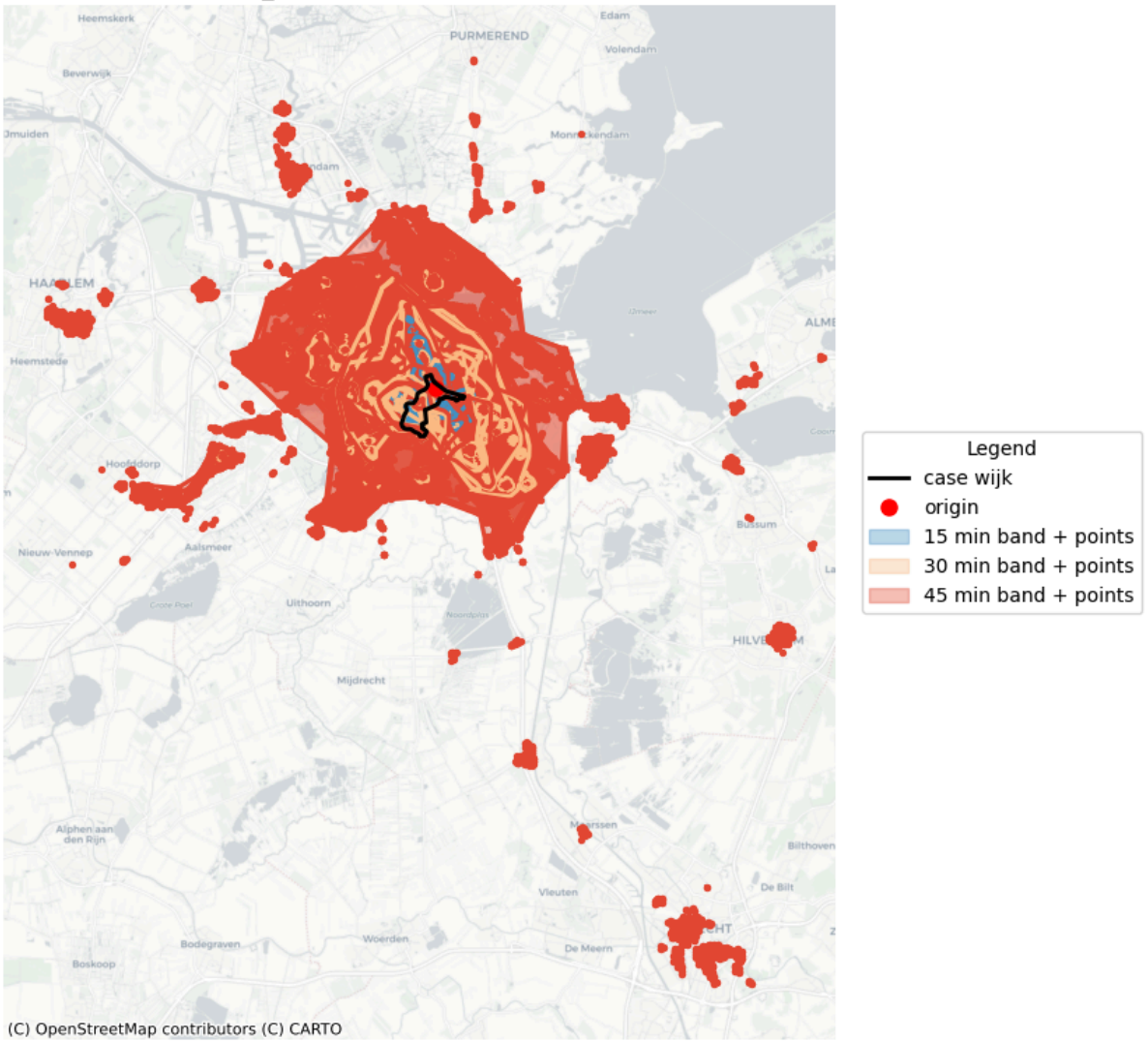
Then, the last step before calculating the SA scores is to calculate the general isochrones/travel distances per mode for the selected neighbourhood. These isochrones are based on the parameters as described in the main report. The visualization of the isochrones is based on a random sample point within the neighbourhood for which the isochrones are drawn on a basemap.



bike isochrone + sampled points



walk_pt isochrone + sampled points



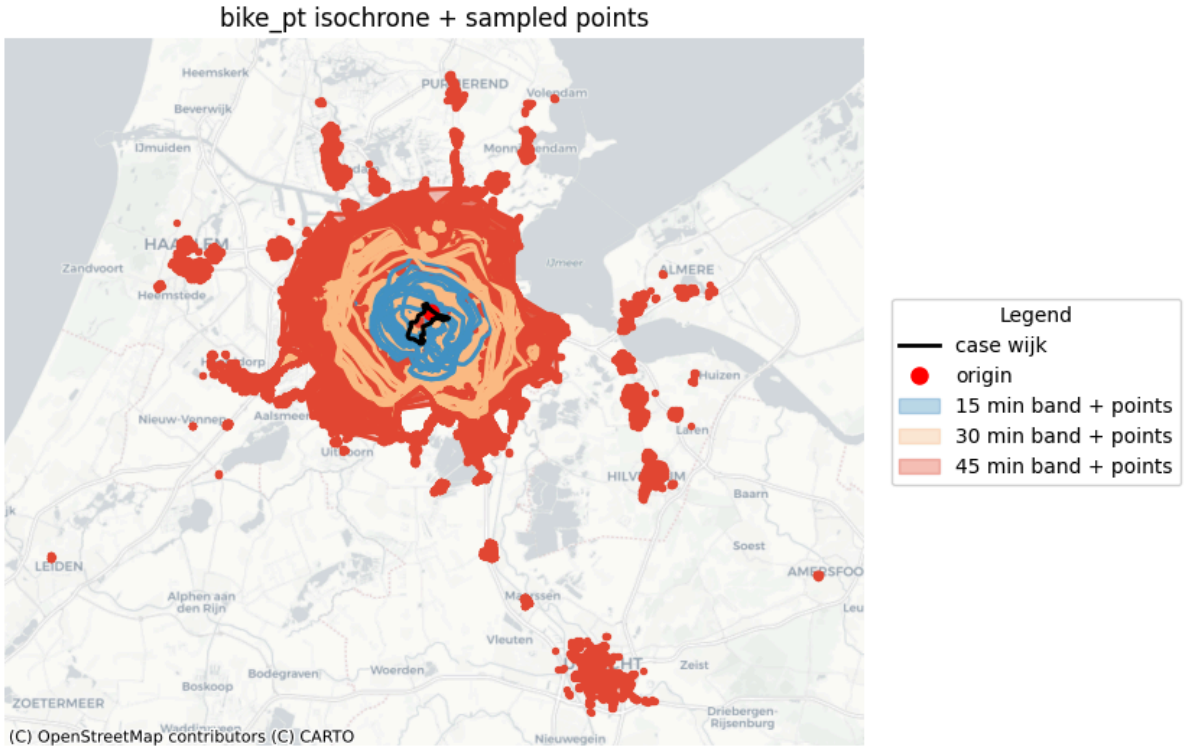


Figure A4: Isochrone maps by mode Amsterdam.

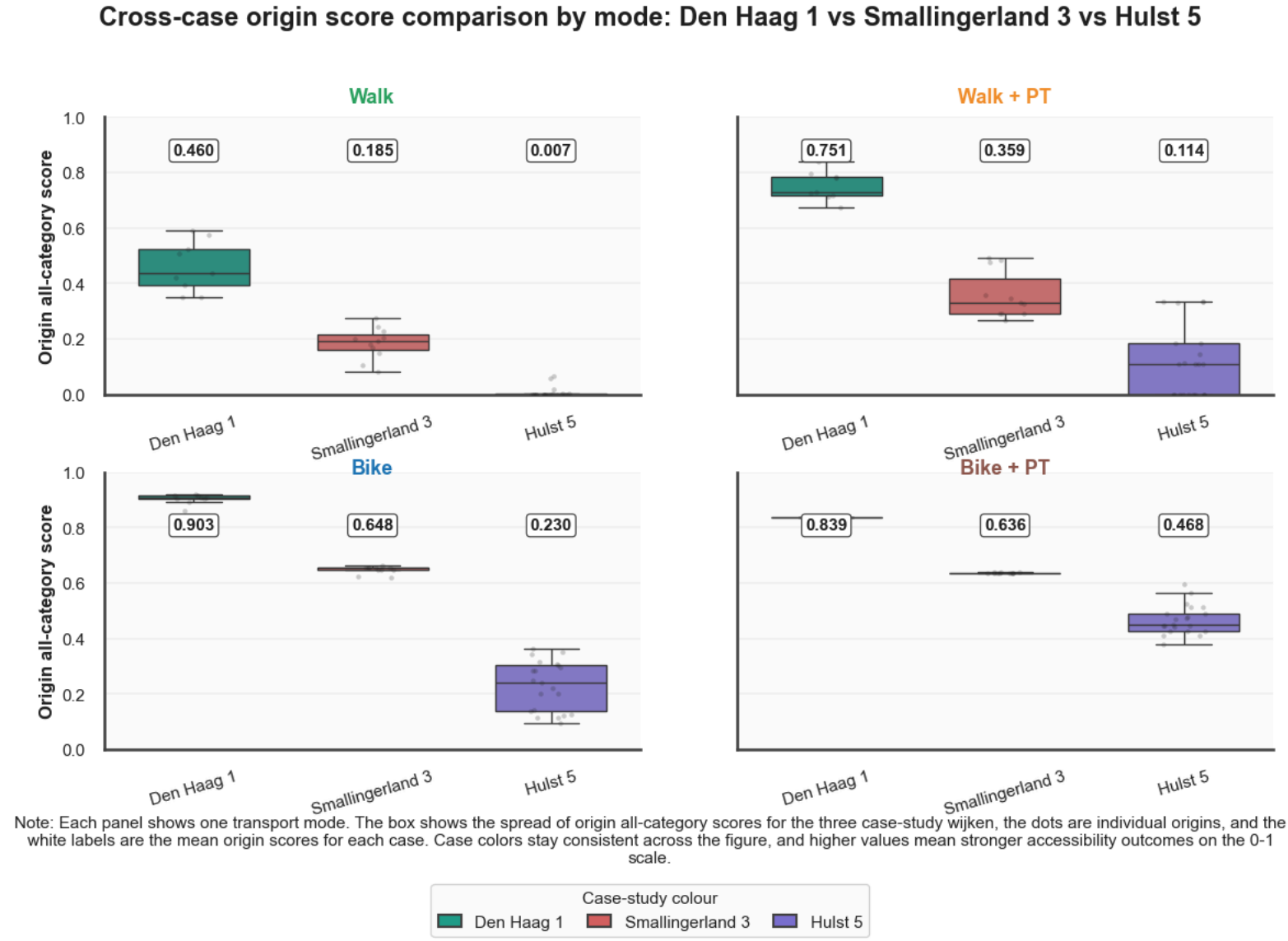
B. Comparative SAI results across all case studies

This appendix presents the complete set of SAI results across all case study areas.

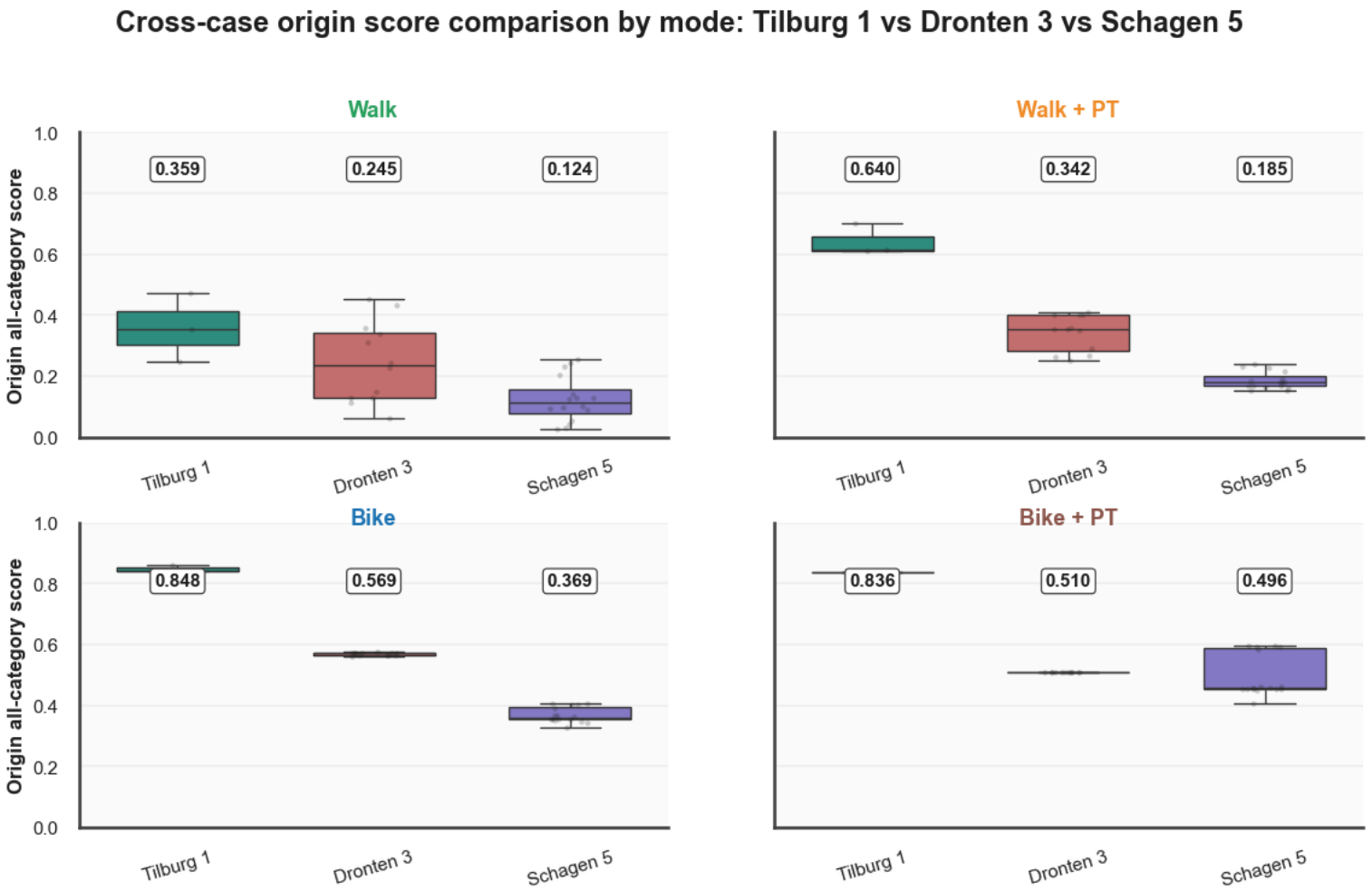
I. Origin-level variation within neighbourhoods

This visualization shows the origin-level variation within neighbourhoods. The boxplots show the variation of neighbourhood-level SA scores per mode.

The first cross-case study set (Amsterdam-Zuthpen-Heusden) can be found in Chapter 4.2.



Note: Each panel shows one transport mode. The box shows the spread of origin all-category scores for the three case-study wijken, the dots are individual origins, and the white labels are the mean origin scores for each case. Case colors stay consistent across the figure, and higher values mean stronger accessibility outcomes on the 0-1 scale.

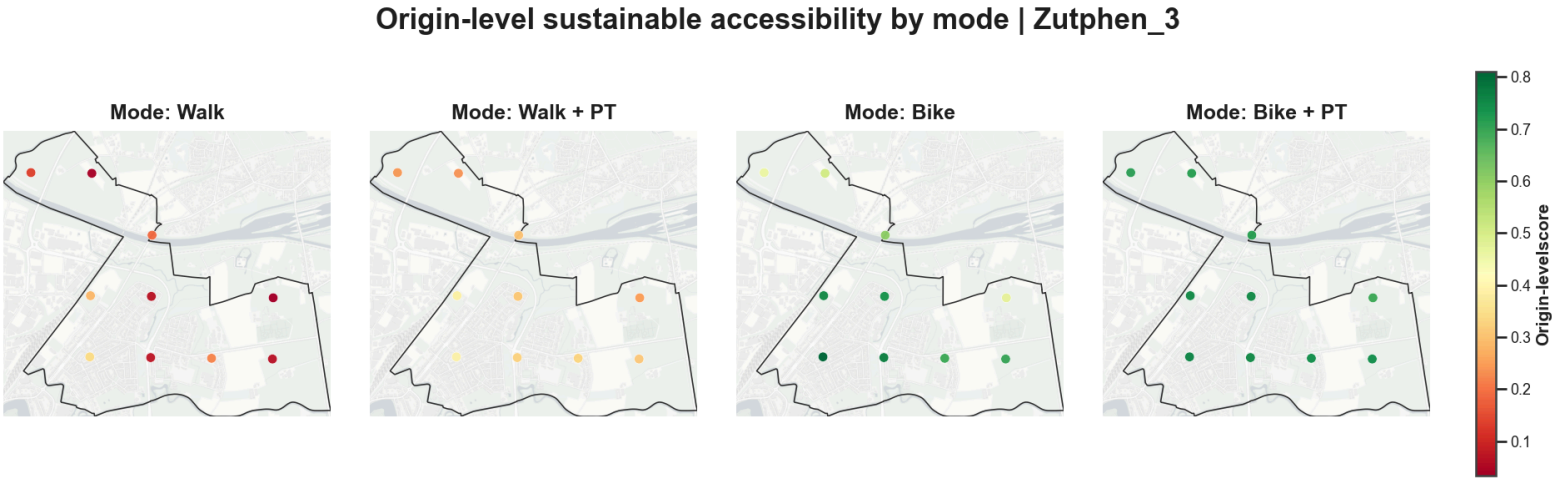


Note: Each panel shows one transport mode. The box shows the spread of origin all-category scores for the three case-study wijken, the dots are individual origins, and the white labels are the mean origin scores for each case. Case colors stay consistent across the figure, and higher values mean stronger accessibility outcomes on the 0-1 scale.

II. Origin-level score map per case study area

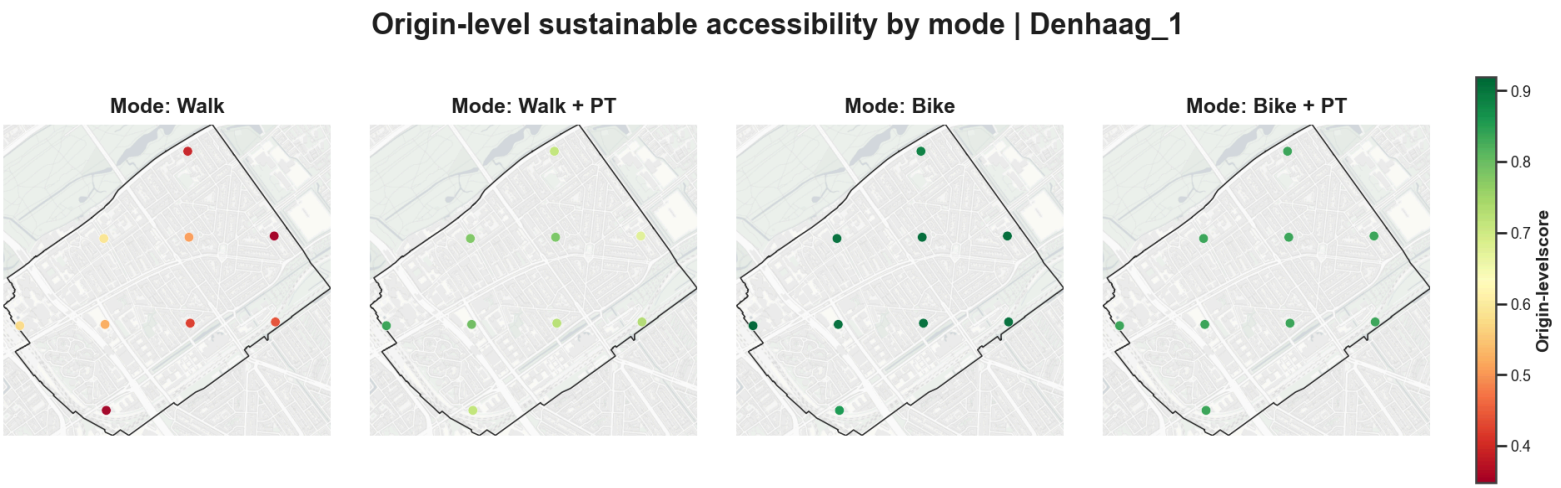
These maps show the origin-level SA score spatially distributed in the selected neighbourhoods.

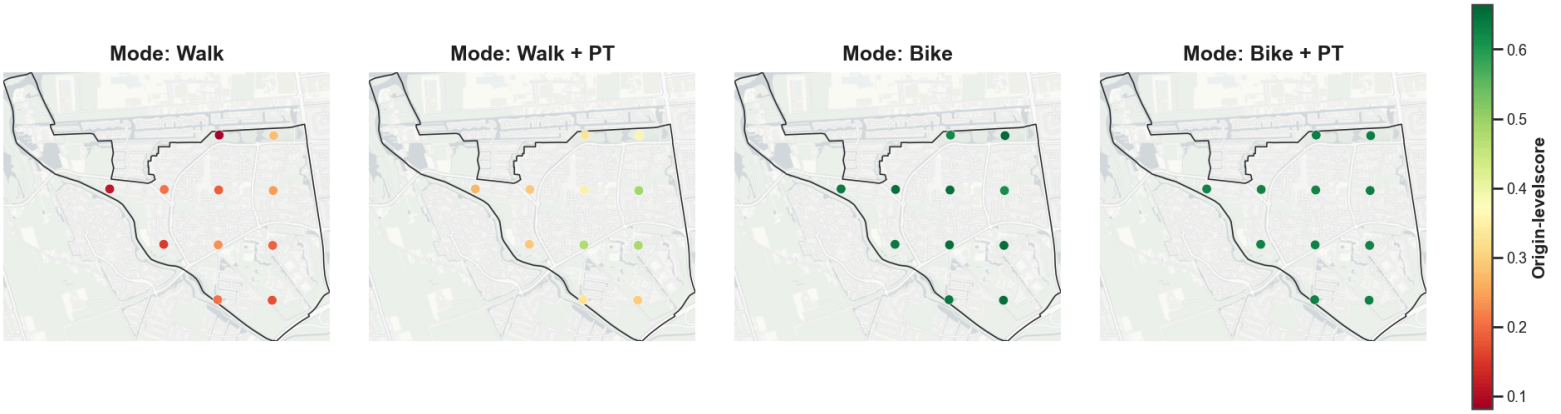
a. Cross case study 1: Amsterdam (Chapter 4) - Zutphen - Heusden

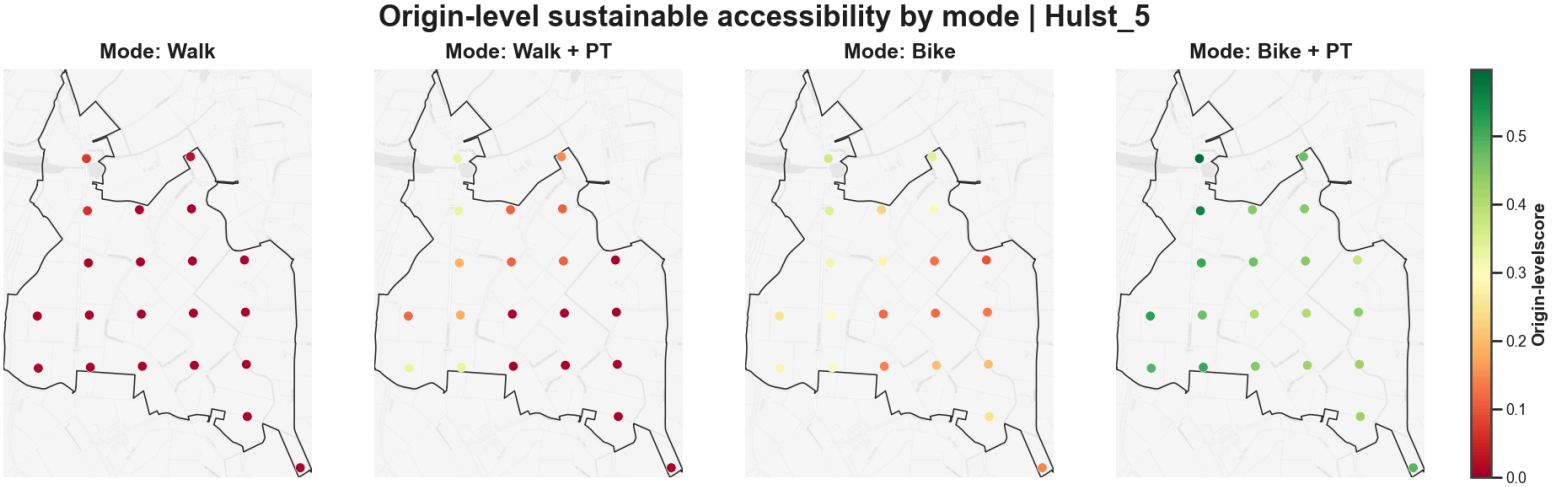




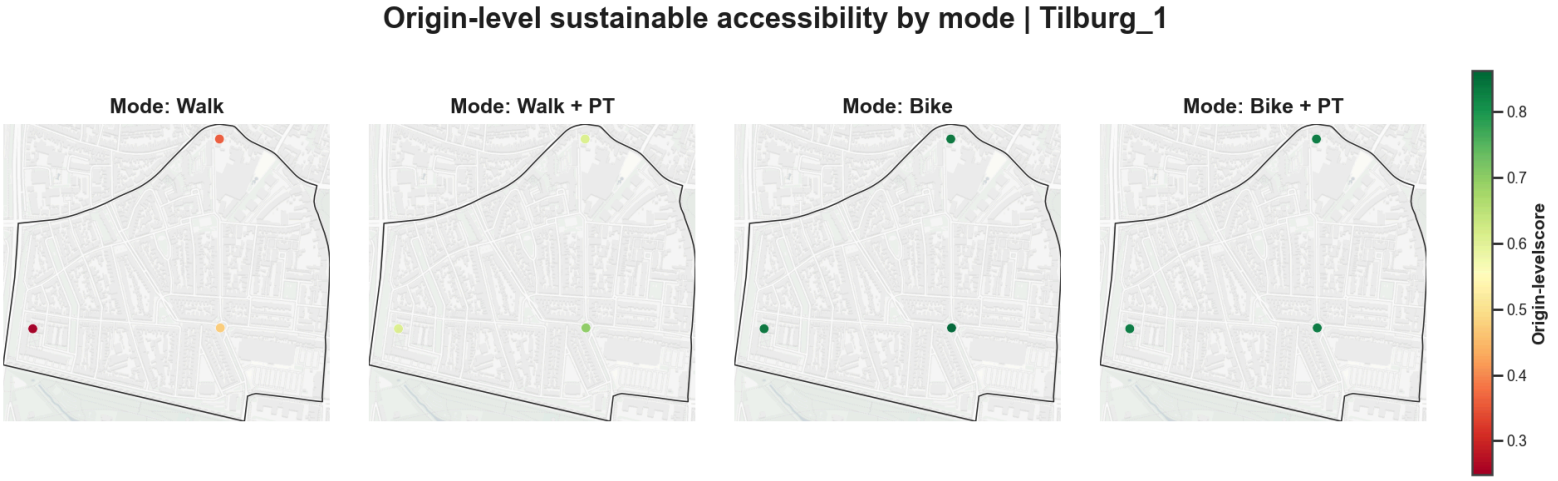
b. Cross case study 2: Den Haag- Smallingerland - Hulst







c. Cross case study 3: Tilburg - Dronten - Schagen



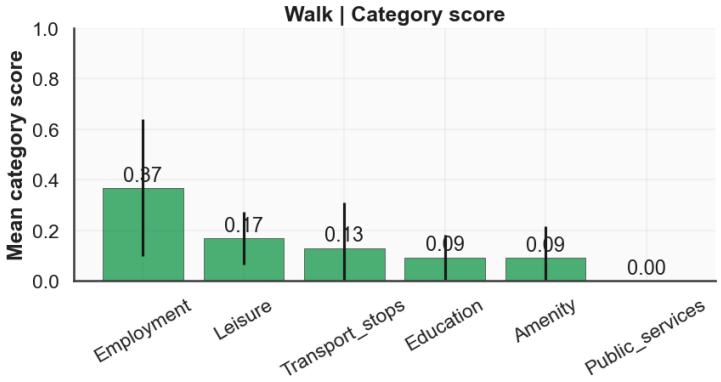


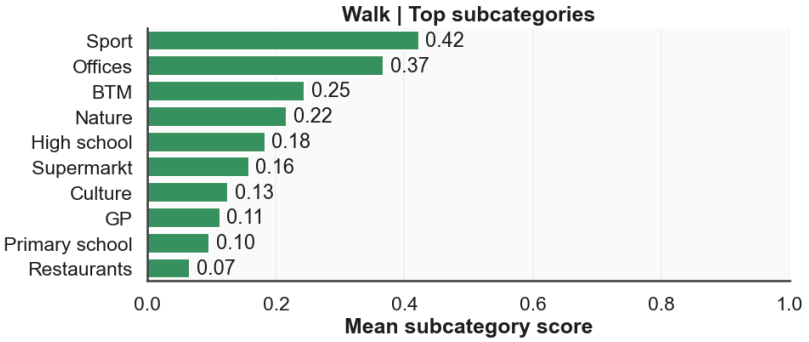


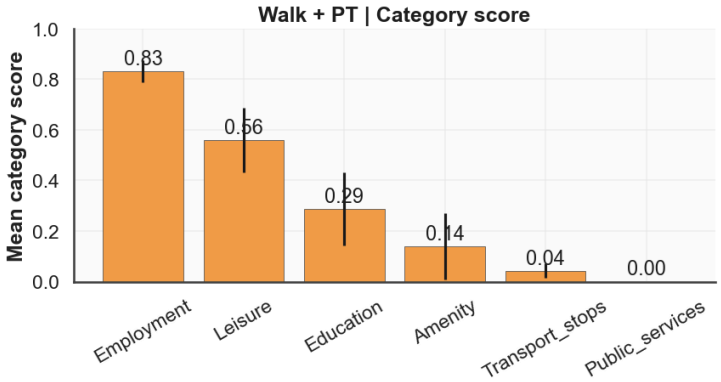
III. Top category and subcategory decomposition for each study area

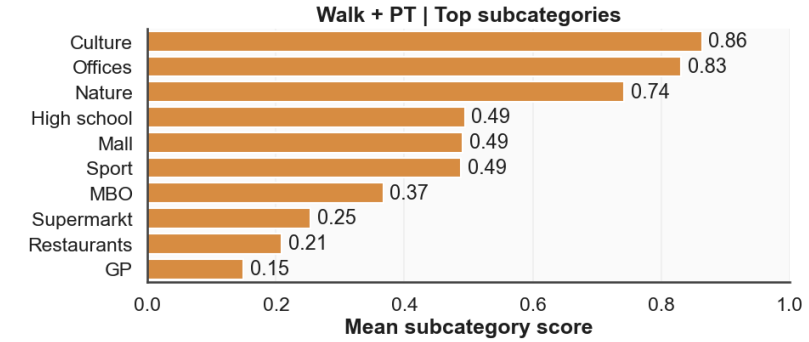
These bar charts show the eight best performing categories (on the leftside) and subcategories (on the rightside) per mode for each neighbourhood.

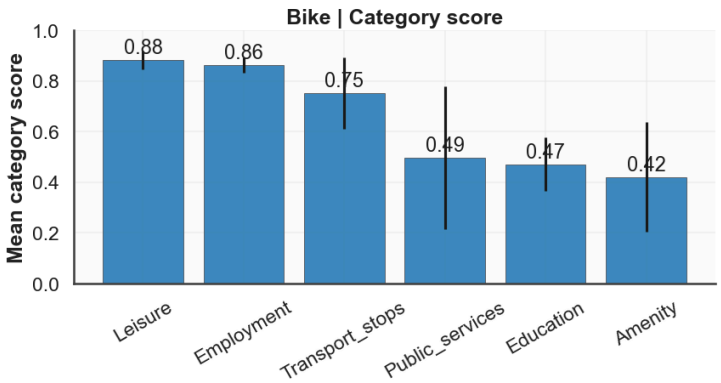
Zutphen_3: category and subcategory decomposition by mode

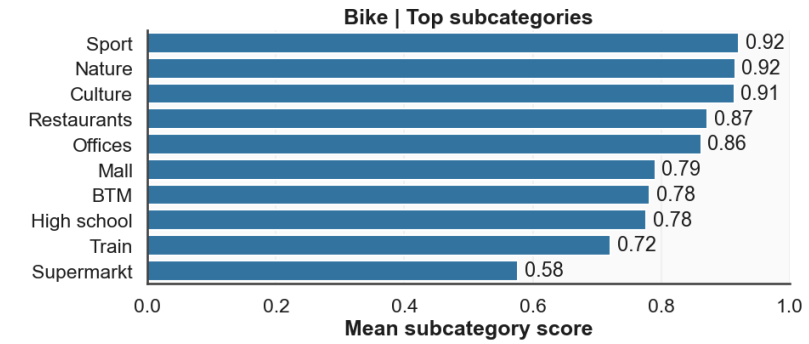


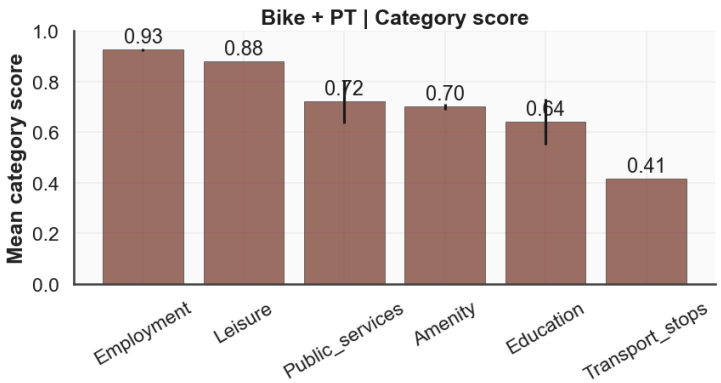


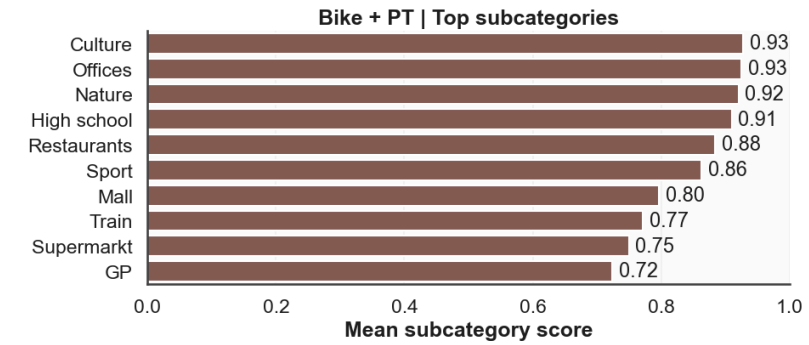




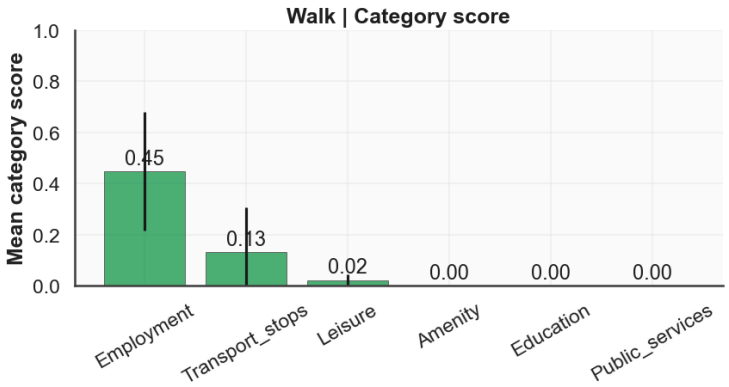


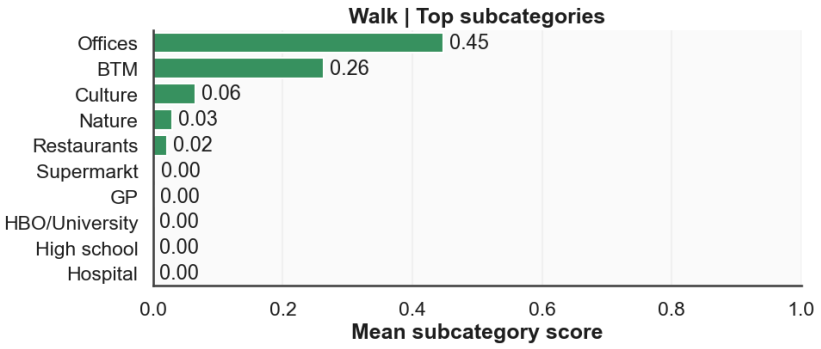


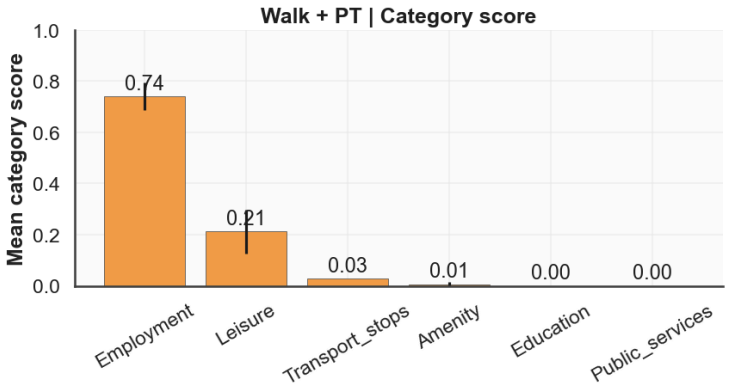


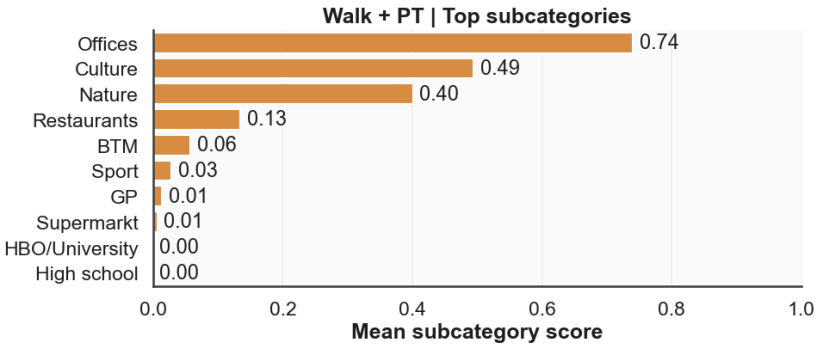


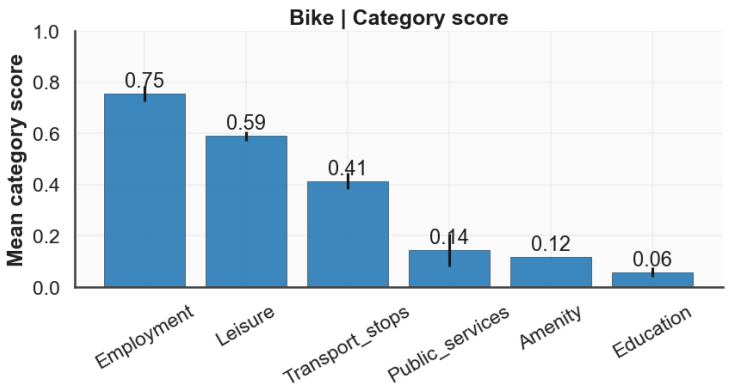
Heusden_5: category and subcategory decomposition by mode

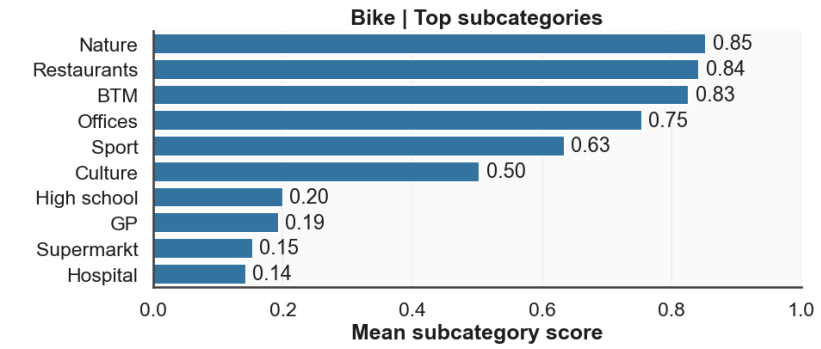


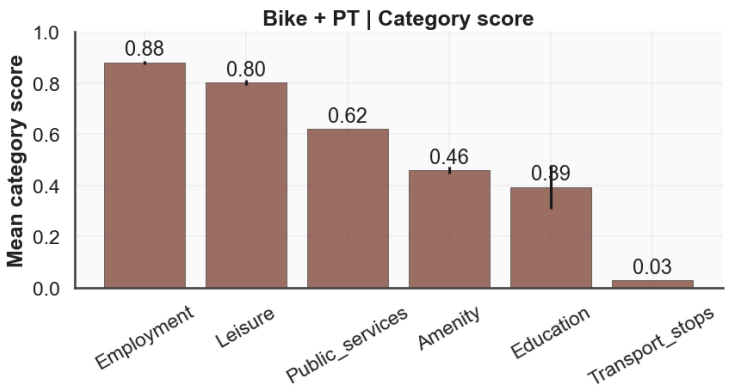


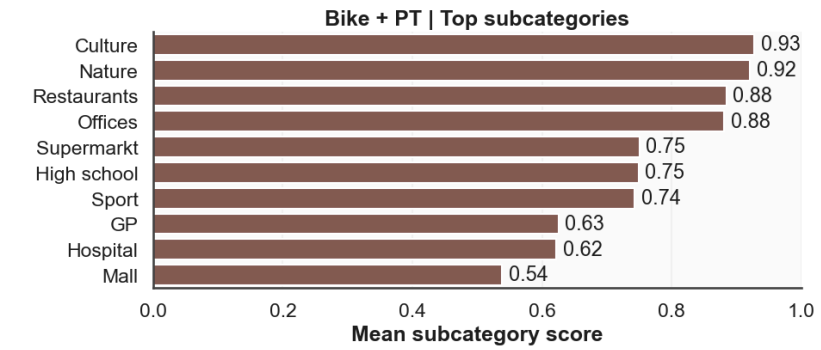




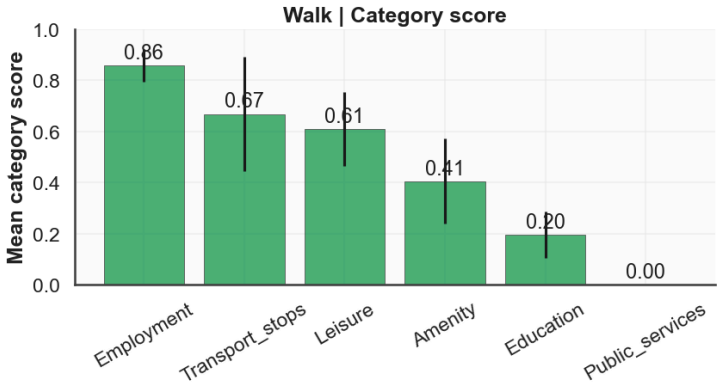


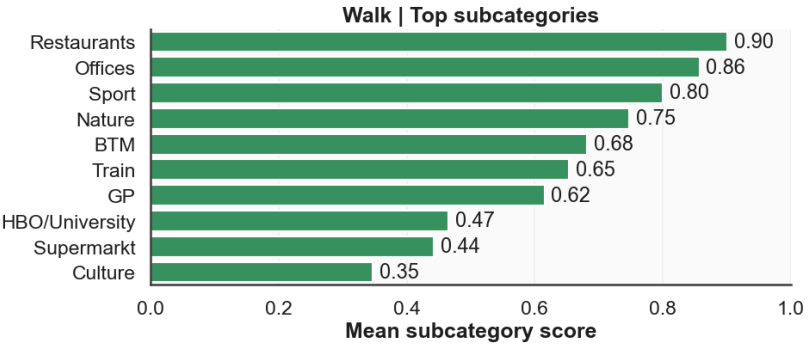


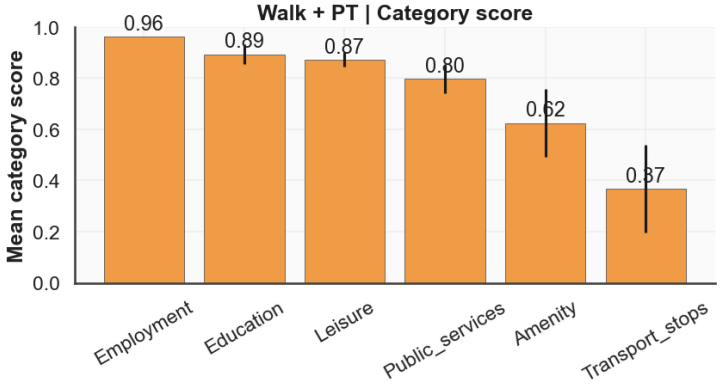


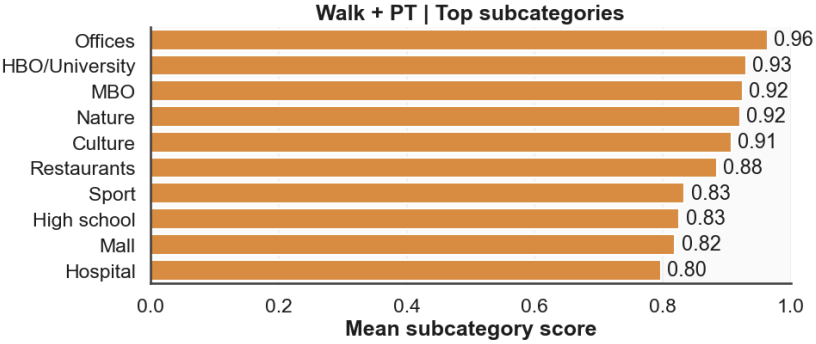


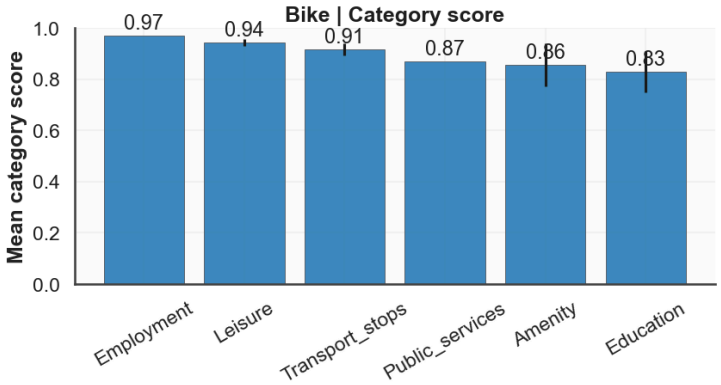
Denhaag_1: category and subcategory decomposition by mode

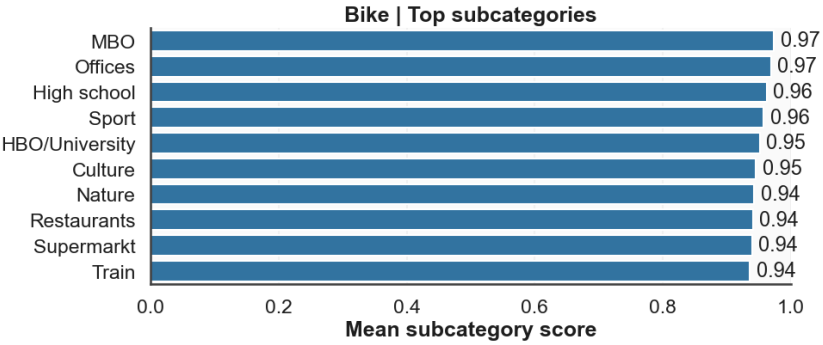


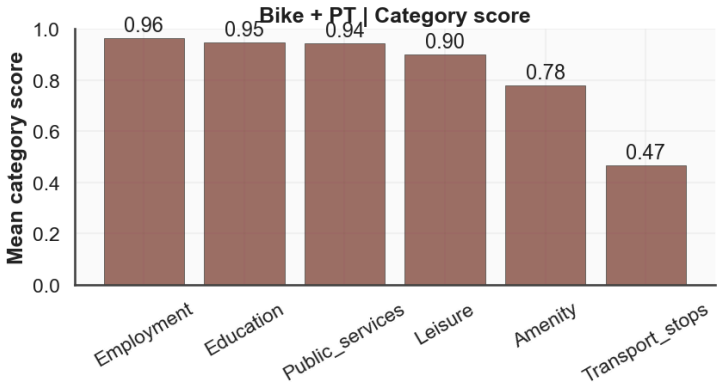


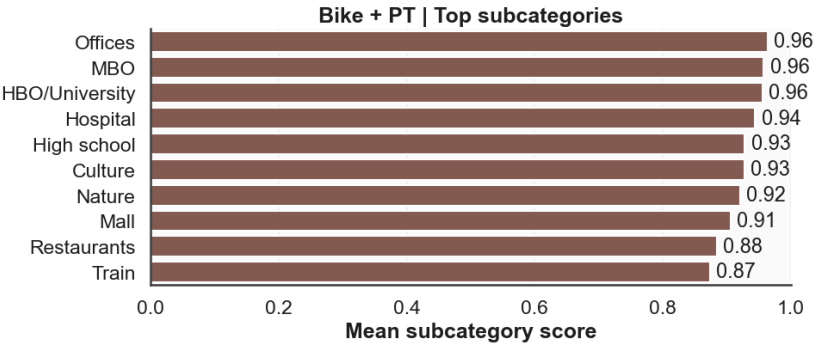




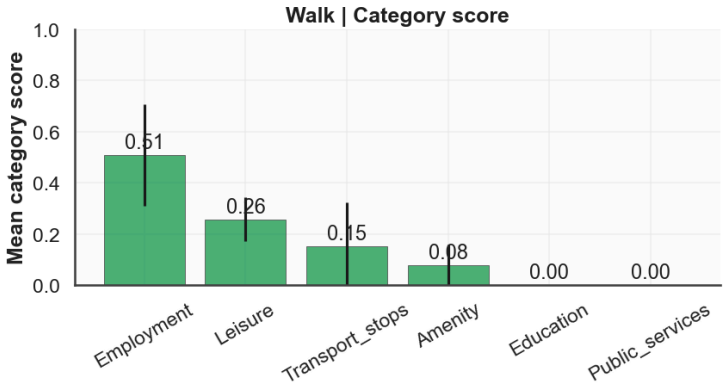


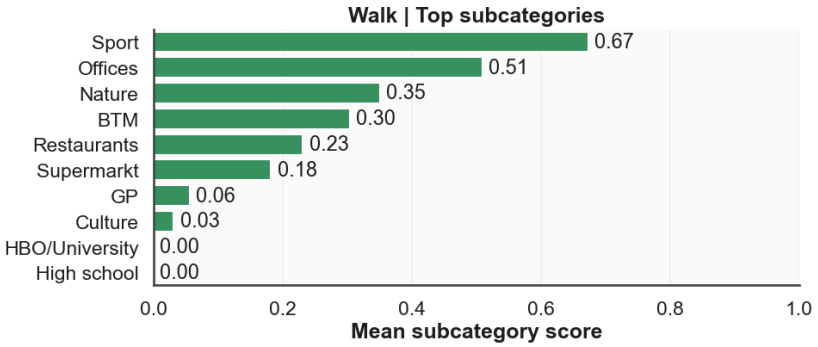


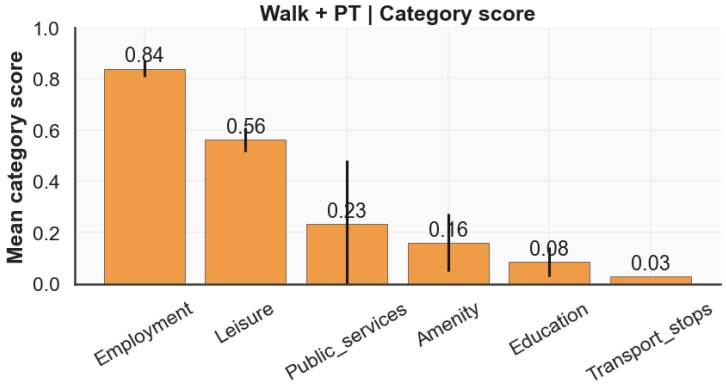


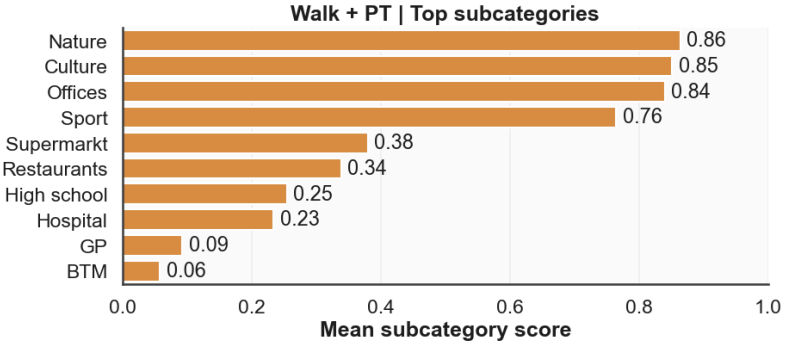


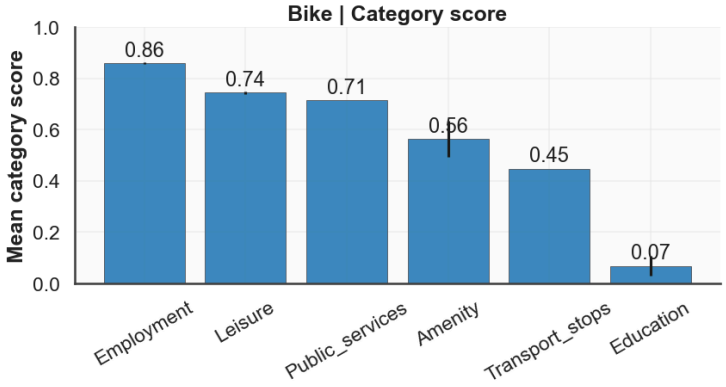
Smallingerland_3: category and subcategory decomposition by mode

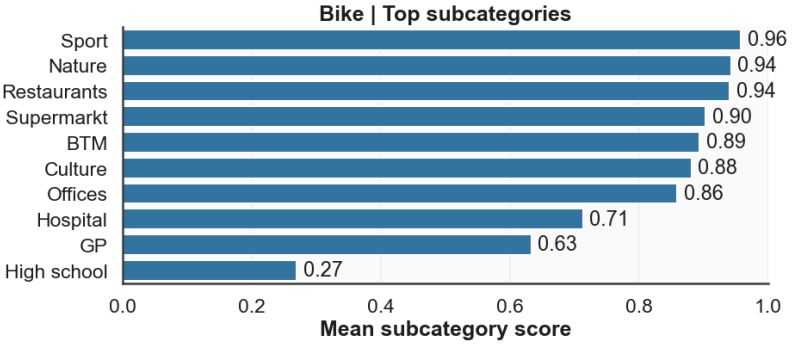


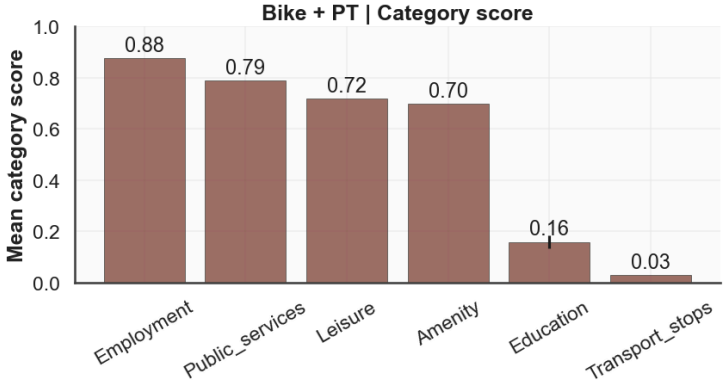


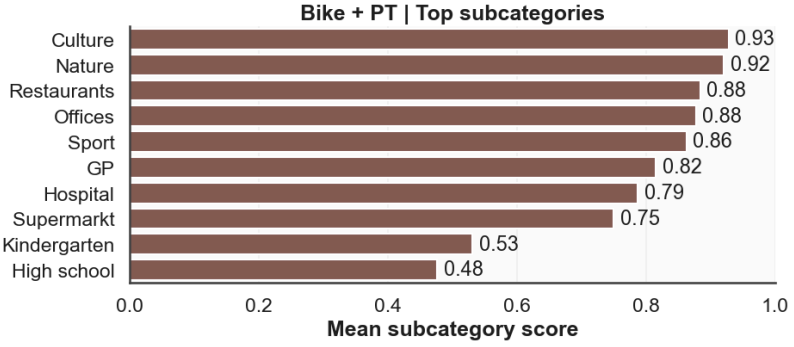




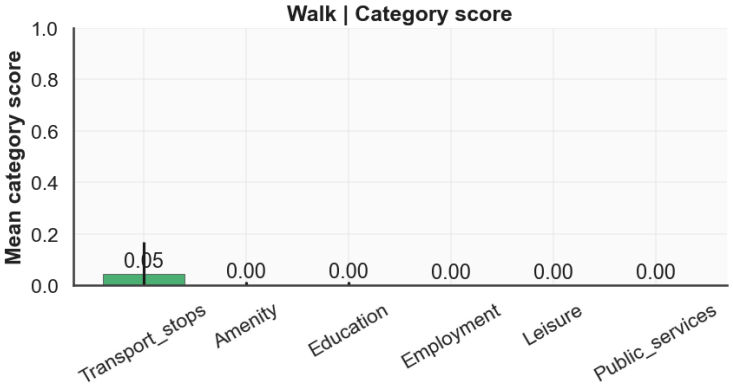


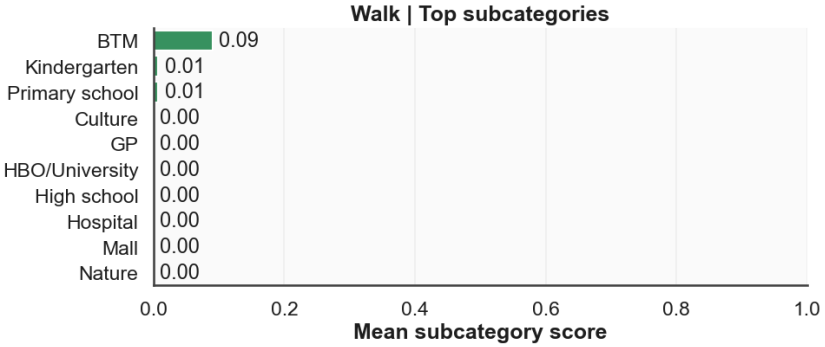


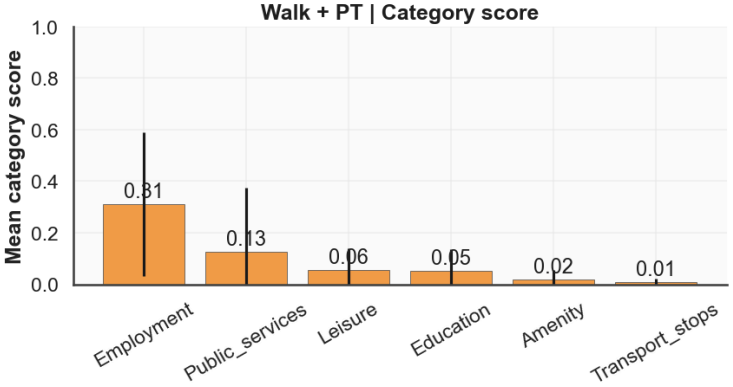


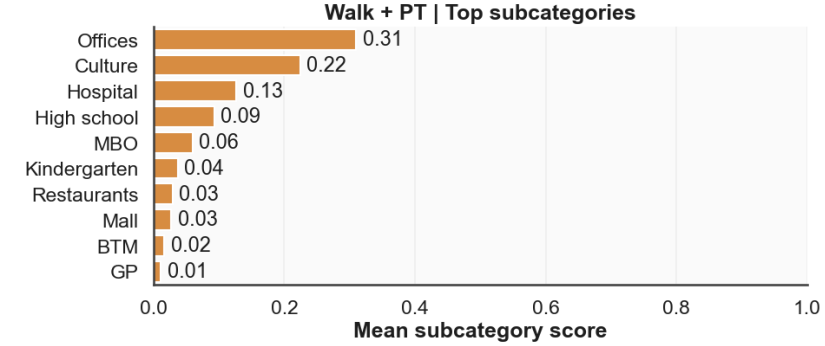


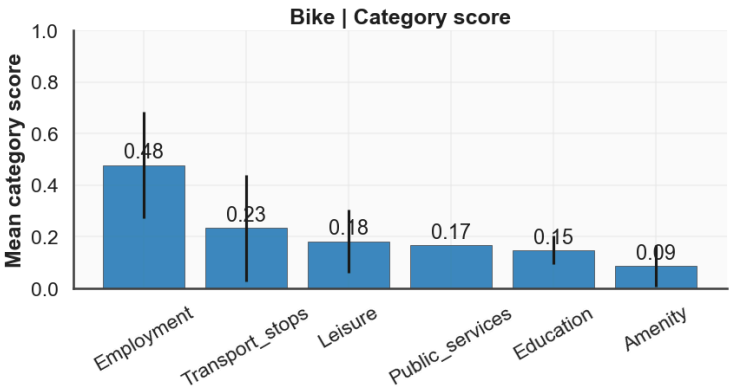
Hulst_5: category and subcategory decomposition by mode

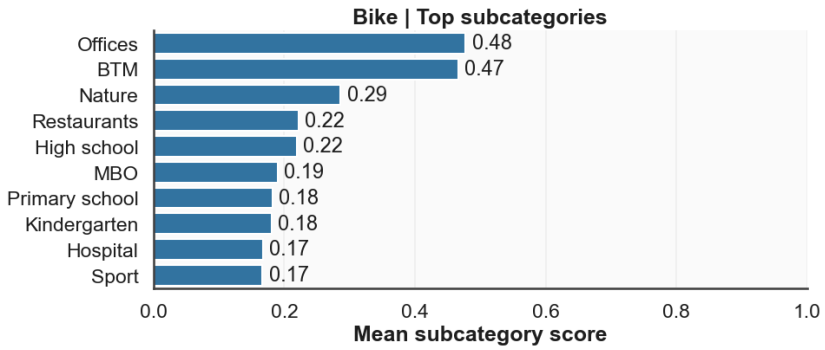


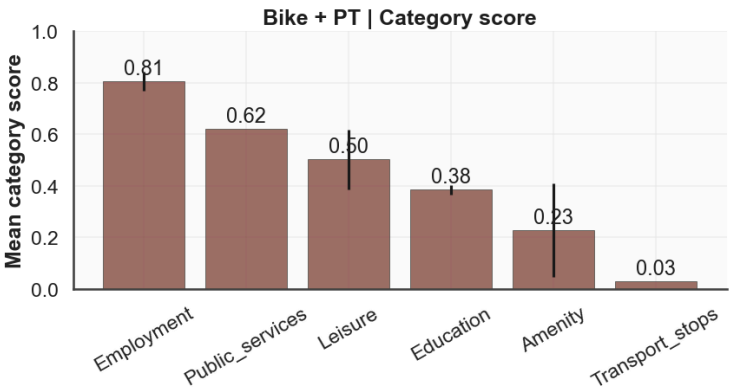


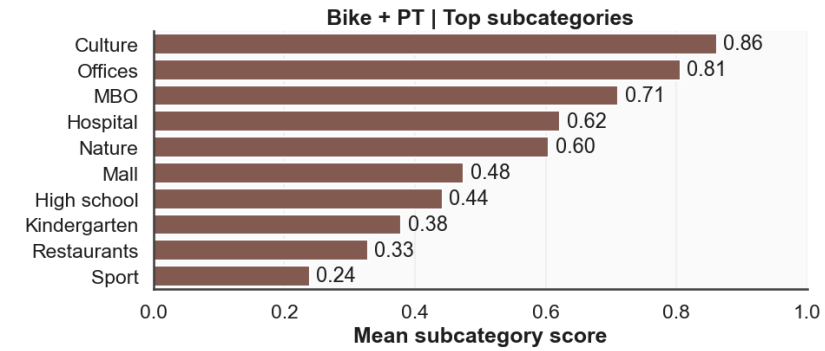




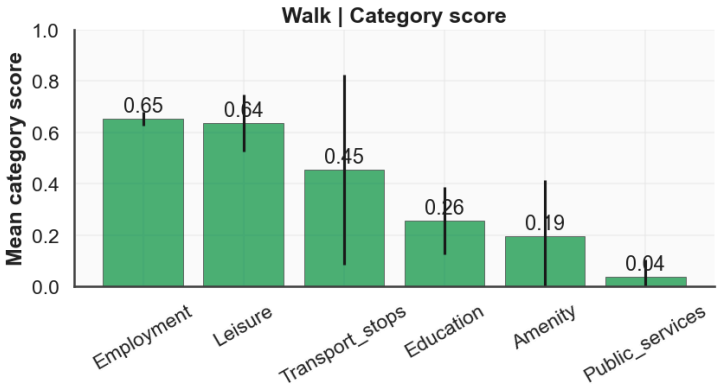


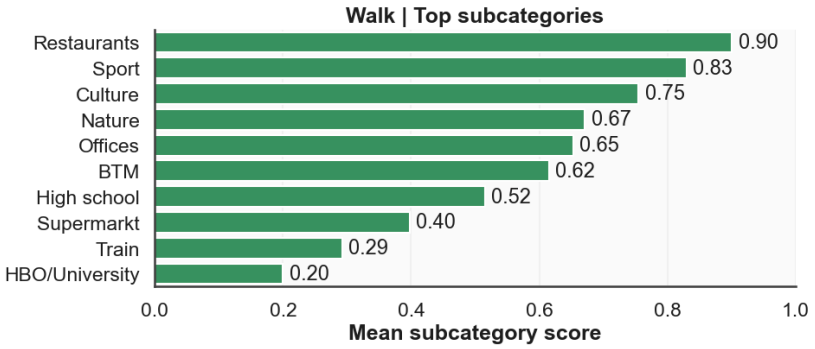


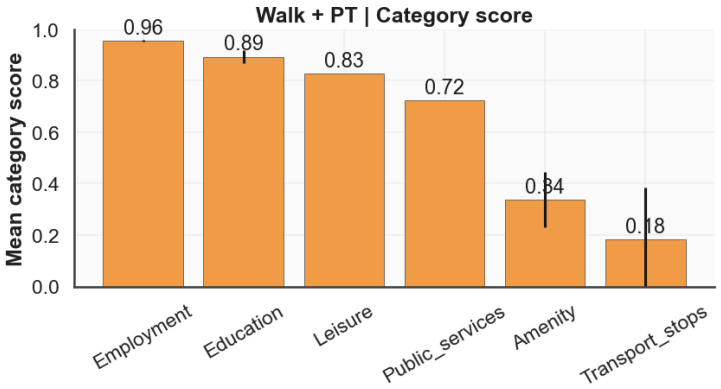


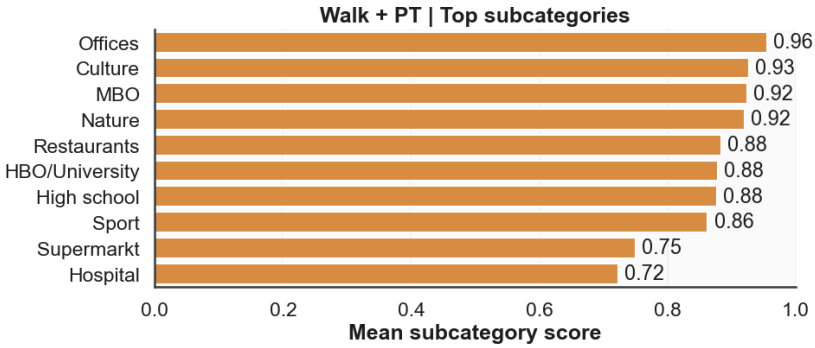


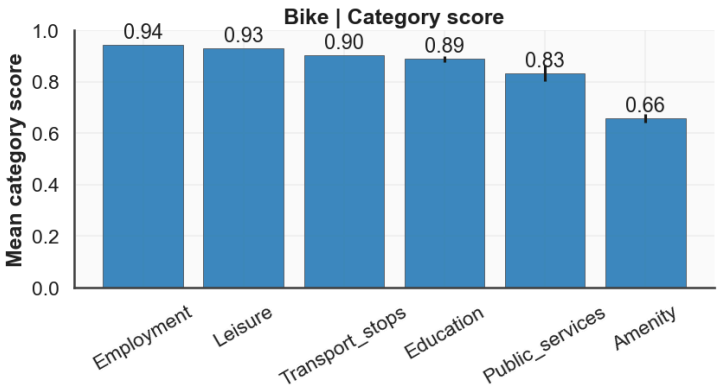
Tilburg_1: category and subcategory decomposition by mode

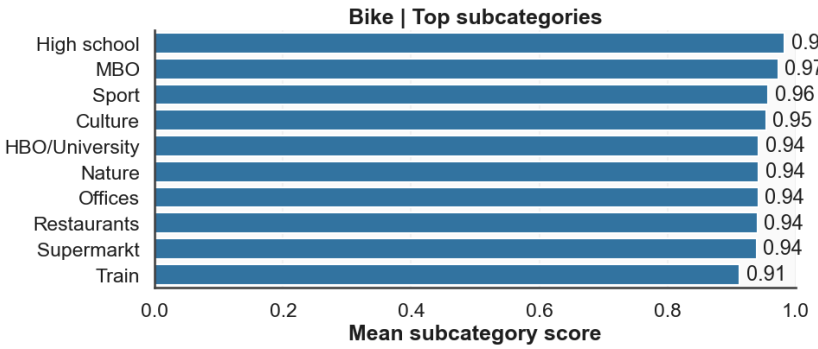


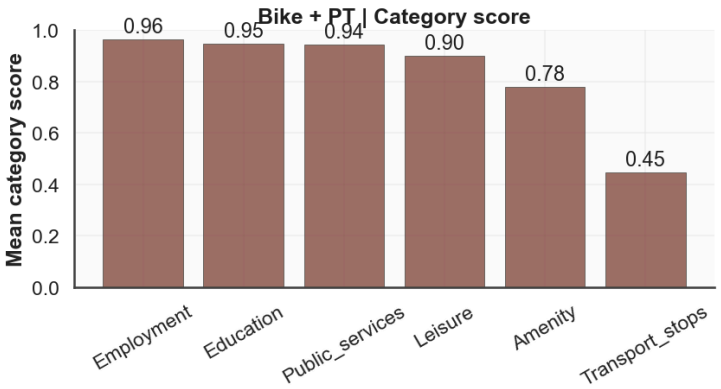


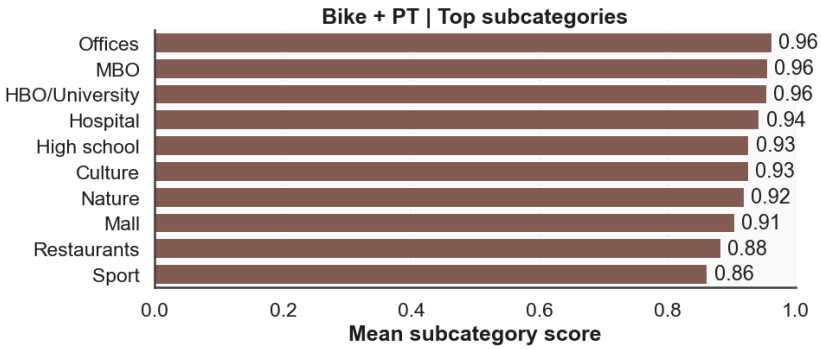




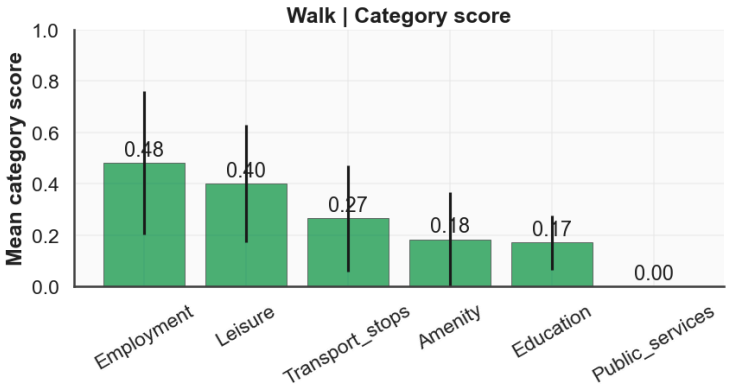


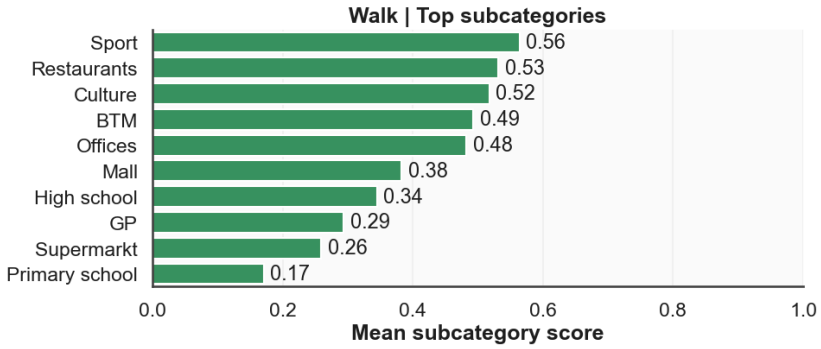


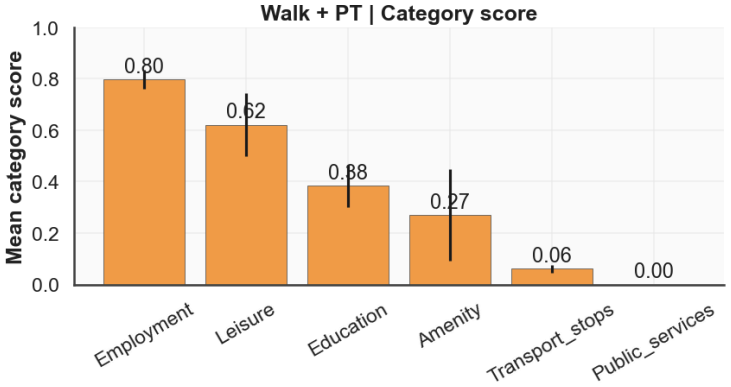


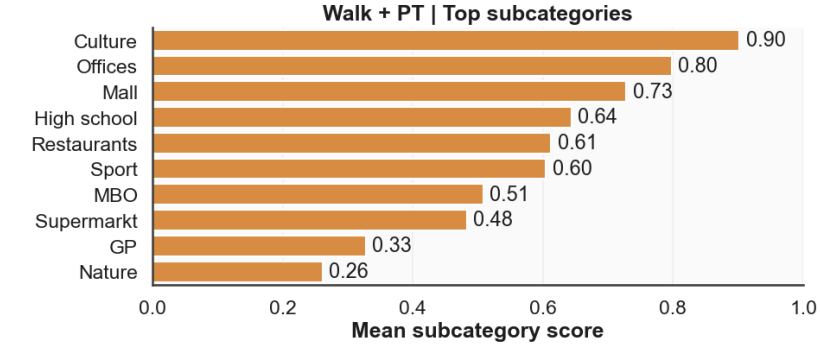


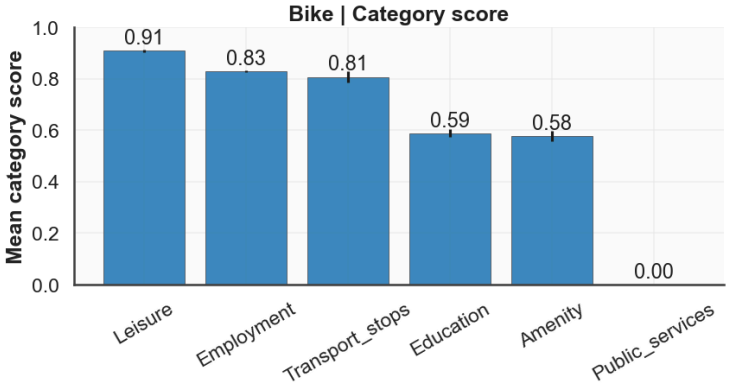
Dronten_3: category and subcategory decomposition by mode

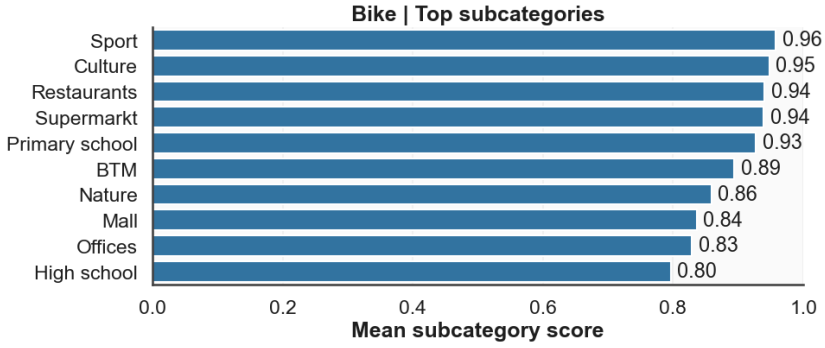


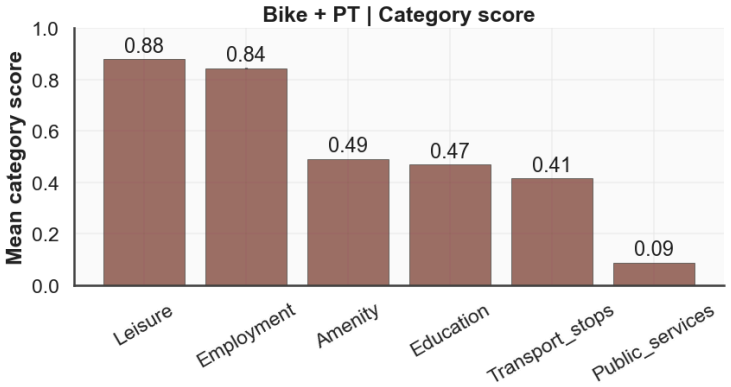


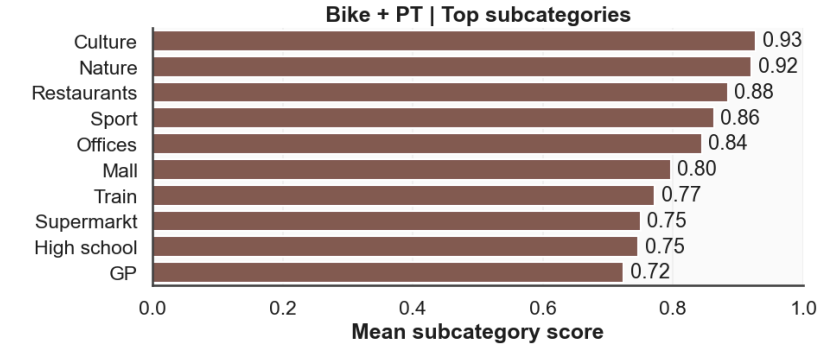




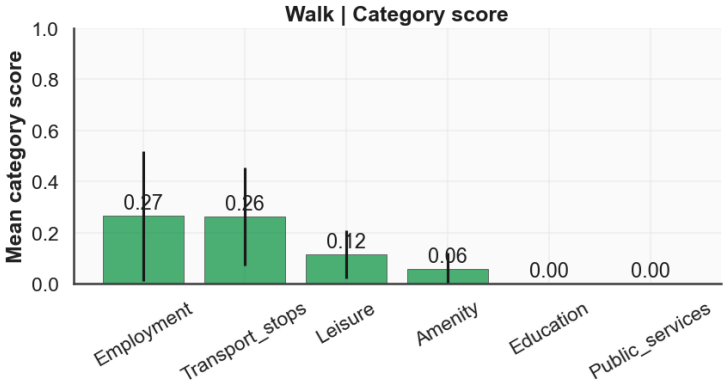


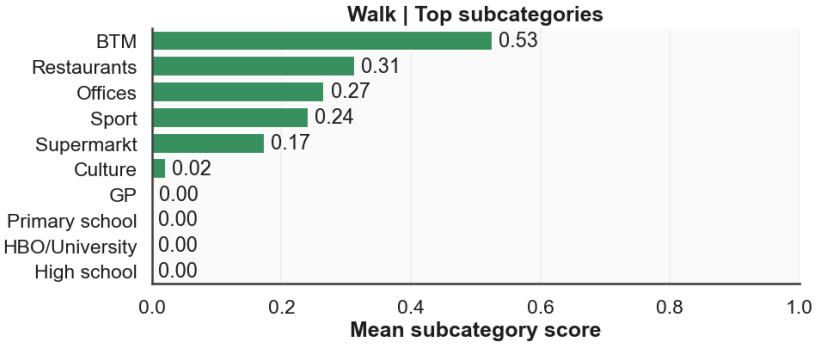


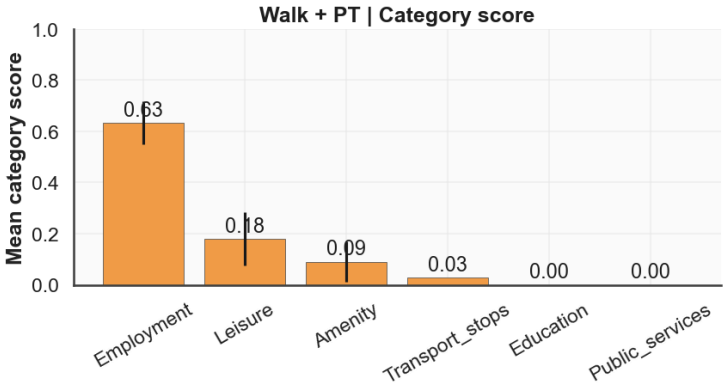


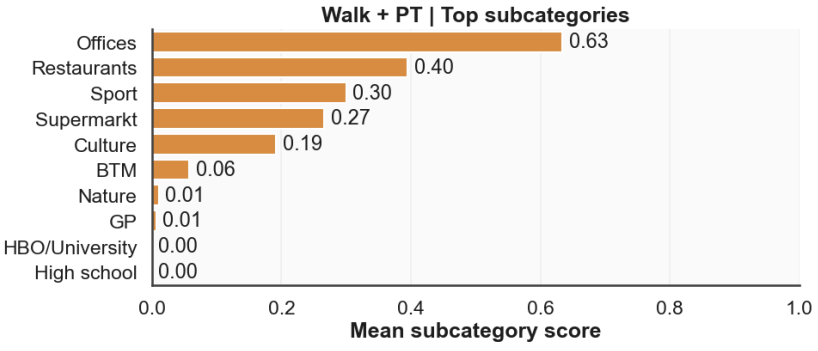


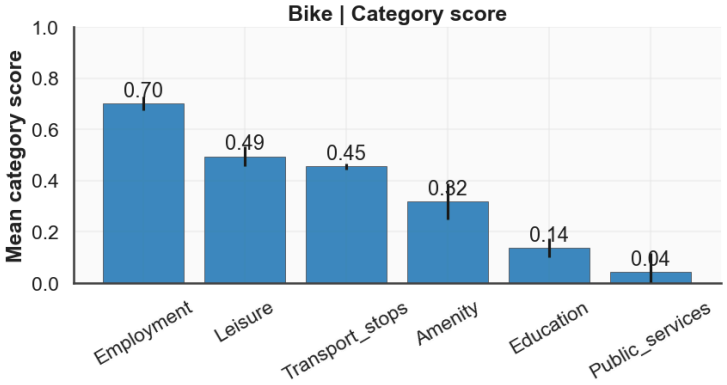
Schagen_5: category and subcategory decomposition by mode

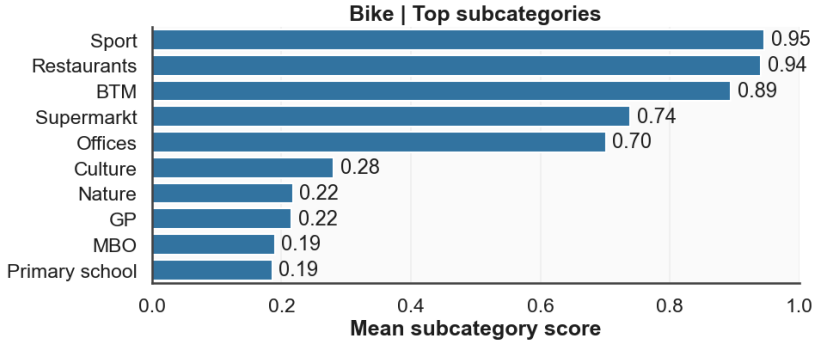


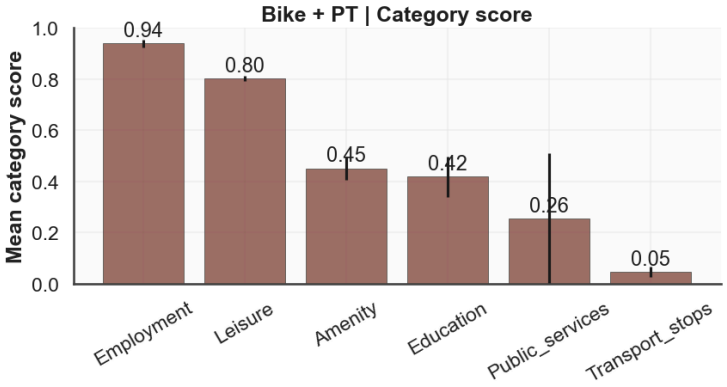


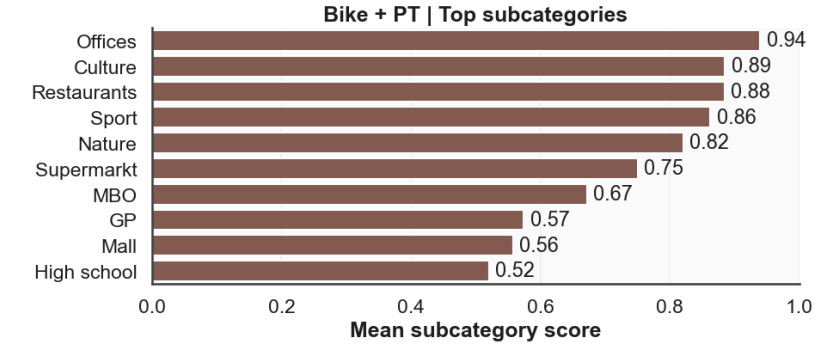








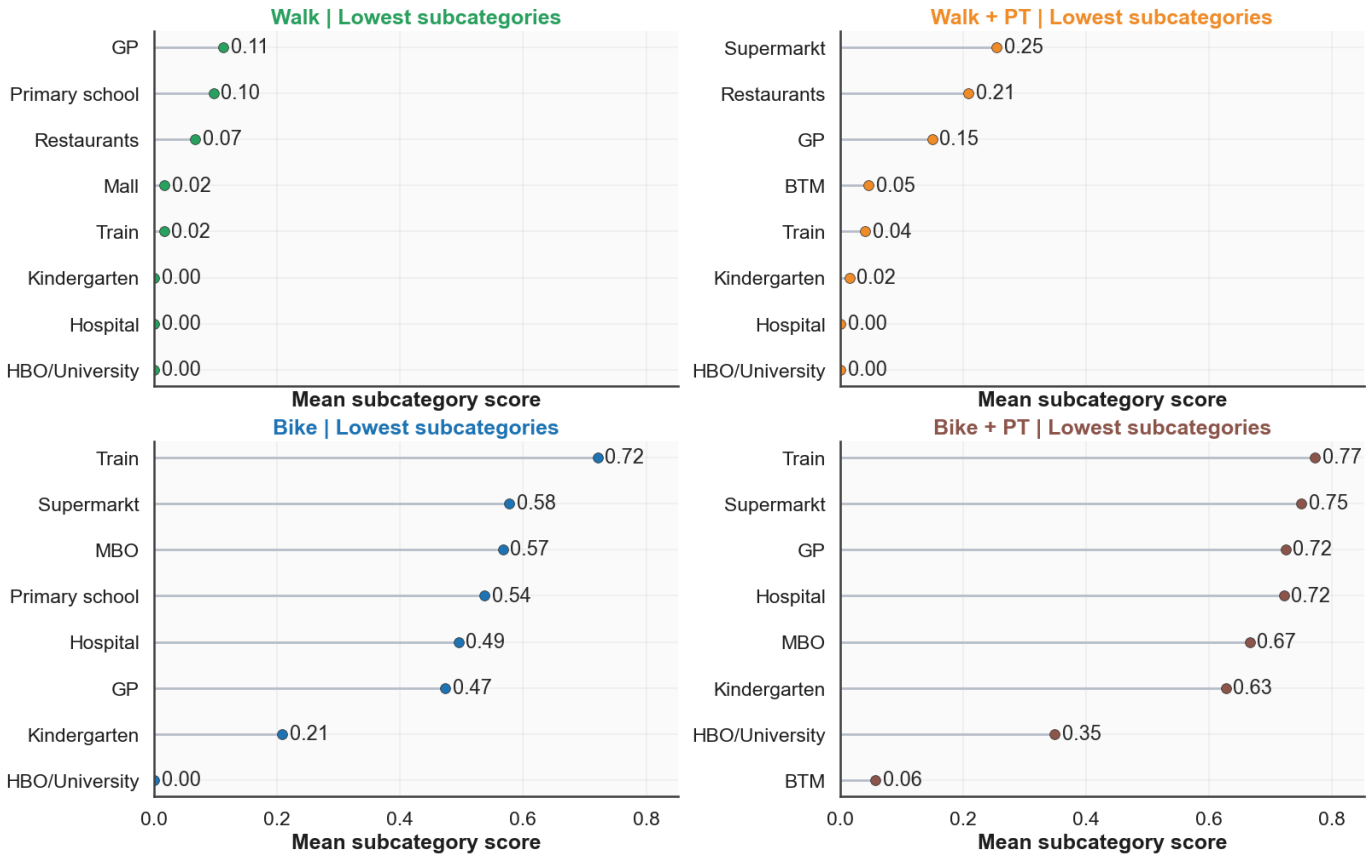




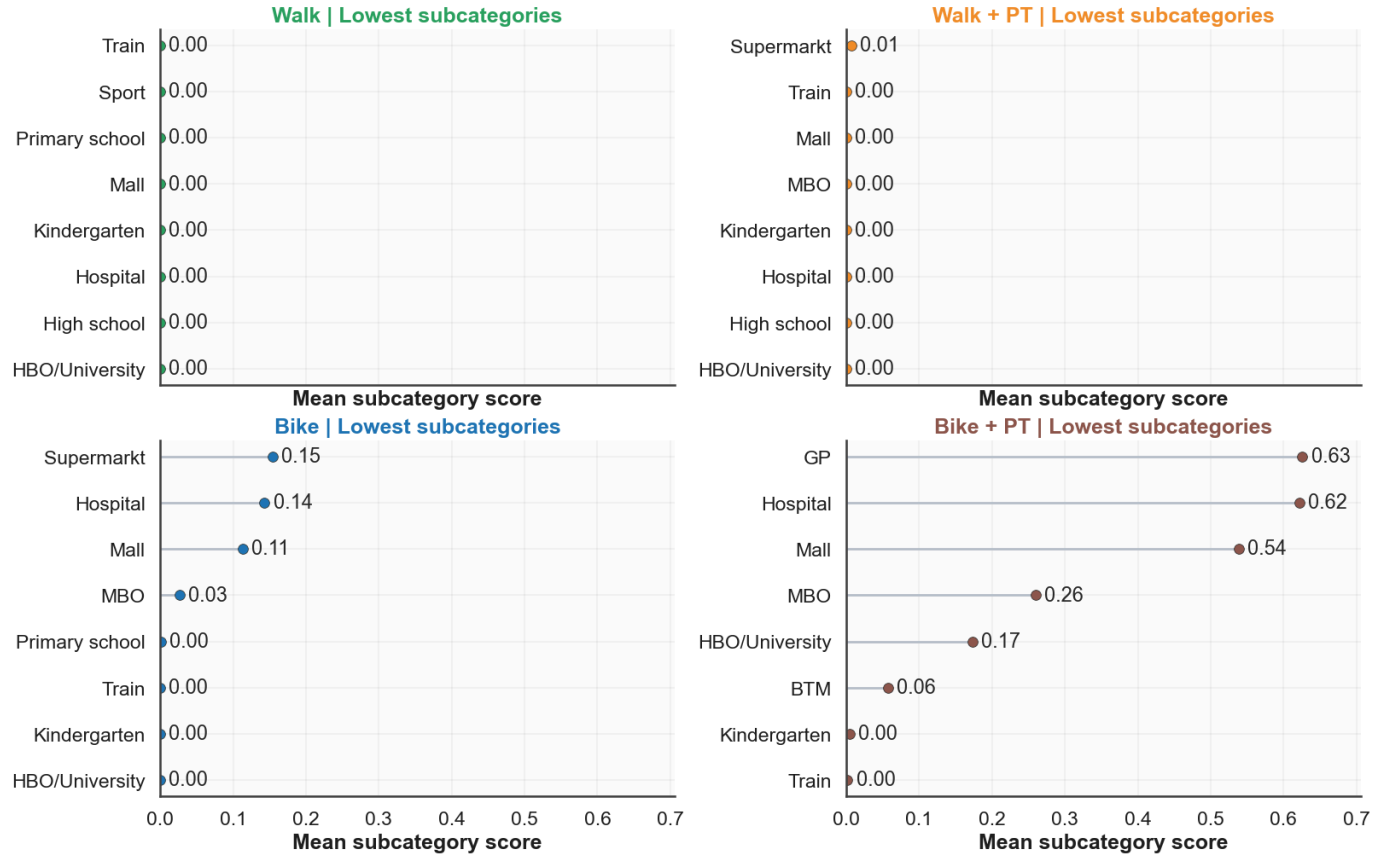
IV. Worst subcategory decomposition for each study area

These figures show the eight worst performing subcategories per mode for each neighbourhood.

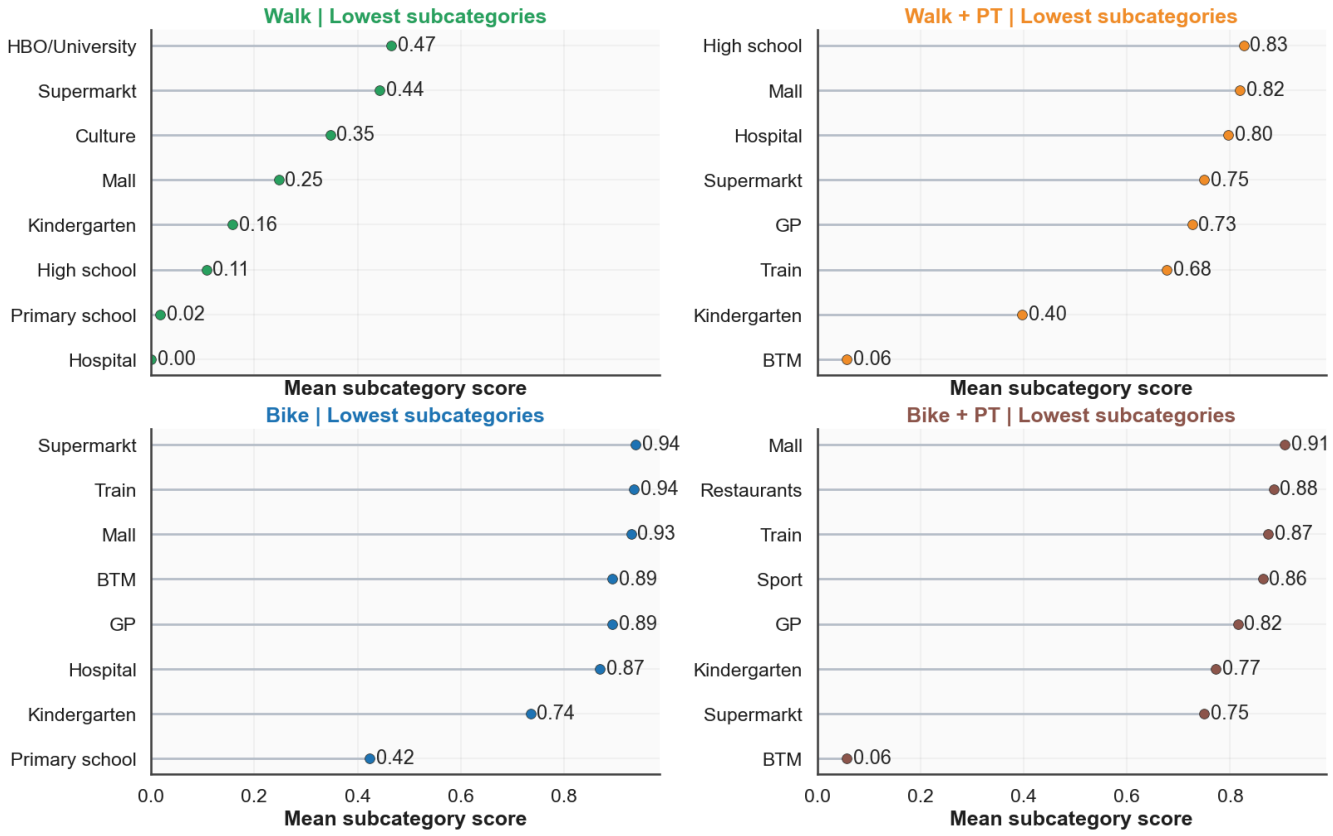
Zutphen_3: weakest subcategories by mode (bottom 8)



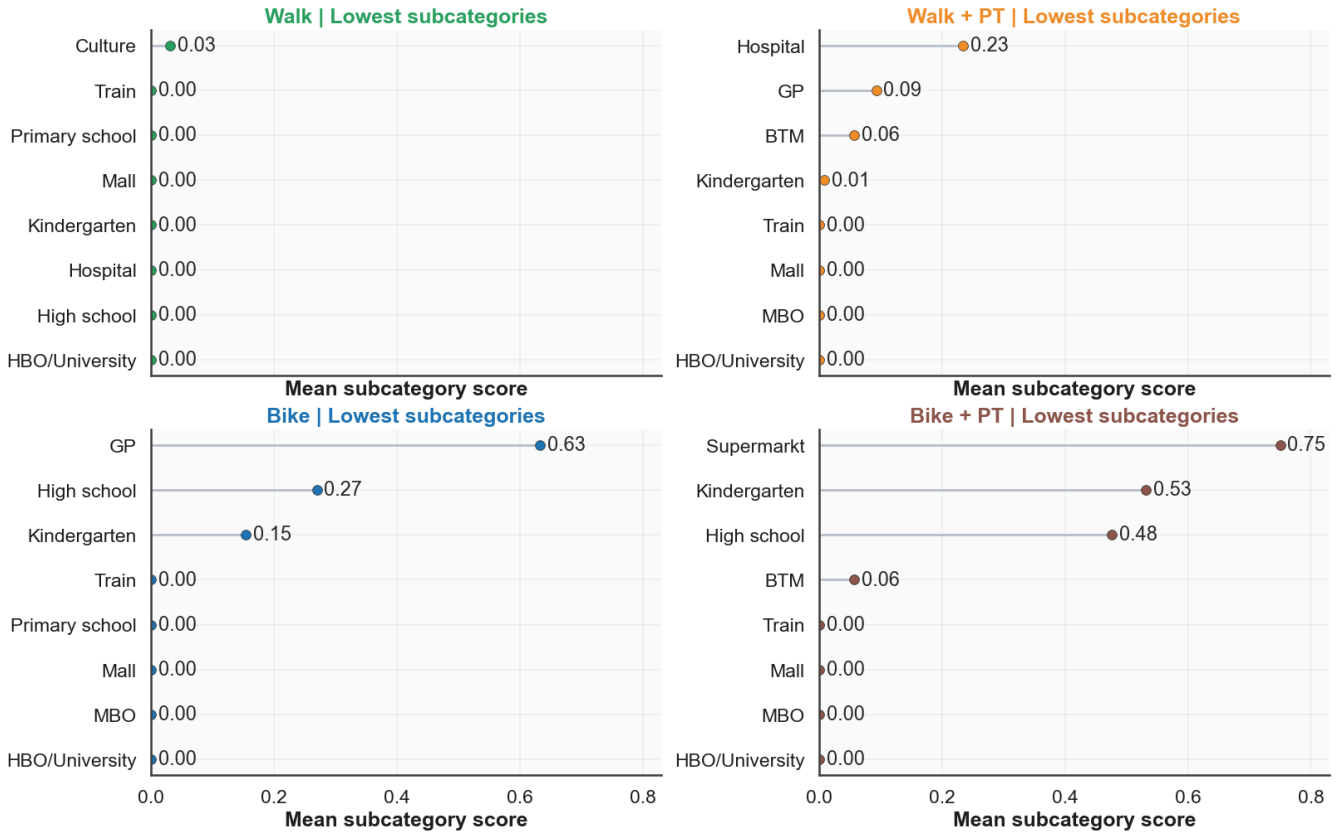
Heusden_5: weakest subcategories by mode (bottom 8)



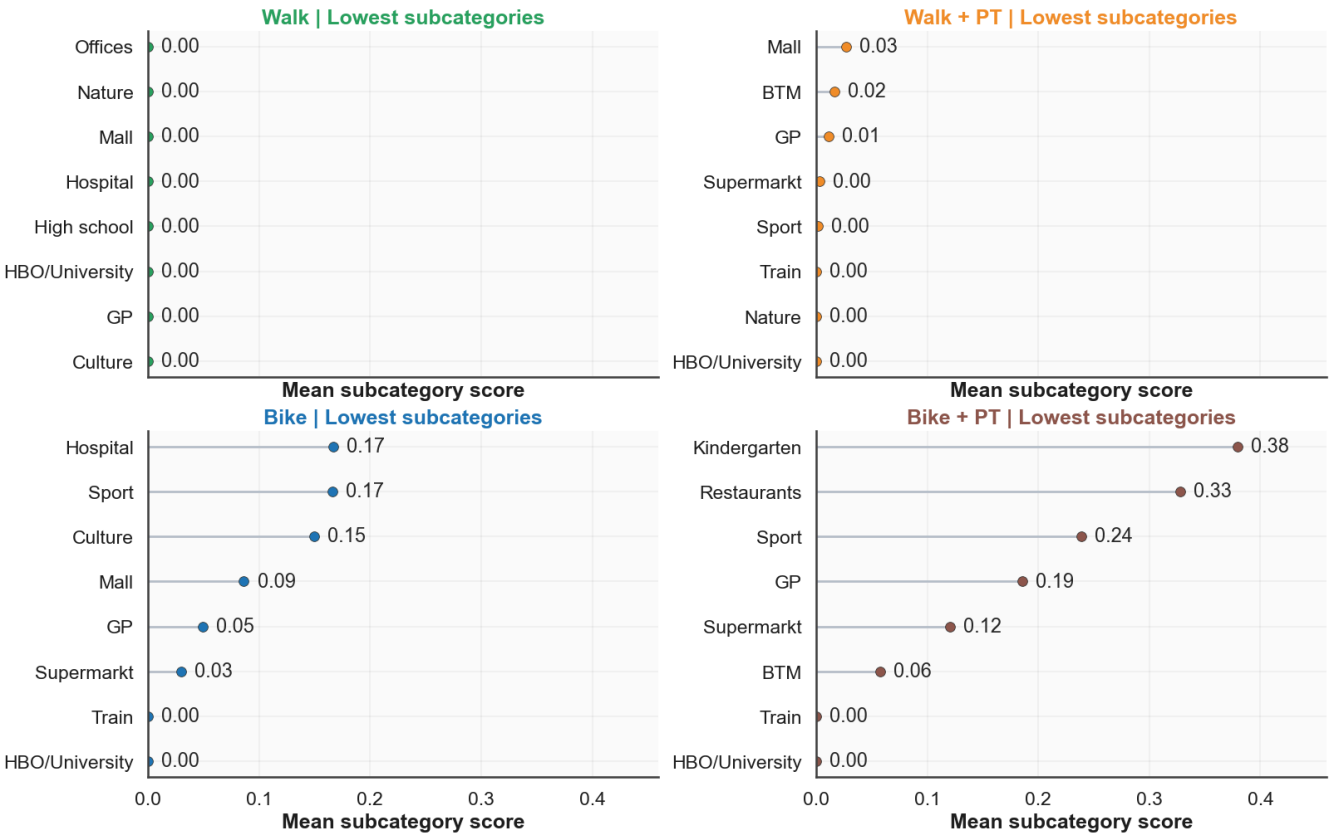
Denhaag_1: weakest subcategories by mode (bottom 8)



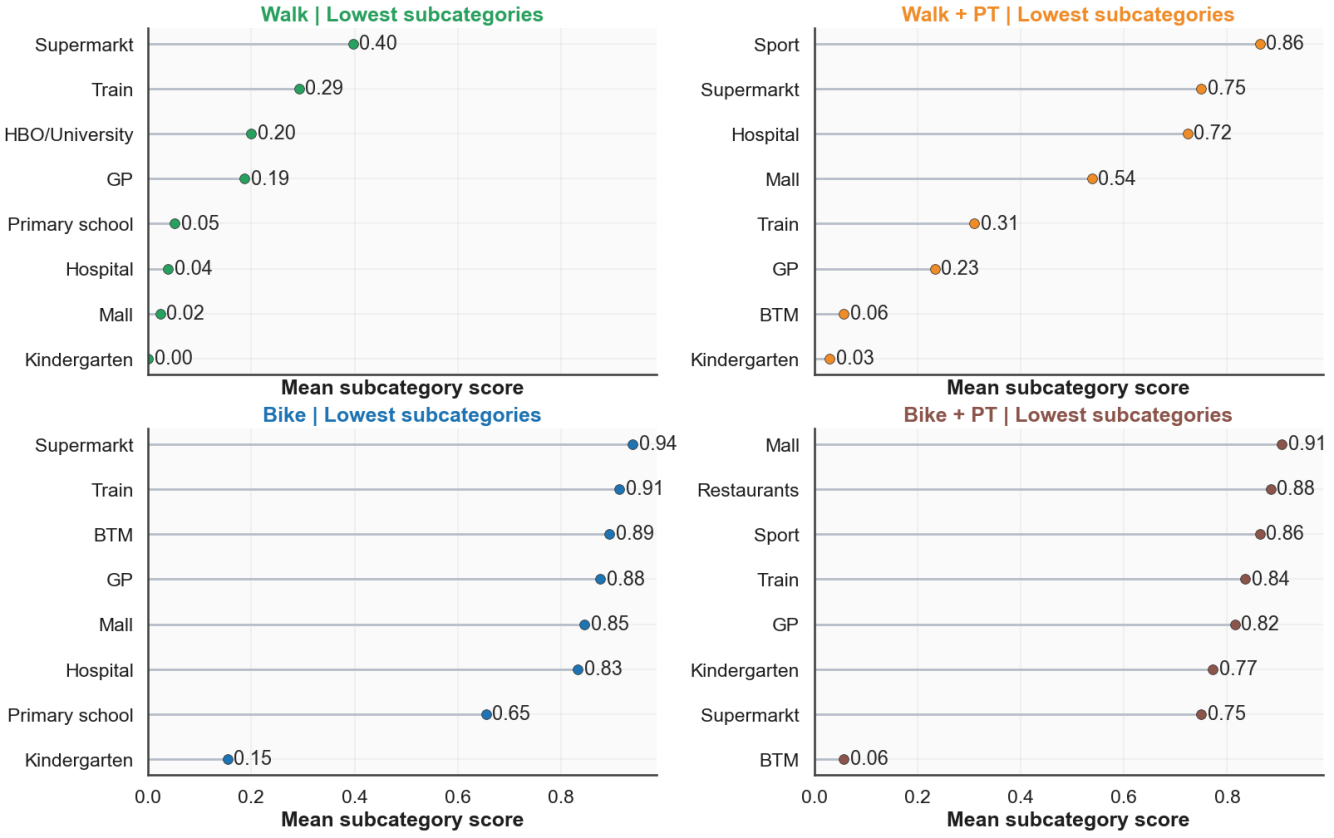
Smallingerland_3: weakest subcategories by mode (bottom 8)



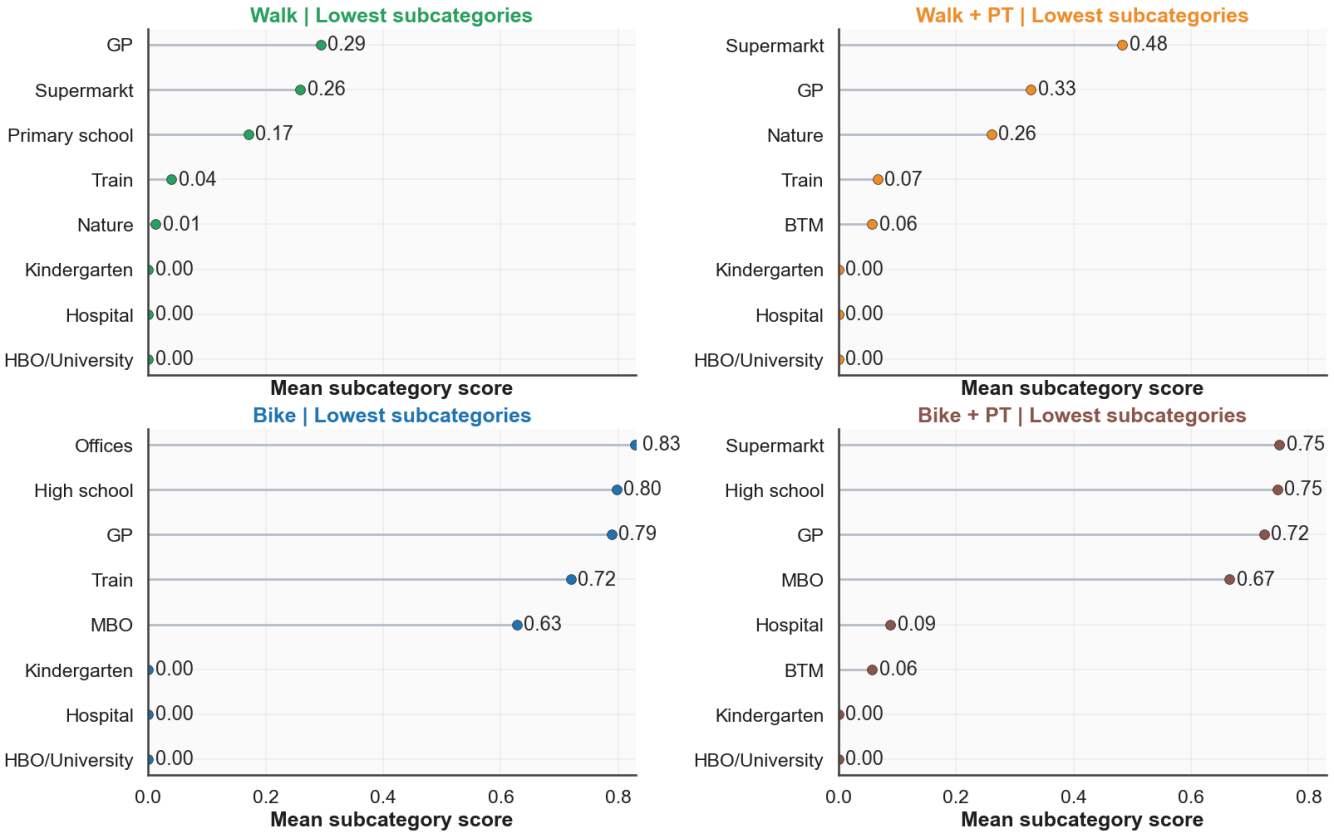
Hulst_5: weakest subcategories by mode (bottom 8)



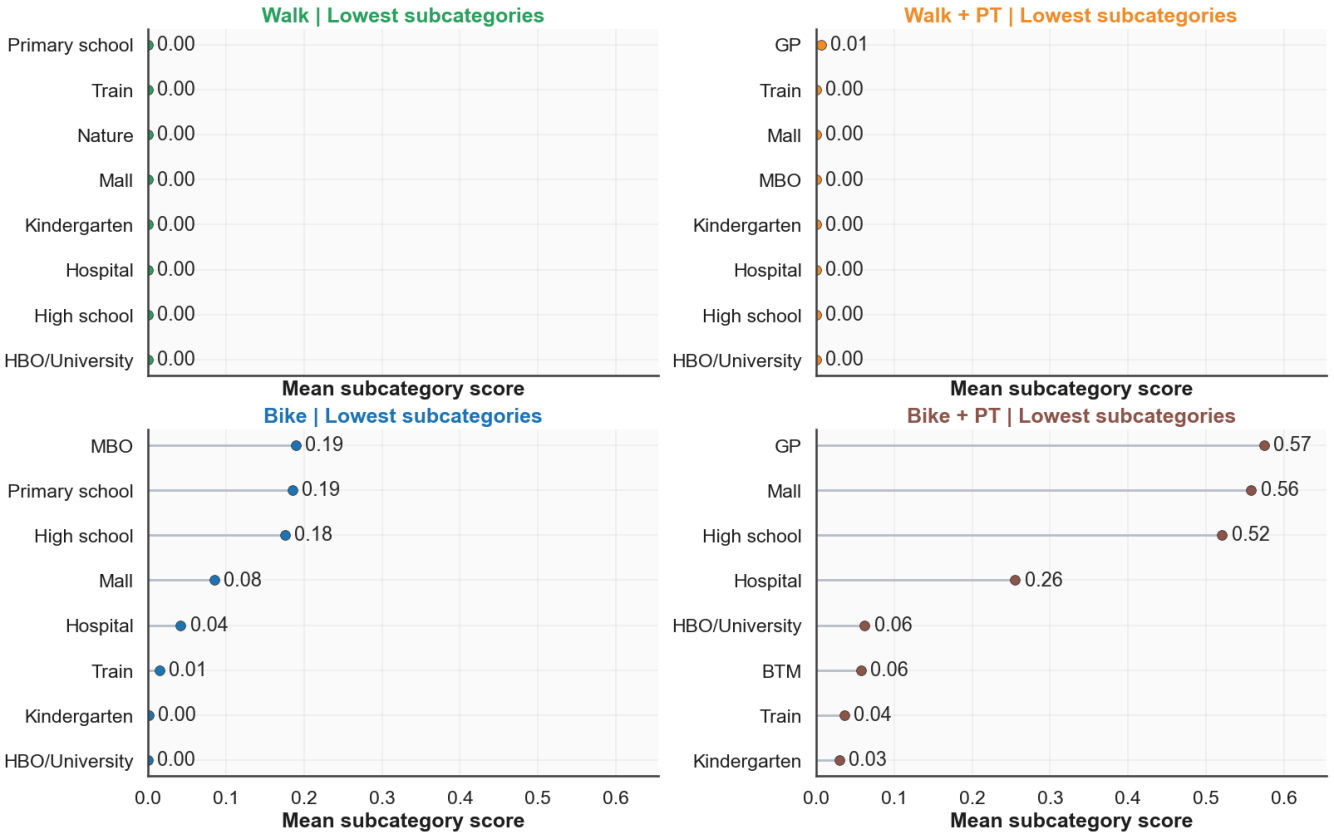
Tilburg_1: weakest subcategories by mode (bottom 8)



Dronten_3: weakest subcategories by mode (bottom 8)



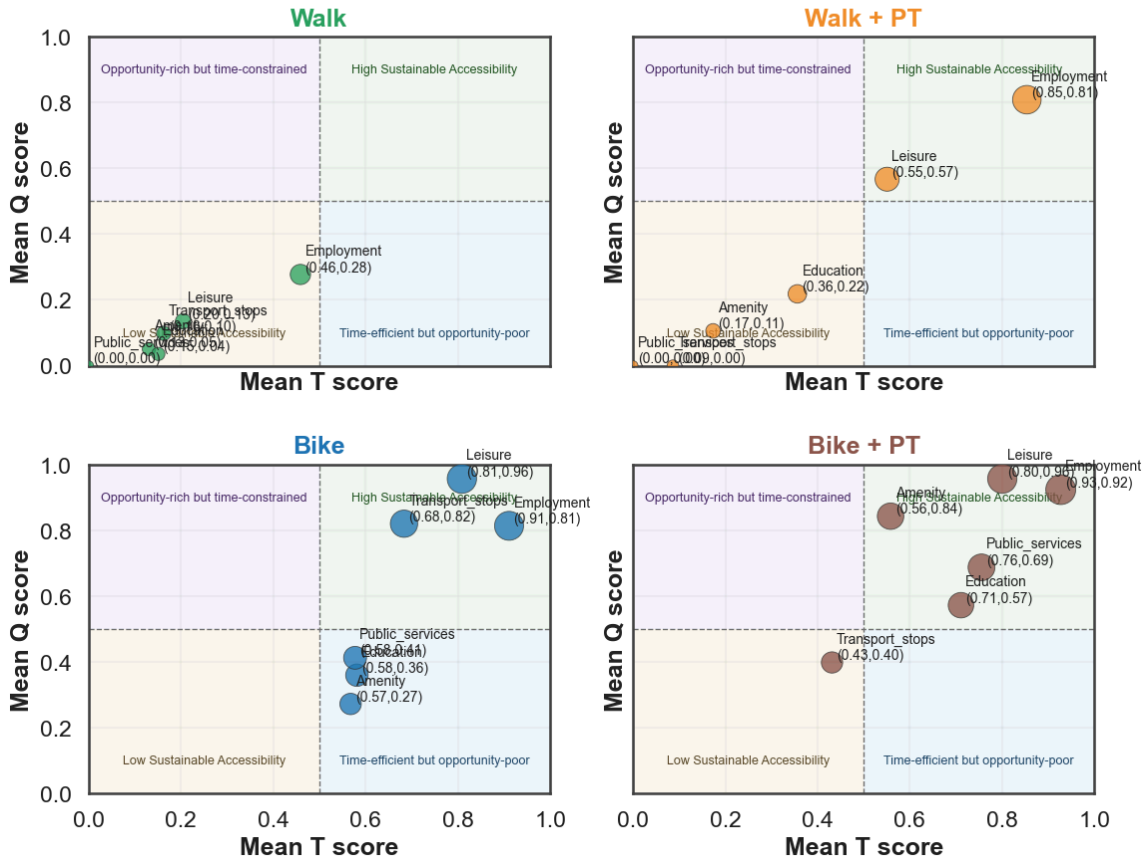
Schagen_5: weakest subcategories by mode (bottom 8)



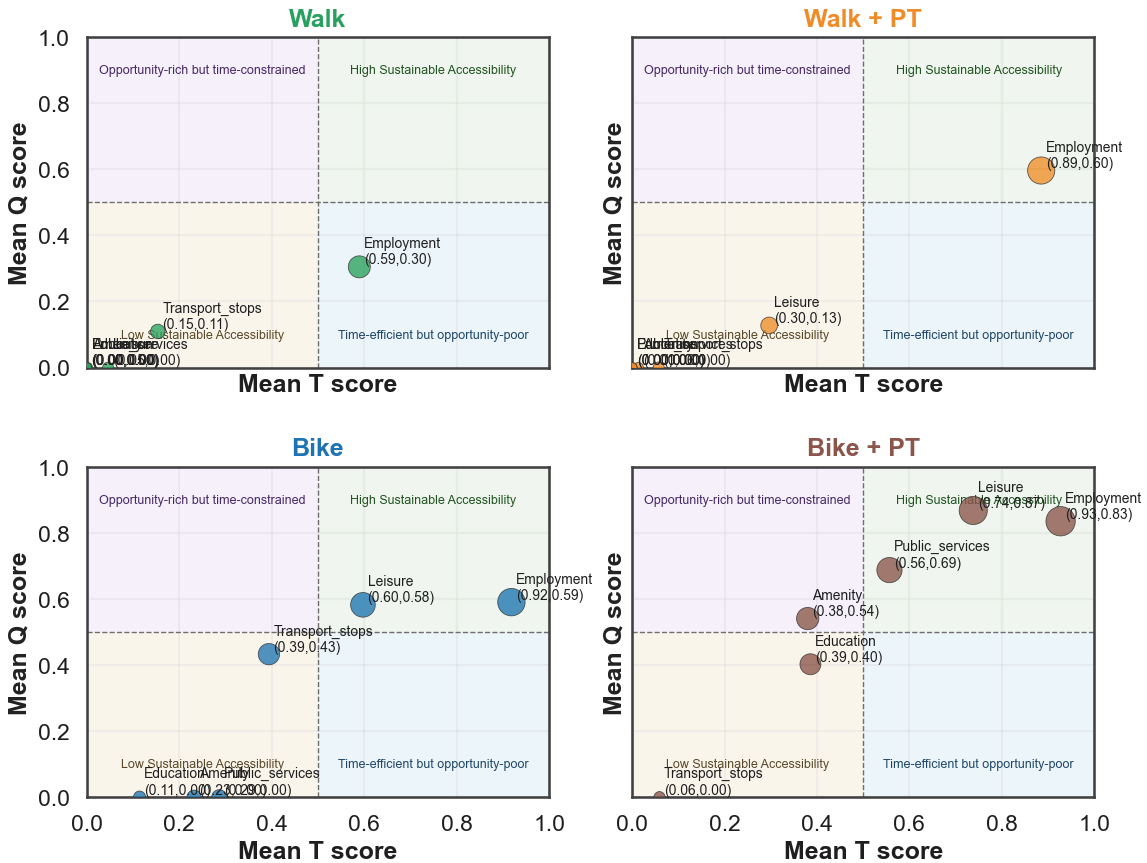
V. T/Q mechanism plot for each study area

These matrices show the average T- and Q-component score of each category by mode for the selected neighbourhoods.

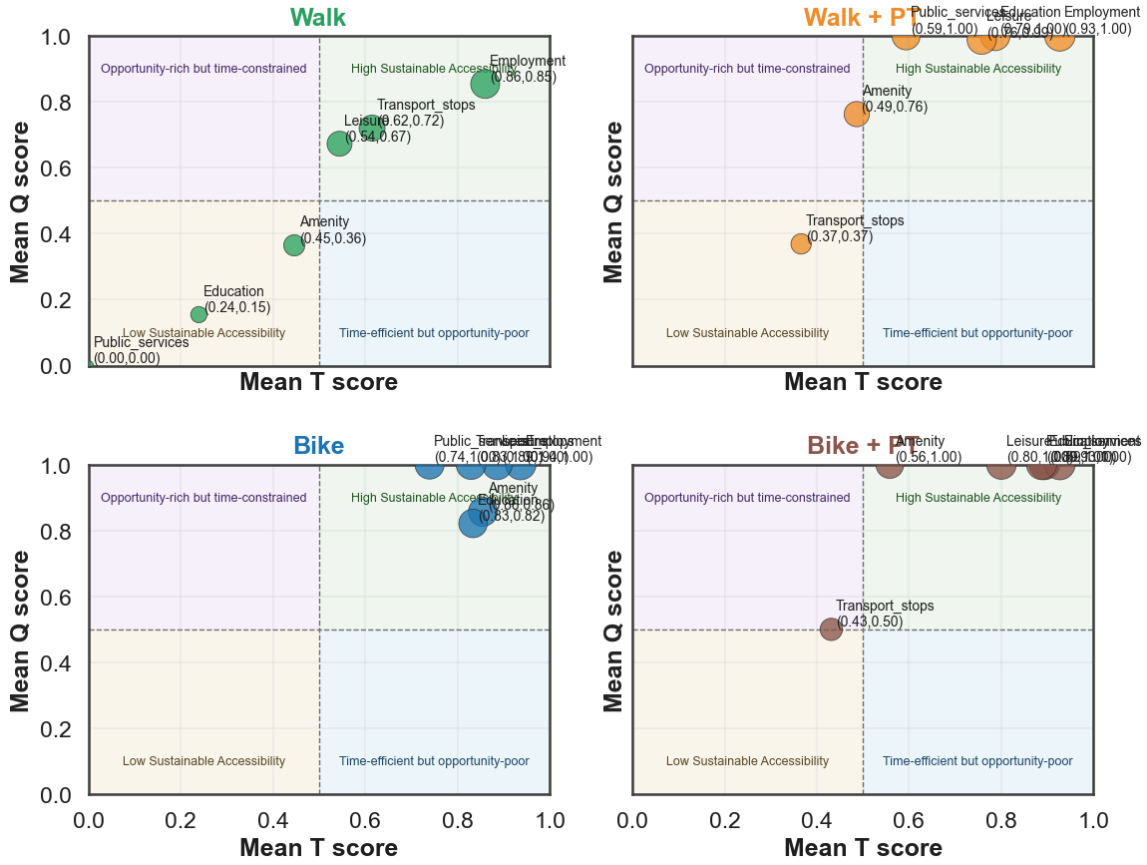
Zutphen_3: Averaged T/Q mechanism by mode and category



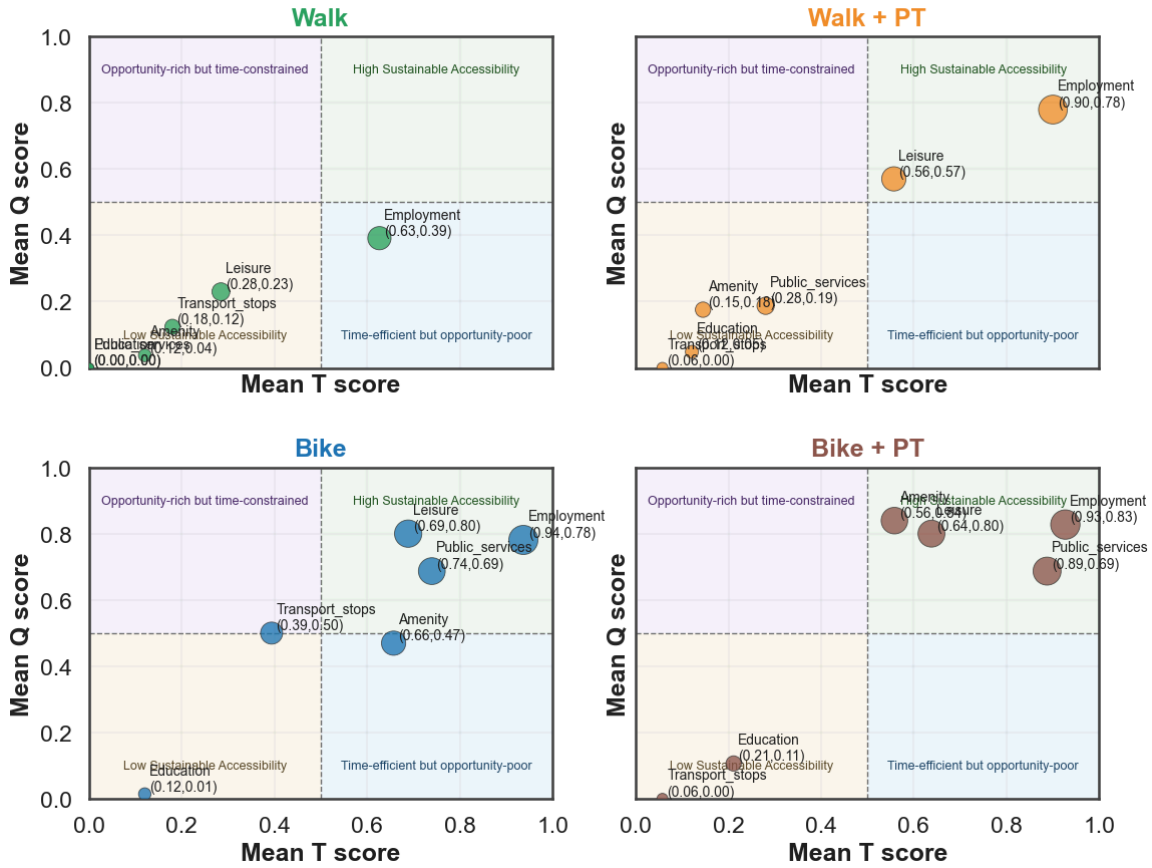
Heusden_5: Averaged T/Q mechanism by mode and category



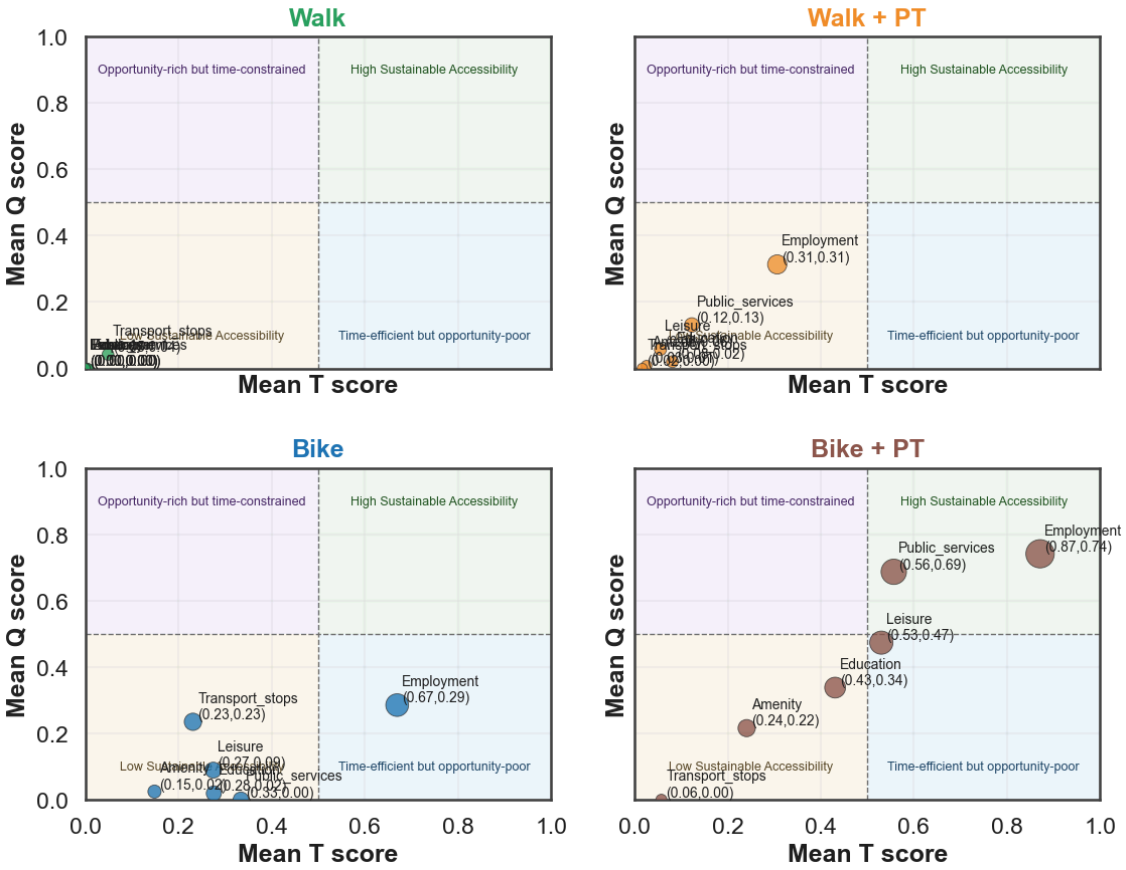
Denhaag_1: Averaged T/Q mechanism by mode and category



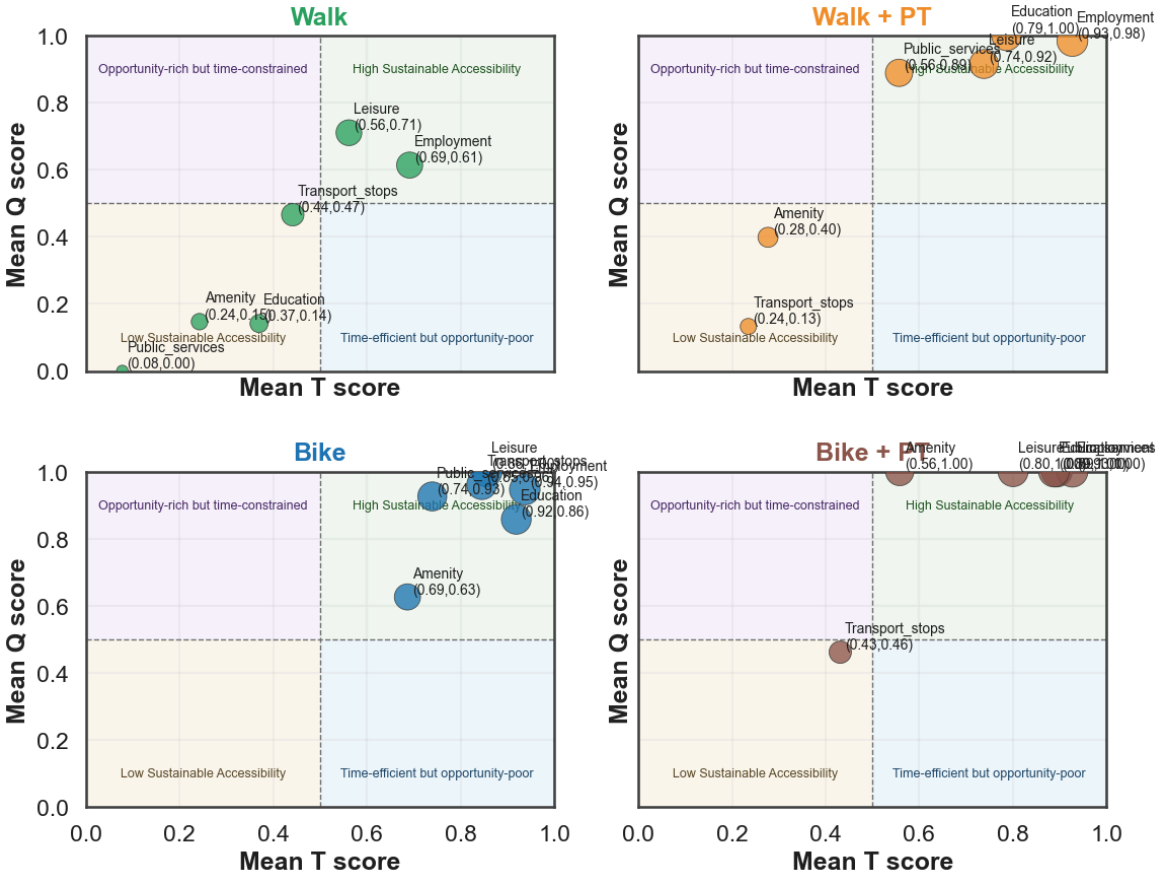
Smallingerland_3: Averaged T/Q mechanism by mode and category



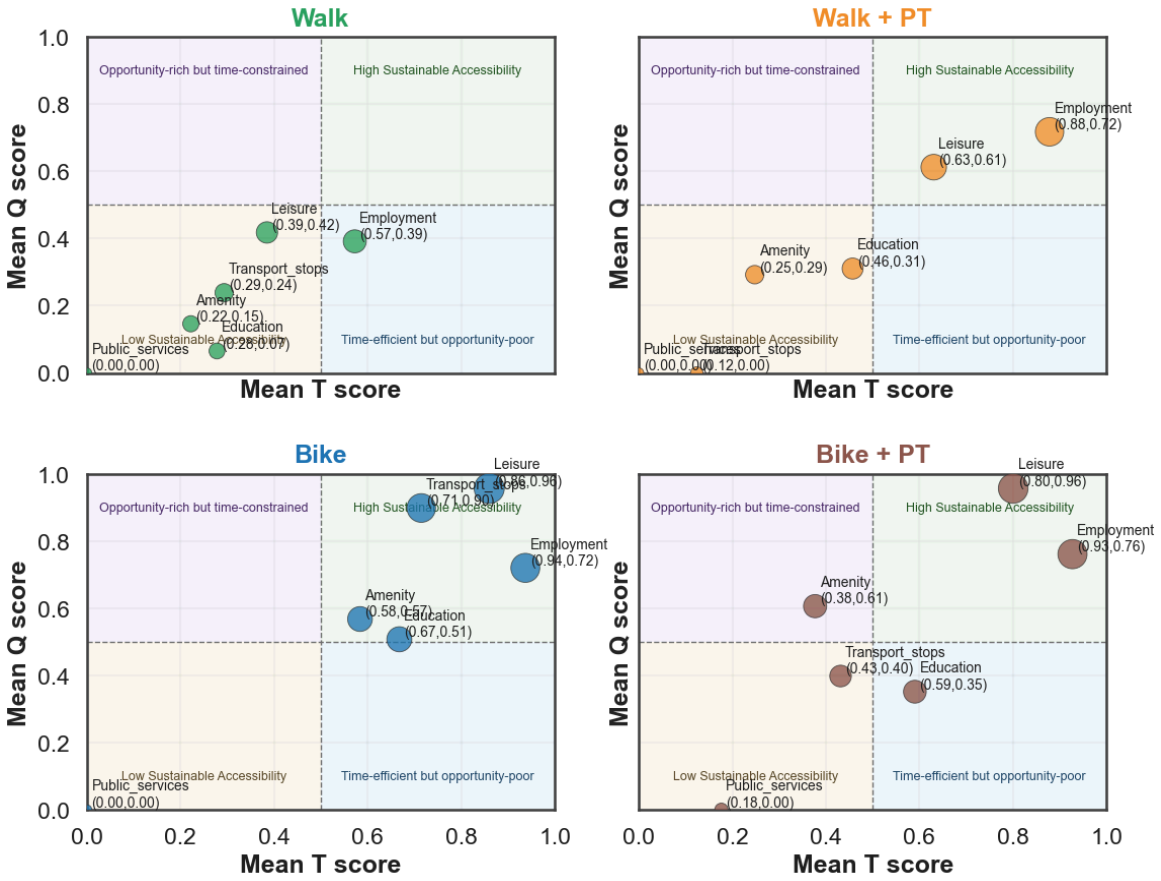
Hulst_5: Averaged T/Q mechanism by mode and category



Tilburg_1: Averaged T/Q mechanism by mode and category



Dronten_3: Averaged T/Q mechanism by mode and category



Schagen_5: Averaged T/Q mechanism by mode and category

